

# Supplementary material (Online Resource 1): the derivations to the article “Universe24: an integer cellular automaton read through clicks, and the formulas it shares with nature”, submitted to Foundations of Physics

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This supplement gives the algebraic steps of every result the paper marks DERIVED or COMPUTED without a proof in its text, numbered as added, each naming the paper’s section it serves; every statement here about the implementation is of the archive’s main branch at the commit d00aa9e2 of 2026-10-04, the pinned commit, the commit the two slits ran at, as the paper’s Section 3.7 says; the theorems the paper states are proved in S.48; the one-line theorems carry their proof in their sentence. References by author and year are the paper’s bibliography’s. Each derivation names its inputs from the paper, its steps, and, where a number was computed, the computation and its check. The notation is the paper’s: Rule3’s coefficients at a Node of clock pace  $p_0$  and Link paces  $p_a$  are  $R_a = 2 \text{ num } p_a^2$  on the two arrivals of the axis  $a$  where the axis’s two Links share one pace (the six Port coefficients  $R_{ij}$  of Eq. (1) of the paper in general),  $S = 12 \text{ den } \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{ num } \sum_a p_a^2$  on the Node’s own level, and the wall  $w = 6 \text{ den } \Gamma^2$ ; the content read at the Node is  $x = c/\Gamma$ , also written  $U$ ;  $g = (\text{den} - \text{num})/\text{den} = 1 - \cos \omega_0$ . The lattice’s own order: every derivation below is exact algebra of the line where it says so; where a continuum tool is used on the lattice (the far field  $3/(4\pi r)$  of a holder’s rest, S.25; the retarded solution of the wave equation, S.42; the shell theorem, S.29; the ray integrals along the straight path, S.20 and S.30; the radial grid, S.60), it holds at long wavelength and large  $r$  with the lattice’s correction of relative order  $(\text{Link}/r)^2$  or  $k^2/12$  along an axis, the order the law’s lines name; and the theorems of mathematics cited by name (the discrete Fourier transform’s resolution  $2\pi/N$  in S.8, Yukawa’s bound-state threshold 1.6798 in S.45, Hull and Dobell’s theorem in S.50, Watson’s constant  $G(0) = 0.252731$  in S.38) are standard results used without proof.

**The use of a large language model.** The derivations of this supplement, like the article’s text, software and computations, were worked out with a large language model (Claude, Anthropic, through Claude Code, 2026) used extensively as a tool under the author’s direction, as the article’s Statements and Declarations state; every step was reviewed by the author, and checked twice and independently where its status says two derivations and once where it says one; the model is not an author, and the author takes full responsibility for the content.

## The path from Rule3 to each result

Every result of the paper is reached from Eq. (1), Rule3’s line with its one write and its three readings, by a chain of the derivations below; where a link of a chain is an assumption of the model, it is named in the chain, so that each path is closed in the algebra with its assumptions explicit.

- *Light’s speed and dispersion:* Eq. (1) at the vacuum’s pace  $\rightarrow$  the band, S.1.
- *Gravity as a slowed clock, light’s index, the bending and Shapiro’s delay:* Eq. (1)  $\rightarrow$  the paces, Eq. (3) (the assumption: the level read once into the clock and twice into the Link)  $\rightarrow$  the band at a pace, S.1  $\rightarrow$  the index, S.2  $\rightarrow$  the ray integral, Eq. (15) and S.20.
- *The redshift of a bound rotation:* Eq. (1)  $\rightarrow$  S.1 at  $k = 0$  with the clock’s pace  $\rightarrow$  S.15.
- *The fall, the Newtonian potential the orbits measure and the post-Newtonian parameters:* Eq. (1)  $\rightarrow$  S.1 at small  $k$   $\rightarrow$  the inertia and the ray equations, S.18  $\rightarrow$  S.19.
- *Lorentz’s factor to the second order:* Eq. (1)  $\rightarrow$  the exact band of a free record, S.1  $\rightarrow$  the rotation at the moving centre, S.21, closed in the algebra for a free quantum of any family; for a body, the body’s own band from its fixed point, S.13, computed by the generator, enters the same S.21.
- *De Broglie’s relation:* Eq. (1)  $\rightarrow$  S.1  $\rightarrow$  S.22.
- *From the Node to us, bodies, clusters and whoever sees:* Eq. (1) with the write  $\rightarrow$  the fixed point and the functional, S.13  $\rightarrow$  the stability, Theorem 4  $\rightarrow$  the functional (derived) and the free fall (S.18)  $\rightarrow$  the clicks between bodies, S.9, S.14 to S.17  $\rightarrow$  the clusters and whoever sees as a cluster (Section 7.5, every body bound and records free, the start declared).

- *The count*: Eq. (1) → the conserved form, Theorem 3 → the share, S.5 and S.6 → the invariant and the drift, S.14.
- *Light born by the write, absorption and emission*: Eq. (1)'s write line → the Wronskian, S.9 → the resonance, S.16.
- *Bell*: Eq. (1) linear → the parts' determinism (Section 9.4) → the declared credit (the assumption) → S.10.
- *The two slits*: Eq. (1) stepped in real arithmetic on the declared file → S.23 and the computed numbers of Section 9.
- *The constants and the bounds*: Eq. (1) → S.1 → S.24 (the Link); the holder's rest → S.25 (Newton's constant); S.11 (the reach), S.12 (the two speeds).

## The model (Section 3 of the paper)

**Derivation S.1** (the band at a pace, Eq. (4) of the paper). Inputs: *Rule3's line at the vacuum's paces, where the two Links of an axis share one  $R_a = 2 \text{ num } \Gamma^2$ ,  $w a_{\text{next}} = \sum_a R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}}$ , with the remainders dropped (they are bounded by  $w$  and do not move a plane wave's rotation).* Steps: for a plane wave  $a_t(\mathbf{x}) = A \cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$  the two arrivals of the axis  $a$  sum to  $2 \cos k_a a_{\text{now}}$  and  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$ , so

$$2 \cos \omega = \frac{2 \sum_a R_a \cos k_a + S}{w} = \frac{4 \text{ num } \sum_a p_a^2 \cos k_a + 12 \text{ den } \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{ num } \sum_a p_a^2}{6 \text{ den } \Gamma^2},$$

that is  $\cos \omega = 1 - g(p_0/\Gamma)^2 - \frac{\text{num}}{3 \text{ den}} \sum_a (p_a/\Gamma)^2 (1 - \cos k_a)$ , which is Eq. (4). At the vacuum's pace,  $p_0 = p_a = \Gamma$ , this is  $\cos \omega = \text{num}/\text{den} - \frac{\text{num}}{3 \text{ den}} \sum_a (1 - \cos k_a) = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$ ; at  $k = 0$ ,  $\cos \omega_0 = \text{num}/\text{den}$ . Light's speed and dispersion: for  $[1, 1]$  along one axis  $\cos \omega = (2 + \cos k)/3$ ; expanding,  $1 - \omega^2/2 + \omega^4/24 = 1 - k^2/6 + k^4/72$ , so  $\omega = (k/\sqrt{3})(1 - k^2/36 + \dots)$  and the group velocity  $d\omega/dk = \sin k/(3 \sin \omega) = (1/\sqrt{3})(1 - k^2/12 + \dots)$ : the speed  $1/\sqrt{3}$  at long wavelength and the group speed falling as  $1 - 0.083 k^2$ . Check: at  $k = 0.05, 0.1$  and  $0.2$  the exact  $\sqrt{3} d\omega/dk - 1$  over  $k^2$  is  $-0.0833, -0.0834$  and  $-0.0835$ .

**Derivation S.2** (light's index at a content, Section 3.4). Inputs: *Eq. (4) with the paces of Eq. (3),  $p_0 = \Gamma(1 - 1/\Gamma)^L = \Gamma N$ ,  $N = e^{-x} + O(1/\Gamma)$  at the content  $x = L/\Gamma$ , and  $p_a = p_0^2/\Gamma = \Gamma N^2$  for a reader at the weight 1 and no tension.* Steps: for light,  $g = 0$ , so  $\cos \omega = 1 - \frac{1}{3} N^4 \sum_a (1 - \cos k_a)$ , and at long wavelength  $\omega = N^2 |\mathbf{k}|/\sqrt{3}$ . Light's speed at the content  $x$  is  $N^2 = e^{-2x}$  times its vacuum speed, so its index is  $n = e^{2x} = 1/(1 - 2x) + O(x^2)$  and  $n - 1 = 2x + O(x^2)$ : the Link's pace is the clock twice, and so is light's slowing.

**Derivation S.3** (the guard's edge and the horizon, Section 3.4). Inputs: *Eq. (4).* Steps: the lowest value of  $2 \cos \omega$  on the band, its top  $\omega$ , is at  $k_a = \pi$  on every axis, where  $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2 - \frac{4 \text{ num}}{3 \text{ den}} \sum_a (p_a/\Gamma)^2$ ; the recurrence  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$  has no exponentially growing solution iff  $|2 \cos \omega| \leq 2$ ; at the edges  $\pm 2$  a linearly growing solution exists ( $u_t = \pm t$  from  $u_0 = 0, u_1 = 1$ ), which the guard admits, light's checkerboard mode sitting at  $-2$ ; for a negative numerator the lowest  $2 \cos \omega$  is at  $k = 0$  and the same bound holds with  $|\text{num}|$ . With  $p_0 = p_a = p$  the condition  $2 \cos \omega \geq -2$  reads  $(p/\Gamma)^2 (2g + 4 \text{ num}/\text{den}) \leq 4$ , that is  $(p/\Gamma)^2 \leq 2 \text{ den}/(\text{den} + \text{num})$ , so the edge is  $P^2 = 2 \text{ den } \Gamma^2 \text{ div } (\text{den} + \text{num})$ , which is  $\Gamma^2$  for light. No horizon: under the composed paces  $p_0 = \Gamma(1 - 1/\Gamma)^c > 0$  at every content and  $p_a = p_0^2/\Gamma > 0$ , so  $R_a = 2 \text{ num } p_a^2$  never vanishes and no Link closes; the quadratic truncation  $p_a = \Gamma(1 - 2x)$  vanished at  $x = 1/2$ , a horizon of the truncation. The frozen clock: at a content  $c$  the clock runs at  $N = (1 - 1/\Gamma)^c$  of the vacuum's,  $e^{-c/\Gamma}$  to the unit, so a deep well freezes a clock to the unit and its Node's share is not a number there.

**Derivation S.4** (the cube's group of 48 and its action, Section 3.1). Inputs: *Rule3's line, whose six arrivals enter through the six Port coefficients  $R_{ij}$ , the two of an axis one  $R_a = 2 \text{ num } p_a^2$  with no coefficient naming an axis beyond its own pace; the readings  $D, F$  and  $W$  of Eq. (1).* The group: a map of the six Ports onto themselves that keeps opposite Ports opposite is a signed permutation of the three axes,  $\mathbf{x} \mapsto \mathbf{P}\mathbf{x}$  with  $\mathbf{P}$  a permutation matrix with signs; there are  $3! \cdot 2^3 = 48$  of them, the group  $O_h$  of the cube, of which the 24 with  $\det \mathbf{P} = +1$  are the rotations and the other 24 reflections and rotatory reflections; with the translations of the lattice they form the space group of the simple cubic lattice. Covariance: let  $\sigma$  be such a map and let the turned GameBoard carry the levels  $a_{\sigma(i)}^\sigma = a_i$  and the paces  $p_{\sigma(a)}^\sigma(\sigma(i)) = p_a(i)$ . The arrivals of the axis  $\sigma(a)$  at  $\sigma(i)$  in the turned GameBoard are the arrivals of the axis  $a$  at  $i$  in the original, in either order, and  $R_a$  is even in the axis's sign; so the sum  $\sum_a R_a (\text{arr}_{+a} + \text{arr}_{-a})$  is the same number at  $\sigma(i)$  as at  $i$ ,  $S$  and  $w$  are the same, and the division gives the same quotient and remainder: the step commutes with  $\sigma$ , and a board turned by  $\sigma$  runs as the turned GameBoard. A translation commutes for the same reason. The readings' kinds:  $D_i$  is one number at a Node with no axis in it, carried to  $\sigma(i)$  unchanged, a scalar. The currents  $F_{ij}$  are attached to the Links and carried with them, and the momentum density  $P_a(i) = (F_{i,i-a} - F_{i,i+a})/\text{num}$  (Section 4.3's orientation, positive for a record moving toward  $+a$ ) changes sign where  $\sigma$  reverses the axis  $a$  and is carried to the image axis otherwise, a vector;  $W_i$  carries no axis and is carried unchanged, a scalar under the cube's 48, odd under the plane's internal reflection  $\text{im} \mapsto -\text{im}$  and under the time reversal. The click  $(R, t)$  names a region and an interval and is carried to the turned region, a scalar. The time reversal: exchanging now and before leaves  $D_i = \text{now}_i^2 - \text{next}_i \text{ before}_i$  unchanged once next and before exchange their roles, since  $D$  is symmetric in the pair (next, before); it reverses  $F_{ij}$  and  $W_i$ , which are antisymmetric in (now, before); and it carries the step to its inverse by Theorem 1. The rational

rotation: the two parts of a plane record obey the same linear line with the same coefficients, so any linear combination of them does; the combination  $(\text{re}', \text{im}') = ((p \text{re} - q \text{im})/r, (q \text{re} + p \text{im})/r)$  with  $p^2 + q^2 = r^2$  carries solutions to solutions, and its Wronskian is  $(p^2 + q^2)/r^2$  times the old one, unchanged. What is not in the group: a boost changes the record's wave number and rotation together, which no permutation of Ports does, and a rotation by an angle other than the cube's carries a Node off the lattice; so neither is a symmetry, and Lorentz's factor, where the paper finds it, is a property of the clicks (Derivation S.21).

## Exact properties (Section 4)

**Derivation S.5** (the share's change is the currents, Eq. (10)). Inputs: the share at the vacuum's paces,  $e_i = 3 \text{den}(\text{now}_i^2 + \text{before}_i^2) - \text{num} \text{now}_i S_6(\text{before})_i$ , with  $S_6$  the six-neighbour sum; Rule3 at the vacuum's paces, where  $S = 0$ ,  $R_a = 2 \text{num} \Gamma^2$  and  $w = 6 \text{den} \Gamma^2$ , so that  $3 \text{den}(\text{next}_i + \text{before}_i) = \text{num} S_6(\text{now})_i + \rho_i$  with  $\rho_i = (r_i - r'_i)/(2\Gamma^2)$  the remainders' term (from  $6 \text{den} \Gamma^2(\text{next} + \text{before}) = 2 \text{num} \Gamma^2 S_6(\text{now}) + r - r'$ ). Steps: after the step the pair (now, before) becomes (next, now), so

$$e'_i - e_i = 3 \text{den}(\text{next}_i^2 - \text{before}_i^2) - \text{num}[\text{next}_i S_6(\text{now})_i - \text{now}_i S_6(\text{before})_i].$$

With  $3 \text{den}(\text{next}_i + \text{before}_i) = \text{num} S_6(\text{now})_i + \rho_i$  the first term is  $(\text{next}_i - \text{before}_i)(\text{num} S_6(\text{now})_i + \rho_i)$ , and the bracket cancels its  $\text{next}_i$  part:

$$\begin{aligned} e'_i - e_i &= \text{num}[\text{now}_i S_6(\text{before})_i - \text{before}_i S_6(\text{now})_i] + (\text{next}_i - \text{before}_i) \rho_i \\ &= \sum_{j \sim i} \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j) + (\text{next}_i - \text{before}_i) \rho_i, \end{aligned}$$

the six currents  $F_{ij}$  into the Node plus the remainder term. At any paces: with the weighted share  $e_i = [w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_j R_{ij} \text{before}_j]/(2p_i^2)$  and the line without its division,  $w \text{next}_i^* = \sum_j R_{ij} \text{now}_j + S_i \text{now}_i - w \text{before}_i$ , the difference  $e_i(\text{next}_i^*, \text{now}) - e_i(\text{now}, \text{before})$  is  $\sum_j (q_{ij}^2/\Gamma^2) \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$  with no anisotropy term, the  $S_i$  terms and the Node's terms cancelling as in Theorem 3's proof, since  $R_{ij}/(2p_i^2) = \text{num} q_{ij}^2/\Gamma^2$  is the same from both ends. The integer step's  $\text{next} = \text{next}_i^* + (r - r')/w$  adds  $-(\text{next}_i - \text{before}_i)(r'_i - r_i)/(2p_i^2)$ , because  $w(\text{next}_i + \text{next}_i^*) - S_i \text{now}_i - \sum_j R_{ij} \text{now}_j = w(\text{next}_i - \text{before}_i)$ . Check: exact fractions on a periodic chain of five Nodes at [2, 3], the two other axes folded (four self-Ports per Node),  $\Gamma = 10$ , with contents 0 to 3, Link factors 5 to 10 and random levels: the identity holds at every Node, with and without the division; and a second witness, a uniform  $\text{now} = 3$  over  $\text{before} = 0$  at  $\Gamma = 10$ ,  $q = 5$ , reads 81 per Node where the mean-of-paces form read 324 (an independent automated check).

**Derivation S.6** (the total share is non-negative, Section 4.4). Inputs:  $e_i$  as above on a closed or periodic GameBoard;  $S_6$  symmetric with eigenvalues  $2 \sum_a \cos k_a \in [-6, 6]$ . Steps:  $\sum_i e_i = 3 \text{den}(|\mathbf{n}|^2 + |\mathbf{b}|^2) - \text{num} \mathbf{n} \cdot S_6 \mathbf{b}$ , and  $|\mathbf{n} \cdot S_6 \mathbf{b}| \leq \|S_6\| |\mathbf{n}| |\mathbf{b}| \leq 6 |\mathbf{n}| |\mathbf{b}| \leq 3(|\mathbf{n}|^2 + |\mathbf{b}|^2)$ , so  $\sum_i e_i \geq 3(\text{den} - |\text{num}|)(|\mathbf{n}|^2 + |\mathbf{b}|^2) \geq 0$  for  $|\text{num}| \leq \text{den}$ . Equality needs  $|\text{num}| = \text{den}$  and  $\mathbf{b}$  an eigenvector of  $S_6$  at  $\pm 6$  with  $\mathbf{n} = \pm \mathbf{b}$ : the static uniform level of light, or on an even periodic board its checkerboard,  $\text{now} = (-1)^{x+y+z}$  with  $\text{before} = -\text{now}$ , whose shares are 0. A Node's share: at fixed  $\text{before}_i$  and  $S_6(\text{before})_i = s$ ,  $e_i$  is a quadratic in  $\text{now}_i$  with the minimum  $3 \text{den} \text{before}_i^2 - \text{num}^2 s^2/(12 \text{den})$ , negative iff  $|s| > 6(\text{den}/\text{num}) |\text{before}_i|$ ; a plane record has  $|s|/|\text{before}_i| = 2 \sum_a \cos k_a \leq 6$  and is never negative, a lay's edge can be. A region: pairing each boundary Link's cross term and bounding  $2|\text{now}_i \text{before}_j| \leq \text{now}_i^2 + \text{before}_j^2$ ,  $\sum_{i \in R} e_i \geq 3(\text{den} - \text{num}) \sum_R (\text{now}_i^2 + \text{before}_i^2) - \frac{\text{num}}{2} \sum_{\text{boundary Ports}} (\text{now}_j^2 + \text{before}_j^2)$  over the outside neighbours  $j$ : a region's share is negative at most by half a level-square per boundary Port in the current's units.

**Derivation S.7** (the tension of a plane wave, Section 4.3). Inputs: the stress reading  $G_{aa}(i) = \text{now}_i(\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a}(\text{now}_{i+a} - \text{now}_{i-a})$ . Steps: for  $\text{now}_i = b \cos(kx_i + \phi)$  along the axis, with  $c_n = \cos(kx_i + nk + \phi)$ ,  $G = b^2[c_0 c_2 - c_0^2 - c_1^2 + c_1 c_{-1}]$ . The identities  $c_0 c_2 = \frac{1}{2}[\cos(2kx + 2k + 2\phi) + \cos 2k]$ ,  $c_1 c_{-1} = \frac{1}{2}[\cos(2kx + 2\phi) + \cos 2k]$ ,  $c_0^2 = \frac{1}{2}[1 + \cos(2kx + 2\phi)]$  and  $c_1^2 = \frac{1}{2}[1 + \cos(2kx + 2k + 2\phi)]$  cancel the phase-dependent parts, leaving  $G = b^2(\cos 2k - 1) = -2b^2 \sin^2 k$  at every Node and interval. So the Link's booking is  $-\text{num} G = 2 \text{num} b^2 \sin^2 k$  and the Node's tension  $T_{aa} = -\text{num} h_a = \text{num} b^2 \sin^2 k$ , and for a wave with the components  $k_a, k_b$  the cross reading is  $G_{ab} = -2b^2 \sin k_a \sin k_b$ .

## Measurement (Section 5)

**Derivation S.8** (the region's reading factor and the draw's scatter, Section 5.3). Steps: a region of  $n$  consecutive rows sums a fringe  $\cos(2\pi y/\Lambda)$  to  $\text{Re} \sum_{y=0}^{n-1} e^{2\pi i y/\Lambda} = \text{Re} \frac{1 - e^{2\pi i n/\Lambda}}{1 - e^{2\pi i/\Lambda}}$ , whose modulus is  $\sin(\pi n/\Lambda)/\sin(\pi/\Lambda)$ ; over  $n$  rows the fringe's modulation is kept with the factor  $\sin(\pi n/\Lambda)/(n \sin(\pi/\Lambda))$ , 0.91 at  $n = 4$ ,  $\Lambda = 16$ , and 0 at  $\Lambda = n$ . A frequency read as a click rate over  $N$  intervals resolves  $\Delta\omega \geq 2\pi/N$ , the window's own resolution.  $N$  quanta drawn independently by the shares  $p_i$  give each region a binomial count,  $N p_i \pm \sqrt{N p_i(1 - p_i)}$ .

**Derivation S.9** (the Wronskian's conservation and its current, Section 5.5). Inputs: *the two parts of a plane record step by the same read matrix  $\mathbf{M}$ , with  $\mathbf{DM}$  symmetric for  $\mathbf{D} = \text{diag}(1/p_i^2)$  (the paper's Theorem 3),  $\mathbf{re}_{\text{next}} = \mathbf{M}\mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}}$  and the same for  $\mathbf{im}$ , with the weights  $\mathbf{D}$  of the conserved form,  $\mathbf{DM}$  symmetric.* Steps: let  $W = \mathbf{re}_{\text{now}}^\top \mathbf{D} \mathbf{im}_{\text{before}} - \mathbf{im}_{\text{now}}^\top \mathbf{D} \mathbf{re}_{\text{before}}$ . After the step it is  $\mathbf{re}_{\text{next}}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{next}}^\top \mathbf{D} \mathbf{re}_{\text{now}} = (\mathbf{M}\mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}})^\top \mathbf{D} \mathbf{im}_{\text{now}} - (\mathbf{M}\mathbf{im}_{\text{now}} - \mathbf{im}_{\text{before}})^\top \mathbf{D} \mathbf{re}_{\text{now}}$ ; the  $\mathbf{M}$  terms are  $\mathbf{re}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{re}_{\text{now}}$ , which vanish since  $\mathbf{M}^\top \mathbf{D} = \mathbf{DM}$  is symmetric, and the rest is  $W$  again. Per Node the same cancellation leaves the Link terms: the weighted Wronskian  $W_i/p_i^2$  changes by  $(2 \text{num}/w)(q_{ij}^2/\Gamma^2)(\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$  summed over the Node's six Ports at the levels now, the Wronskian's current, which for the integer  $W_i$  at the vacuum's paces is  $(\text{num}/3 \text{den})(\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$  per Port. Check: a periodic chain of three Nodes at  $\Gamma = 2$  with the two other axes folded,  $[1, 1]$ ,  $\mathbf{re}_{\text{now}} = (0, 3, 0)$ ,  $\mathbf{im}_{\text{now}} = (3, 0, 0)$  and zeros before steps to  $(1, 4, 1)$  and  $(4, 1, 1)$  with no remainder;  $W_0$  goes from 0 to 3,  $W_0/p_0^2$  by  $3/4$ , and the current is  $(2/24) \cdot 9 = 3/4$  (an independent automated check's witness); exact fractions on a chain of five Nodes with contents and Link factors confirm the identity at every Node. For the circular record  $\mathbf{re} = A \cos \omega t$ ,  $\mathbf{im} = eA \sin \omega t$  at one Node,  $W = A \cos \omega t \cdot eA \sin \omega(t-1) - eA \sin \omega t \cdot A \cos \omega(t-1) = -eA^2 \sin \omega$ , constant. Domain: the identity is the line's,  $\mathbf{re}_{\text{next}} = \mathbf{M}\mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}}$  exactly. The integer step's division adds  $\epsilon = (r - r')/w$  to each part, and the remainder term  $\epsilon_{\text{re}} \mathbf{im}_{\text{now}} - \epsilon_{\text{im}} \mathbf{re}_{\text{now}}$  is not zero in general: at the vacuum's pace  $S = 0$  and  $w a_{\text{next}} = 12 \text{num} \Gamma^2 a_{\text{now}} - w a_{\text{before}}$  for a uniform level, so at  $[2, 3]$  the parts  $(1, 3)$  with zeros before step to  $([4/3], [4]) = (1, 4)$  and  $W$  goes from 0 to  $1 \cdot 3 - 4 \cdot 1 = -1$ . An exactly conserved integer charge is open.

**Derivation S.10** (Bell's correlation under the declared credit, Section 5.7 and Section 9.4). Inputs: *two parts with the products  $M_1, M_2$  of their amplitudes at the two sides' ports; the port coefficients  $e(+) = (\cos a, \sin a)$ ,  $e(-) = (-\sin a, \cos a)$  on side A and the same with  $b$  on side B; the joint share  $J(p, q) = (\sum_k e_k(p) e_k(q) M_k)^2$ ;  $E = \sum_{p,q} s(p)s(q) J(p, q) / \sum_{p,q} J(p, q)$  with  $s(\pm) = \pm 1$ .* Steps: write  $c_a = \cos a$ ,  $s_a = \sin a$  and the same for  $b$ . The four sums are  $M_1 c_a c_b + M_2 s_a s_b$ ,  $M_1 s_a s_b + M_2 c_a c_b$ ,  $-M_1 c_a s_b + M_2 s_a c_b$  and  $-M_1 s_a c_b + M_2 c_a s_b$  for  $(+, +)$ ,  $(-, -)$ ,  $(+, -)$  and  $(-, +)$ . Their squares sum to  $(M_1^2 + M_2^2)(c_a^2 c_b^2 + s_a^2 s_b^2 + c_a^2 s_b^2 + s_a^2 c_b^2) = M_1^2 + M_2^2$ , the cross terms cancelling in pairs. The signed sum is  $(M_1^2 + M_2^2)(c_a^2 - s_a^2)(c_b^2 - s_b^2) + 8M_1 M_2 c_a s_a c_b s_b$ , so

$$E(a, b) = \cos 2a \cos 2b + \rho \sin 2a \sin 2b, \quad \rho = \frac{2M_1 M_2}{M_1^2 + M_2^2},$$

and the marginal  $P(A+) = [(M_1 c_a c_b + M_2 s_a s_b)^2 + (-M_1 c_a s_b + M_2 s_a c_b)^2] / (M_1^2 + M_2^2) = (M_1^2 c_a^2 + M_2^2 s_a^2) / (M_1^2 + M_2^2)$ , independent of  $b$ . For equal parts  $\rho = 1$  and  $E = \cos 2(a - b)$ , the known form for parallel polarisations, derived here under the declared credit by Lagrange's identity. The integer settings: with  $e(+) = (p, q)$  and  $e(-) = (-q, p)$  unnormalised and equal parts,  $J(+, +) = J(-, -) = (p_{APB} + q_{AQB})^2$  and  $J(+, -) = J(-, +) = (p_{AQB} - q_{APB})^2$ , whose sum is  $2(p_A^2 + q_A^2)(p_B^2 + q_B^2)$  by Lagrange's identity, so  $E = [(p_{APB} + q_{AQB})^2 - (p_{AQB} - q_{APB})^2] / [(p_A^2 + q_A^2)(p_B^2 + q_B^2)]$ , a rational number. At  $a = (1, 0)$ ,  $a' = (1, 1)$ ,  $b = (12, 5)$ ,  $b' = (5, 12)$ :  $E(a, b) = (144 - 25)/169 = 119/169$ ,  $E(a, b') = (25 - 144)/169 = -119/169$ ,  $E(a', b) = (17^2 - 7^2)/338 = 120/169$ ,  $E(a', b') = 120/169$ , and  $S = E(a, b) - E(a, b') + E(a', b) + E(a', b') = 478/169 = 2.8284$ . At the ideal angles  $S = \sqrt{2}(1 + \rho)$ ; at the integer settings  $S = (238 + 240\rho)/169$ . The two local credits: with the pair's phase  $\theta$  uniform, a credit of each side by the shares of its ports gives  $E = \langle \cos 2(a - \theta) \cos 2(b - \theta) \rangle_\theta = \frac{1}{2} \cos 2(a - b)$  and  $S = \sqrt{2}$ ; a credit by the sign of the larger port gives  $E = \langle \text{sgn} \cos 2(a - \theta) \text{sgn} \cos 2(b - \theta) \rangle_\theta = 1 - 4|a - b|/\pi$ , the sawtooth, whose sum at the four settings is exactly 2. Check: the averages over  $\theta$  on a grid of  $2 \times 10^5$  points give  $S = 1.4142$  and  $2.0000$ , and the contrast credit, one side's count weighted by its contrast  $|\cos 2(a - \theta)|$  and the other side's sign always counted, gives  $E = \cos 2(a - b)$  exactly by  $\langle \cos x \text{sgn} \cos(x - 2\delta) \rangle = (2/\pi) \cos 2\delta$ ,  $S = 2.8284$  at one side's efficiency  $2/\pi$ ; the symmetric readings do not ( $|\cos|$  on both sides:  $E(\pi/8) = 0.88$ ,  $S = 3.52$ ).

## The families (Section 6)

**Derivation S.11** (the reach of a holder with a gap, Section 6.4). Inputs: *a holder's rest outside its source, the line with  $a_{\text{next}} = a_{\text{before}} = a_{\text{now}} = a$ :  $6 \text{den } a = \text{num } S_6(a)$  at the vacuum's pace, that is  $\text{num}(S_6 - 6)a = 6(\text{den} - \text{num})a$ .* Steps: for a rest falling along one axis as  $e^{-\kappa r}$  and uniform across,  $S_6(a) = (2 \cosh \kappa + 4)a$ , so  $6 \text{den} = \text{num}(2 \cosh \kappa + 4)$  and  $\cosh \kappa = 3 \text{den}/\text{num} - 2$ . For  $\text{num}$  just under  $\text{den}$ ,  $\cosh \kappa = 1 + \kappa^2/2 + \dots$ , so  $\kappa^2 = 6 \text{den}/\text{num} - 6 = 6(\text{den} - \text{num})/\text{num}$  and the range is  $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ . Links: 20.00 at  $[2400, 2401]$ ;  $[1, 2]$  has  $\cosh \kappa = 4$ ,  $e^{-\kappa} = 0.127$ . The gap is also the rest rotation of the holder's own travelling events,  $\cos \omega_g = \text{num}/\text{den}$ ,  $\omega_g^2 = 2(\text{den} - \text{num})/\text{den} + \dots$ , whose Compton length at light's speed,  $c/\omega_g = \sqrt{\text{den}/(6(\text{den} - \text{num}))} = R\sqrt{\text{den}/\text{num}}$ , is the range to the gap's first order: Yukawa's identity with no input.

**Derivation S.12** (the two speeds, Section 6.5). Steps: *light at a content  $x$  travels at  $(1 - 2x)/\sqrt{3}$  (Derivation S.2). Under the law's line the massless holder reads the content as every family does, its own level among it, so its travelling events have the same paces and the two arrive together at every content, the difference of their Shapiro delays 0. With the massless holder reading no content, as a model whose massless holder read no content would have it, its events travel*

at  $1/\sqrt{3}$  and over  $L$  Links the two arrive  $L\sqrt{3}[1/(1-2x)-1] = 2xL\sqrt{3} + O(x^2)$  intervals apart at long wavelength, against GW170817's bound on  $\gamma_{\text{GW}} - \gamma_{\text{EM}}$ , between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$ ; under the line the measurement bounds no vacuum content.

## Bodies (Section 7)

**Derivation S.13** (the bound body: the fixed point and the functional, Section 7.2). Inputs: *Rule3's line at the paces of the wells the record sources; a standing record  $a_{i,t} = \phi_i \cos(\omega_b t)$  with one rotation at every Node; the clock's pace  $(p_0/\Gamma)^2 = 1 - 2\Phi_i/\Gamma + \dots$  at a well  $\Phi_i$  (in levels).* The fixed point's equation: *with the remainders dropped the line gives at every Node*

$$2 \cos \omega_b \phi_i = \frac{1}{w} \left[ \sum_a R_a(i) (\phi_{i+a} + \phi_{i-a}) + S_i \phi_i \right] = (\mathbf{M}\phi)_i,$$

so  $\phi$  is an eigenvector of the read matrix  $\mathbf{M}$  and the standing record is its top mode, the maximiser of  $\phi^\top \mathbf{D}\mathbf{M}\phi$  on the weighted sphere  $\phi^\top \mathbf{D}\phi = 1$ ,  $\mathbf{D}\mathbf{M}$  symmetric under the law's read coefficient (S.5) (the generator of Section 7.1 iterates this read act). To the first order in  $\Phi/\Gamma$ ,  $S_i/w = 2 - 2g(1 - 2\Phi_i/\Gamma) - \frac{2 \text{num}}{3 \text{den}} \sum_a (p_a/\Gamma)^2 = 2 \cos \omega_0 + 4g\Phi_i/\Gamma - \frac{2 \text{num}}{3 \text{den}} \sum_a (p_a/\Gamma)^2$  and  $R_a/w = \frac{\text{num}}{3 \text{den}} (p_a/\Gamma)^2$ , so, keeping the band's term at the vacuum's Link pace (its product with the well is of the second order),

$$(\mathbf{M}\phi)_i = 2 \cos \omega_0 \phi_i + \frac{4g}{\Gamma} \Phi_i \phi_i - \frac{\text{num}}{3 \text{den}} (-\Delta^2 \phi)_i, \quad (-\Delta^2 \phi)_i = 6\phi_i - \sum_a (\phi_{i+a} + \phi_{i-a}).$$

The functional: *the well is sourced by the record itself,  $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$  for a body of  $M$  quanta with the counts  $c_j = M\phi_j^2$ .* Let

$$F[\phi] = \frac{\text{num}}{3 \text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2,$$

the paper's Eq. (12). Its gradient is  $\partial F/\partial \phi_i = \frac{2 \text{num}}{3 \text{den}} (-\Delta^2 \phi)_i - \frac{8g}{\Gamma} \Phi_i \phi_i$ , using the symmetry of  $G_r$  (each well counted once, Hartree's form, is what makes the factor  $2g$  in  $F$  give  $4g$  in the gradient). So  $\partial F/\partial \phi = 2\lambda\phi$  on the sphere is exactly the fixed point's equation with  $\lambda = 2 \cos \omega_0 - 2 \cos \omega_b$ : a body is a critical point of  $F$  on the sphere, and the virial and the second variation along the dilation are Theorem 4 of the paper. The scaling powers: under  $\phi_R(\mathbf{x}) = R^{-d/2} \phi(\mathbf{x}/R)$  the sum over Links of  $(\phi_i - \phi_j)^2$  is the lattice form of  $\int |\nabla \phi|^2$  and scales as  $R^{-2}$ , the pressure  $K/R^2$ . The well energy  $\sum_{ij} \phi_i^2 G_{ij} \phi_j^2$  is  $\iint \rho(\mathbf{x}) G(\mathbf{x} - \mathbf{y}) \rho(\mathbf{y})$  with  $\rho = \phi^2$  of unit mass, which scales as  $R^{-(d-2)}$  for the massless kernel  $G \sim 1/r$  and for a screened kernel within its reach, and as  $R^{-d}$  for a screened kernel beyond its reach, where  $G$  acts as a point kernel and the energy is  $\int \rho^2$ : the powers  $s_r$  of Theorem 4. The cloud's size: for the massless kernel in three dimensions  $F(R) = a/R^2 - c/R$  with  $a = \frac{\text{num}}{3 \text{den}} \int |\nabla \phi|^2$  and  $c = \frac{2g}{\Gamma} \frac{M}{E} \iint \frac{\rho \rho'}{4\pi |\mathbf{x} - \mathbf{y}|} \cdot 3$  at the unit size, so the minimum is at  $R = 2a/c \propto \Gamma E/(gM)$ : the radius falls as  $1/M$  and grows with  $\Gamma$ ; the continuum constant of the rms radius is  $(2\pi \cos \omega_0/(9g)) r_P \Gamma E_g/M$  with Pekar's  $r_P = 4.635$  (the Choquard minimiser's rms radius, at Lieb's energy  $-0.1085$  and the virial  $1/2$ ),  $(4\pi/9) r_P = 6.47$  at  $[2, 3]$  where  $\cos \omega_0/(2g) = 1$ , the bracket  $\cos \omega_0/(2g)$  tending to  $1/\omega_0^2$  as the gap closes (4.5 at  $[9, 10]$ , 500 at  $[999, 1000]$ ). The hollow's barrier (Section 7.2 (b)): a hollow beyond the body's reach adds the point-kernel term  $-b/R^3$  (the power  $R^{-d}$  above at  $d = 3$ ),  $F(R) = a/R^2 - b/R^3 - c/R$ ;  $F'(R) = (cR^2 - 2aR + 3b)/R^4$  vanishes at  $R_\pm = (a \pm \sqrt{a^2 - 3bc})/c$ , two critical points where  $a^2 > 3bc$  and none where  $3bc > a^2$ ; at a root  $F'' = 2(aR - 3b)/R^5$ , positive at  $R_+ \geq a/c > 3b/a$  and negative at  $R_- = 3b/(a + \sqrt{a^2 - 3bc}) < 3b/a$ , so the wide critical point is the minimum and the narrow one the saddle. Status: the equation a theorem (the functional's variational derivative); its solution computed; GameBoard.

**Derivation S.14** (the invariant and the count's drift, Eq. (13)). Inputs: *the one-Node line  $a_{t+1} + a_{t-1} = 2 \cos \omega_t a_t$  with  $\omega_t$  changing slowly.* Steps: *at fixed  $\omega$  the form  $D_t = a_t^2 - a_{t+1}a_{t-1} = a_t^2 + a_{t-1}^2 - 2 \cos \omega a_t a_{t-1}$  is conserved and equals  $A^2 \sin^2 \omega$  for  $a_t = A \cos(\omega t + \phi)$ ; in the plane  $(a_t, a_{t-1})$  the step is a linear map of determinant 1 whose orbits are the ellipses  $D = \text{const}$ , of area  $\pi D/\sin \omega$ . A slow change of  $\omega$  keeps the area enclosed by the orbit, the action of the map, so  $D/\sin \omega = A^2 \sin \omega$  is the adiabatic invariant: the discrete form of  $E/\omega$ . Equivalently, the ansatz  $a_t = C \cos(\sum_{s \leq t} \omega_s + \phi)/\sqrt{\sin \omega_t}$  satisfies the line to the first order in  $\omega_{t+1} - \omega_t$ . Check: the recurrence stepped 20,000 intervals with  $\omega$  lowered from 0.841 to 0.600 keeps  $D/\sin \omega$  within  $4 \times 10^{-5}$  of its value while  $D$  falls by a quarter. The drift: the share at a Node in quanta is  $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b/T$  (Eq. (9) at the Node's paces), so with  $A^2 \sin \omega_b$  fixed,  $N(t_2)/N(t_1) = [\sin \omega_b(t_2)/\sin \omega_b(t_1)] [p_i^2(t_1)/p_i^2(t_2)]$ , Eq. (13). Planck's relation: the share  $A^2 \sin^2 \omega/T$  is the invariant  $I = A^2 \sin \omega/T$  times  $\sin \omega$ . The body's click: under the declaration that one click is one quantum's change in the absorbing body (Section 7.4), the NodeReader's own record answers at its rotation by the two-mode line of S.16, one transition per click from the mode  $\omega_i$  to the mode  $\omega_j$ , and the share it takes is  $T(\sin \omega_j - \sin \omega_i) = T \sin(\omega_j - \omega_i) [1 - \omega_i \omega_j/2 + O(\omega^4)]$ , by  $\sin \omega_j - \sin \omega_i = 2 \cos \frac{\omega_i + \omega_j}{2} \sin \frac{\omega_j - \omega_i}{2}$ : one photon's share*

$T \sin \omega_L$  at the line  $\omega_L = \omega_j - \omega_i$ , to a relative  $\omega_0^2/2 < 10^{-18}$  at nature's gap (S.27). So a body counts photons,  $I$  per click, and reads  $\hbar\omega$  per click with  $T$  for  $\hbar$ ; the share is the energy a click carries, and through a deepening well the number of quanta,  $I$ , is kept.

**Derivation S.15** (the clock's click rate, Section 7.4). Inputs: Eq. (4) at  $k = 0$  with  $p_0^2 = \Gamma^2(1 - 2U + 2U^2)$ :  $\cos \omega(U) = 1 - g(1 - 2U + 2U^2) = \cos \omega_0 + 2gU - 2gU^2$ . Steps: to first order  $d(\cos \omega) = 2g dU = -\sin \omega_0 d\omega$ , so  $d\omega/\omega_0 = -[2g/(\omega_0 \sin \omega_0)] dU = -f(\omega_0) dU$  with  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ ;  $f = 1.063$  at  $[2, 3]$ , and as  $\omega_0 \rightarrow 0$ ,  $1 - \cos \omega_0 \rightarrow \omega_0^2/2$  and  $f \rightarrow 1$ , the rate  $1 - U$  of the clock's pace alone. The light clock: a packet between two declared faces  $L$  Links apart at the content  $x$  has the round trip  $2L/v_g = 2L\sqrt{3}/N^2$  (Derivation S.2), so its rate falls by  $2x$  to the first order; between two bodies, which stand  $L/h$  Links apart under the principle, the round trip is  $2L\sqrt{3}/N$  (Derivation S.31).

**Derivation S.16** (the resonance at twice the rotation, Section 7.4). Inputs: the one-Node line with the rotation modulated by a level  $b_L \cos \omega_L t$  read into the content, the law's two-mode line; light's declared read is the holder's rotation (Section 5.5), under which S.43 derives the transfer line:  $2 \cos \omega(t) = 2 - 2g(1 - 2x + 2x^2)$  with  $x = x_0 + b_L \cos(\omega_L t)/\Gamma$ , so  $d(2 \cos \omega)/dx = 4g(1 - 2x)$  and  $a_{\text{next}} + a_{\text{before}} = [2 \cos \omega_b + \epsilon \cos(\omega_L t + \vartheta)] a_{\text{now}}$  with  $\epsilon = 4g(1 - 2x_0) b_L/\Gamma$ . Steps: write  $a_t = \text{Re}[z_t e^{i\omega_b t}]$  with  $z$  slowly varying,  $z_{t \pm 1} = z_t \pm z'$ . Then  $a_{t+1} + a_{t-1} = \text{Re}[e^{i\omega_b t} (2 \cos \omega_b z_t + 2i \sin \omega_b z')]$ , and the product  $\epsilon \cos(\omega_L t + \vartheta) \text{Re}[z_t e^{i\omega_b t}]$  has, at  $\omega_L = 2\omega_b$ , the resonant part  $(\epsilon/2) \text{Re}[\tilde{z}_t e^{i\vartheta} e^{i\omega_b t}]$ . Equating the slowly varying coefficients of  $e^{i\omega_b t}$ ,  $2i \sin \omega_b z' = (\epsilon/2) e^{i\vartheta} \tilde{z}$ , so  $z' = -i\epsilon e^{i\vartheta} \tilde{z}/(4 \sin \omega_b)$ ; this antilinear map  $z \mapsto \lambda \tilde{z}$  with  $|\lambda| = \mu$  has the real eigenvalues  $\pm \mu$  on two lines of the  $z$  plane, so the amplitude grows as  $e^{\mu t}$  along one line and the share, quadratic in the amplitude, as  $e^{2\mu t}$ . Detuned by  $\Delta = \omega_L - 2\omega_b$ , the substitution  $z = \zeta e^{i\Delta t/2}$  gives  $\zeta' = -i(\Delta/2)\zeta - i\mu e^{i\vartheta} \tilde{\zeta}$ , whose exponents are  $\pm \sqrt{\mu^2 - \Delta^2/4}$ : growth only for  $|\Delta| < 2\mu$ , a full width  $\epsilon/\sin \omega_b$ . Check: the recurrence stepped 20,000 intervals at  $2 \cos \omega_b = 1.4317$  and  $\epsilon = 0.013$  grows at 0.004653 to 0.004656 per interval over seven modulation phases, the formula's first-order 0.004654 to four digits. The two-mode line: with two bound modes the same modulation multiplies  $a_t = \text{Re}[z_i e^{i\omega_i t} + z_j e^{i\omega_j t}]$ , and its product with  $\cos \omega_L t$  carries the frequencies  $\omega_i \pm \omega_L$ . The term resonant with the mode  $j$  from the mode  $i$  is at  $\omega_L = \omega_j - \omega_i$ , a transfer between the two modes, and the term that drives both from the conjugate of the other is at  $\omega_L = \omega_i + \omega_j$ , of which the one-mode case above is the degenerate case  $i = j$  (at  $i = j$  the difference line is  $\omega_L = 0$ , a static shift and no line). The transfer's rate,  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin \omega_i \sin \omega_j})$ , follows by the same slowly varying coefficients with the two modes'  $2i \sin \omega_i z'_i$  and  $2i \sin \omega_j z'_j$  coupled through  $\epsilon_{ij}$ , a Rabi oscillation; under light's declared read, the rotation, S.43 derives the transfer line.

**Derivation S.17** (the cross current of two standing records, Section 7.5). Inputs: two standing records  $a_t = A \cos(\omega_1 t + \alpha)$  and  $b_t = B \cos(\omega_2 t + \beta)$  at the two ends of a Link. Steps: the current  $\text{num}(a_t b_{t-1} - a_{t-1} b_t)$  is  $\text{num} AB [\cos(\omega_1 t + \alpha) \cos(\omega_2 t - \omega_2 + \beta) - \cos(\omega_1 t - \omega_1 + \alpha) \cos(\omega_2 t + \beta)]$ , a sum of cosines of  $(\omega_1 - \omega_2)t$  and  $(\omega_1 + \omega_2)t$  with the constant parts  $\frac{1}{2} \cos(\alpha - \beta + \omega)$  and  $-\frac{1}{2} \cos(\alpha - \beta - \omega)$  only when  $\omega_1 = \omega_2$ ; for  $\omega_1 \neq \omega_2$  its mean over the beat is 0, so no quantum passes at the first order. The reach: two bodies' tails are  $a_1 e^{-\kappa d_1}$  and  $a_2 e^{-\kappa d_2}$  at a Link  $d = d_1 + d_2$  Links from the two centres, so the current there is of the order  $a_1 a_2 e^{-\kappa d}$  per interval and falls under one quantum's measure of a current,  $W_c = 3 \text{ den } T$ , at  $d = \ln(\text{num } a_1 a_2 / (3 \text{ den } T)) / \kappa$ .

## The forms of nature (Section 8)

**Derivation S.18** (the fall and the Newtonian potential the orbits measure, Eq. (14)). Inputs: Eq. (4) at a content  $U$  for a reader at the weight 1,  $\cos \omega(k, U) = 1 - g(1 - 2U + 2U^2) - \frac{\text{num}}{3 \text{ den}} (1 - 2U)^2 (1 - \cos k)$  along the motion. Steps: at small  $k$  and  $U$ ,  $\omega(k, U) = \omega_0(U) + k^2/(2m^*)$  with the inertia  $m^* = 3 \text{ den } \sin \omega_0 / \text{num} = 3 \tan \omega_0$  (from  $\sin \omega_0 d\omega = \frac{\text{num}}{3 \text{ den}} k dk$ ), so the group velocity is  $v = k/m^*$ . The rays of  $\omega(k, x)$  give  $dk/dt = -\partial\omega/\partial x = -(\partial\omega_0/\partial U) \nabla U = (2g/\sin \omega_0) |\nabla U|$  toward the well (Derivation S.15), so the acceleration is

$$a = \frac{1}{m^*} \frac{dk}{dt} = \frac{\text{num}}{3 \text{ den } \sin \omega_0} \cdot \frac{2g}{\sin \omega_0} |\nabla U| = \frac{2 \cos \omega_0 (1 - \cos \omega_0)}{3 \sin^2 \omega_0} |\nabla U| = \frac{2 \cos \omega_0}{3(1 + \cos \omega_0)} |\nabla U|,$$

which in the gap  $\epsilon = 1 - \cos \omega_0$  is  $[(1 - \epsilon)/(3(1 - \epsilon/2))] |\nabla U|$ : 0.2667 at  $[2, 3]$  and  $1/3$  as the gap closes. Nature's fall is  $a = c^2 |\nabla U_K|$  with  $U_K = GM_K/(c^2 r)$  and  $c^2 = 1/3$  here, so  $U_K = [2 \cos \omega_0 / (1 + \cos \omega_0)] U$ , Eq. (14), 0.800 at  $[2, 3]$ .

**Derivation S.19** (the post-Newtonian parameters against the Newtonian potential the orbits measure, Section 8). Inputs: Eq. (14),  $U = \lambda U_K$  with  $\lambda = (1 + \cos \omega_0)/(2 \cos \omega_0) = (\text{den} + \text{num})/(2 \text{ num})$ ; the composed clock's metric  $N^2 = e^{-2U}$ ,  $h^2 = e^{2U}$  (S.34); the three forms in  $U$ : the bending  $4U_b$ , the periapsis  $6\pi U_1/(a(1 - e^2))$ , the redshift  $f(\omega_0) U$ . Steps: (the orbit step) a slow packet of matter moves by the geometric optics of its band at the composed paces, whose rays are the geodesics of the metric  $N^2 = e^{-2U}$ ,  $h^2 = e^{2U}$ ; its expansion,  $g_{00} = -(1 - 2U + 2U^2)$  to the second order and  $g_{ab} = (1 + 2U) \delta_{ab}$  to the first, the orders the periapsis reads, is the post-Newtonian metric with  $\gamma = 1$  exactly and  $\beta = 1$  in  $U$  at the small gap, so the periapsis advances per orbit by Einstein's  $6\pi U_1/(1 - e^2)$  with  $U_1 = \mu/a$ , that is  $6\pi\mu/(a(1 - e^2))$ . (The three parameters against  $U_K$ :) the bending is  $4\lambda U_{K,b} = 2(1 + \text{den}/\text{num}) U_{K,b}$ , and the post-Newtonian bending is

$(1 + \gamma) 2GM/(c^2 b) = (1 + \gamma) 2U_{K,b}$ , so  $1 + \gamma_K = 1 + \text{den}/\text{num}$ ,  $\gamma_K = \text{den}/\text{num}$ . The periapsis is  $\lambda$  times Einstein's  $6\pi U_{K,1}/(a(1 - e^2))$ , and the post-Newtonian advance is  $(2 - \beta + 2\gamma)/3$  of Einstein's, so  $(2 - \beta_K + 2 \text{den}/\text{num})/3 = (\text{den} + \text{num})/(2\text{num})$ , whence  $\beta_K = 2 + 2 \text{den}/\text{num} - 3(\text{den} + \text{num})/(2\text{num}) = (\text{den} + \text{num})/(2\text{num})$ . The redshift is  $f(\omega_0)\lambda U_K$ , and  $f\lambda = \frac{2(1 - \cos \omega_0)}{\omega_0 \sin \omega_0} \cdot \frac{1 + \cos \omega_0}{2 \cos \omega_0} = \frac{\sin^2 \omega_0}{\omega_0 \sin \omega_0 \cos \omega_0} = \frac{\tan \omega_0}{\omega_0}$ , so  $1 + \alpha_K = \tan \omega_0/\omega_0$ . At  $[2, 3]$ : 1.5, 1.25 and 1.329; all three tend to 1 as  $\omega_0 \rightarrow 0$ , and  $\gamma_K - 1 < 2 \times 10^{-5}$  requires  $\text{den}/\text{num} - 1 < 2 \times 10^{-5}$ . The push on a moving record, of which the bending is the  $\beta = 1$  case, is derived from the same dispersion in S.58; the orbit step above holds for the quadratic band, and at the computed pair the band's  $k^4$  term, of the same post-Newtonian order, adds 13 to 15 percent to the advance for an orbit in a coordinate plane and 8 to 10 percent in the (111) plane, the band's fourth-order term of S.21 against its quadratic one, computed on the exact band twice ( $U_1/p$  from  $5 \times 10^{-4}$  to  $1.1 \times 10^{-2}$ ,  $e$  from 0.25 to 0.33; the pure-axis coefficient's 17 percent realised by no orbit), vanishing as  $\tan^2 \omega_0$  (two recomputations on the exact band).

**Derivation S.20** (the bending of light and Shapiro's delay, Section 8 (b) and (c)). Inputs: light's index  $n = 1 + 2U$  with  $U = \mu/r$  (Derivation S.2). Steps: (a) The delay. Along a straight path at the impact parameter  $b$  from  $-x_1$  to  $x_2$  is  $\int (n - 1) dl/c = 2\sqrt{3} \mu \int_{-x_1}^{x_2} dl/\sqrt{l^2 + b^2} = 2\sqrt{3} \mu [\text{arsinh}(x_2/b) + \text{arsinh}(x_1/b)]$ , which for  $x_1, x_2 \gg b$  is  $2\sqrt{3} \mu \ln(4x_1 x_2/b^2)$ : Shapiro's delay with  $\gamma = 1$  in  $U$ , and with  $\gamma_K$  against  $U_K$  by the same factor as the bending, since the bending is the transverse derivative of the same integral. (b) The bending. With  $n - 1 = 2U = 3\sigma_{\text{tot}}/(2\pi\Gamma r)$ ,  $r^2 = b^2 + l^2$ , the transverse gradient is  $\partial n/\partial b = -3\sigma_{\text{tot}} b/(\pi\Gamma(b^2 + l^2)^{3/2})$ , and  $\int_{-\infty}^{\infty} b dl/(b^2 + l^2)^{3/2} = 2/b$ , so the turn is  $3\sigma_{\text{tot}}/(\pi\Gamma b) = 4U_b$  with  $U_b = 3\sigma_{\text{tot}}/(4\pi\Gamma b)$ , the index's  $2U$  giving twice the Newtonian  $2U_b$ : Eq. (15) of the paper.

**Derivation S.21** (the moving clock to the second order, Section 8 (e)). Inputs: the exact band of a free record of the pair  $[\text{num}, \text{den}]$  along an axis, from Eq. (1) at the vacuum's pace (S.1):  $\cos \omega = c_0(2 + \cos k)$  with  $c_0 = \text{num}/(3 \text{den})$ ,  $\cos \omega_0 = 3c_0$ ; the rotation at the moving centre,  $\Omega(k) = \omega(k) - k\omega'(k)$ , the phase rate less  $k$  times the group velocity  $v = \omega'(k)$ , which is what a clock carried by the packet reads. Steps: write  $s = \sin \omega_0$ ,  $c = \cos \omega_0$  and  $u = 1 - \cos k = k^2/2 - k^4/24 + \dots$ ; then  $\cos \omega = c - c_0 u$ , and with  $\omega = \omega_0 + \delta$ ,  $\cos \omega = c - s\delta - c\delta^2/2 + \dots$ , so  $s\delta + c\delta^2/2 = c_0 u$  and

$$\omega(k) = \omega_0 + \alpha k^2 - \beta k^4 + O(k^6), \quad \alpha = \frac{c_0}{2s}, \quad \beta = \frac{c_0}{24s} + \frac{c c_0^2}{8s^3}.$$

Hence  $v = \omega' = 2\alpha k - 4\beta k^3$ ,  $\Omega = \omega - k\omega' = \omega_0 - \alpha k^2 + 3\beta k^4$ , and inverting  $v(k)$ ,  $k^2 = v^2/(4\alpha^2) + \beta v^4/(4\alpha^5) + \dots$ , so

$$\frac{\Omega}{\omega_0} = 1 - \frac{v^2}{4\alpha\omega_0} - \frac{\beta v^4}{16\alpha^4\omega_0} + O(v^6) = 1 - \frac{v^2}{2c_m^2} - \frac{2\omega_0}{\sin 2\omega_0} \cdot \frac{v^4}{8c_m^4} + O(v^6), \quad c_m^2 = 2\alpha\omega_0 = \frac{c_0 \omega_0}{s} = \frac{\omega_0}{3 \tan \omega_0},$$

where the fourth-order coefficient is  $\beta/(16\alpha^4\omega_0)$  over Lorentz's  $1/(8c_m^4)$ , which reduces with  $3c_0 = c$  to  $\omega_0(s/c + c/s) = \omega_0/(\sin \omega_0 \cos \omega_0) = 2\omega_0/\sin 2\omega_0$ . So the tick of a free quantum is Lorentz's factor at the family's own light speed  $c_m$  to the second order in  $v$  exactly, and at the fourth order, along an axis, it carries the factor  $2\omega_0/\sin 2\omega_0$ , 1.69 at  $[2, 3]$  and 1 as the gap closes; along the unit vector  $\mathbf{n}$  the band's  $k^4$  term carries  $\sum_a n_a^4$ , so the factor is  $\omega_0 [(\sum_a n_a^4) \tan \omega_0 + \cot \omega_0]$ , 1.2224 along a face diagonal and 1.0657 along the body diagonal, where the group velocity is parallel to  $\mathbf{k}$ : the full factor is reached only in the limit of a small rest rotation, the same limit in which the post-Newtonian parameters reach 1 (S.19). Check: with the exact band at  $[2, 3]$  the residual  $(f - 1 + v^2/(2c_m^2))/v^4$  at  $v = 0.02$  is  $-3.37$  along an axis against 1.6926 times Lorentz's  $-1.988$ , that is  $-3.365$ , and  $-2.12$  along the body diagonal against  $1.0657 \times (-1.988) = -2.119$ . A body: a bound body's centre moves on the body's own band,  $\omega_b(k)$ , whose rest rotation and curvature are computed from the fixed point (S.13); the same steps on that band give the same form with the body's  $c_b^2 = \omega_b/m_b^*$ , and the body's band is a computed input, not derived in closed form here. The largest group speed: on the full band  $v_g$  is largest at  $\mathbf{k} = (\pi/2, \pi/2, \pi/2)$ ,  $(\text{num}/\text{den})/\sqrt{3} = c \cos \omega_0$ , 0.3849 at  $[2, 3]$ .

**Derivation S.22** (de Broglie's relation, Section 8 (h)). Steps: from Derivation S.18,  $v = k/m^*$  with  $m^* = 3 \tan \omega_0$  at the vacuum's content, so  $k = m^* v = \omega_b v/c_m^2$  with  $c_m^2 = \omega_0/m^* = \omega_0/(3 \tan \omega_0)$ , the family's own light speed; 0.2508 at  $[2, 3]$ .

## The two slits (Section 9)

**Derivation S.23** (the fringe's spacing and the near-field minima, Section 9.2). Inputs: the gaps centred at the rows 18 and 30,  $d = 12$ ; the screen  $L = 24$  Links from the wall;  $\lambda = 8$ . Steps: the far-field spacing is  $\lambda L/d = 16$  Links. In the near field the minima are where the two gaps' paths to the screen differ by half a wavelength,  $\sqrt{L^2 + (y - 18)^2} - \sqrt{L^2 + (y - 30)^2} = \pm 4$ , whose solutions are  $y = 15.28$  and  $y = 32.72$ : the rows 15 and 33. Check: the two roots computed from the equation above.

## Calibration (Section 10)

**Derivation S.24** (the Link's bound from light's dispersion, Section 10.2). Inputs: the group speed  $v_g = (1/\sqrt{3})(1-k^2/12)$  (Derivation S.1); the bound on a quadratic dependence of light's speed on its energy, MAGIC's subluminal  $E_{QG,2}$  above  $6.3 \times 10^{10}$  GeV (Acciari et al. 2020; the lattice band is subluminal,  $v_g = (1/\sqrt{3})(1-k^2/12)$ , so the subluminal limit applies), that is  $|\delta v/v| = (3/2)(E/E_{QG,2})^2 < 3.8 \times 10^{-16}$  at  $E = 1$  TeV in the standard quadratic form. Steps:  $k^2/12 < 3.8 \times 10^{-16}$  gives  $k < 6.7 \times 10^{-8}$  per Link at 1 TeV; with  $k = 2\pi \text{Link}/\lambda$  and  $\lambda = 1.24 \times 10^{-18}$  m at 1 TeV, the Link is below  $k\lambda/(2\pi) = 1.3 \times 10^{-26}$  m along an axis, where the band's  $k^4$  coefficient is  $-1/54$  ( $-1/135$  averaged over directions,  $2.1 \times 10^{-26}$  m; 0 along a diagonal, no bound); over the unit of S.27,  $6.69 \times 10^{-13}$  m, the axis bound gives  $\omega_e < 2.0 \times 10^{-14}$  ( $3.1 \times 10^{-14}$  on the average) and  $\gamma_K - 1 = \omega_e^2/2 < 2.0 \times 10^{-28}$ .

**Derivation S.25** (Newton's constant as a reading, Eq. (20)). Inputs: the massless holder's rest far from a static source of  $s$  quanta per interval,  $3s/(4\pi r)$  over the divisor  $E_g$ ; the clock's share  $U = c/\Gamma$ . Steps:  $U = 3s/(4\pi r \Gamma E_g)$ , and nature's  $U = GM/(c^2 r)$  with  $M = s$  in quanta and  $c^2 = 1/3$  gives  $G/c^2 = 3/(4\pi \Gamma E_g)$ , that is  $G_{\text{clock}} = 1/(4\pi \Gamma E_g)$ ; Kepler's  $G$ , read from the orbits of matter, is  $[2 \cos \omega_0/(1 + \cos \omega_0)] G_{\text{clock}}$  by Eq. (14).

**Derivation S.26** (the fine-structure constant as a coefficient of the file, Section 10.1). Inputs: the holder of the sign under its rotation (Section 5.5) with its declared write weight  $k$ , the written level  $k$  times the plain one and the turn  $\theta = \text{level}/\Gamma$ ,  $k = 1$  in the rule's file; a circular record of a family at the vacuum's paces,  $z = Ae^{i\omega_s t}$ , the quantum of a family one unit of the invariant,  $2A^2 \sin \omega_s = T$ , with the share  $T \sin \omega_s$  and the Wronskian  $A^2 \sin \omega_s = T/2$  (S.9); the holder's rest far from a static source of  $s$  per interval,  $3s/(4\pi r)$  over its weight  $E_s$  (Derivation S.25's line); de Broglie's relation and light's  $c = 1/\sqrt{3}$  (S.1, S.22). Steps: (a) One quantum writes  $s = W/(E_s T) = 1/(2E_s)$  per interval, for every family, since the sourcing family's denominator enters no write (the count is the share over the wall  $3 \text{den}_s T$  and the source the form over  $E_s T$ , the same number); so  $|q_p| = |q_e|$  with nothing declared. The angle at  $r$  is  $\theta(r) = 3ks/(4\pi r \Gamma) = 3k/(8\pi r \Gamma E_s)$ . (b) A reader of unit sense gains the angle per interval, so its wave number changes by the gradient,  $dk_{\text{dB}}/dt = -\nabla \theta = 3k/(8\pi r^2 \Gamma E_s)$ ; Coulomb's law in nature's units,  $dk_{\text{dB}}/dt = \alpha c/r^2$ , gives  $\alpha_{\text{law}} = \sqrt{3} r^2 dk_{\text{dB}}/dt = (3\sqrt{3}/8\pi) k/(\Gamma E_s) = 0.20675 k/(\Gamma E_s)$ ; the holder's weight enters the source and the read alike as  $k$  and  $E_s$ , and the family's gap cancels. (c) The reader's shift: the step carries  $e^{-i\theta_0}$  on  $z_{\text{next}}$ , so a sense + quantum goes from  $\omega$  to  $\omega - \theta_0$  at its fixed Wronskian (S.33), its energy from  $T \sin \omega$  to  $T \sin(\omega - \theta_0)$ ,  $\delta E = -T\theta_0 \cos \omega$  per quantum; a body of  $n$  quanta and sense  $\pm$  shifts by  $\mp n T \theta_0 \cos \omega = -2\theta_0 \cos \omega W$  with  $W = \pm n T/2$ , the force  $\pm n \nabla \theta$ , Coulomb's magnitude with  $q = \pm ne$  and  $e = T/2$ ; which sense moves apart is the declared bit of Section 5.5 (as the chain is written like senses attract, and nature's measurement declares that like charges repel), the magnitude untouched; the reader sources the holder it reads at the same weight, so the pair is reciprocal (the reciprocity Theorem 3 needs). (d) The number: nature's  $\alpha = 1/137.036$  fixes  $k/(\Gamma E_s) = 1/28.33$  and nothing else;  $k$  is the file's integer read from this one click as  $E_g$  is from  $G$ , and at  $k = 1$  no integer  $\Gamma E_s$  gives it (28 gives  $1/135.43$  and 29 gives  $1/140.27$ ). (d') The energy line. The weight is the product of the write's and the read's,  $k = k_w k_r$ , with the read's  $k_r = 1$  on the integer lattice (a read weight would multiply a level the write already rounded away). A plane record under the turn loses  $2W \cos \omega \Delta \theta$  of form per Node per step, the plane's form at the turned levels, per quantum  $T \cos \omega_s (k_r/\Gamma) \Delta \ell$ ; the holder's own form gains  $\sum_i s_i (\ell_{t+1} - \ell_{t-1})$  per step under the write, per quantum  $(k_w/E_s) \Delta \ell$ , the two factors 2 cancelling; so the loss over the gain is  $\kappa = k_r E_s T \cos \omega_s / (k_w \Gamma)$  at every step and for every motion under the turn's convention, a level built by the body, an identity checked in floating point at  $-0.996 \pm 0.009$  for  $\kappa = 1$ ,  $-0.498$  at 0.5 and  $-1.993$  at 2, and a charged body and its light conserve if and only if  $k_r E_s T \cos \omega_s = k_w \Gamma$ , the law's line. Under it, with  $k_r = 1$ ,  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) T \cos \omega_s / \Gamma^2$ , at nature's gap  $T = \Gamma^2/28.33$ ,  $E_s$  a unit of the level and the file one integer fewer; the line is per family through  $\cos \omega_s$ , which in nature is 1 to  $10^{-18}$  for every family, so one write weight serves all; light and every real-line family read no turn and stand outside it. The rule's own universe has no plane family and keeps  $T = 2^{15}$ ; the loader's gate,  $E_h T \text{num} = k_w \Gamma \text{den}$  for a plane family reading a holder under the rotation, enforces the line. The per-step identity above is the one-row case, a level built by the body itself; under the rule in force, no record reading its own write, the reader's loss is by the other record's row and the writer's gain its own row's, and what holds per step is the two-record balance at the same coefficient per quantum, not run; the loader's gate is unchanged by this. The radiated share of a charged body (S.45 (c)): the time level carries half of Larmor's power in form; the odd lines' part depends on their wall and coupling, the fork open; no rate and no lifetime is claimed. Two derivations on the identity and its number; the line a decision of the model (assumption). (e)  $\Gamma$  is bounded by no click of the clocks (the gap bound binds the matter family's own denominator,  $\text{den} > 1.5 \times 10^{19}$  for the least pair, S.27, and  $\Gamma$  only in a file that writes that pair over  $\Gamma$ , as the shipped file does): the content is an integer and the clock reads it as  $p_0 = \Gamma(1 - 1/\Gamma)^c$ , so the finest level at one Node and one interval is  $1/\Gamma$ ; but the write adds  $(\sum wq + r) \text{div } E$  with the remainder  $r$  kept, so a sub-unit source does not vanish but writes one level once every  $1/s$  intervals, the level at a Node a sequence of 0 and 1 whose mean is the exact rest  $3s/(4\pi r)$ , a first-order sigma-delta modulation; a clock in such a level reads the level 1 for the fraction  $\Delta U \Gamma$  of its intervals and its mean redshift is  $\Delta U$ , finer than  $1/\Gamma$ , and a bound body spanning many Nodes averages their independent dithers, so the gravitational clocks of nature bound the mean potential and not  $\Gamma$ ; nothing bounds  $\Gamma$  from above and no line fixes it:  $\Gamma$  is free and large (two derivations). Check:  $3\sqrt{3}/(8\pi) = 0.20675$ ;  $137.036 \times 0.20675 = 28.33$ ; the share's formula at  $[2, 3]$  and  $\Gamma = 6,000$  gives  $|W|/(\text{count } T) = 0.6708 = 1/(2 \sin \omega_0)$  for a record laid by share

( $A^2 = T/(2\sin^2 \omega_0)$ , the unit S.61 rows 12 and 16 name for such a lay) and  $1/2$  for a record of count 1, one quantum of the family under the record's own unit (S.49 (5), S.61 row 16);  $3\sqrt{3}/(8\pi)/28.33 = 0.20675/28.33 = 1/137.0$ .

**Derivation S.27** (the units, the share's ceiling and the matter pair's gap, Section 10.1). Inputs: the electron as the free quantum of matter, rotating at  $\omega_0$  per interval, and nature's electron at  $m_e c^2/\hbar$  per second; light's band  $\cos \omega = 1 - \frac{1}{3} \sum_a (1 - \cos k_a)$  (S.1); the highest photon observed, 1.4 PeV; the share  $T \sin \omega$  a click reads (S.14). Steps: one interval is  $\omega_0 \hbar / (m_e c^2) = \omega_0 \times 1.288 \times 10^{-21}$  s, and one Link, light moving  $1/\sqrt{3}$  Link per interval at  $c$ , is  $\sqrt{3} \omega_0 \bar{\lambda}_e = \omega_0 \times 6.69 \times 10^{-13}$  m with  $\bar{\lambda}_e = \hbar / (m_e c) = 3.862 \times 10^{-13}$  m. The band's top along the body diagonal,  $k_a = \pi$  for all  $a$ , is  $\cos \omega = -1$ ,  $\omega = \pi$ , and along an axis,  $k = \pi$  alone,  $\cos \omega = 1/3$ ,  $\omega = 1.231$ . Along an axis the group velocity vanishes at the edge; along the diagonal,  $k_a = q$  on every axis, the band is  $\omega = q$  exactly, so the speed stays  $1/\sqrt{3}$  up to the corner, where the band has a cusp and no one gradient. That is a GameBoard statement of the highest rotation, which no click reads as an energy: what a click reads of a quantum of frequency  $\omega$  is its share,  $T \sin \omega$  (S.14), at most  $T$ , reached at  $\omega = \pi/2$ , which the band contains along the diagonal (along an axis it ends at 1.231,  $\sin \omega = 0.943$ ), and an energy measurement of nature, a calorimeter or an air shower, is a sum of transitions' shares. The electron's rest energy is  $T \sin \omega_0$ , so a photon of 1.4 PeV read as a share gives  $\sin \omega_0 \leq 0.511 \text{ MeV} / 1.4 \text{ PeV}$ ,  $\omega_0 < 3.7 \times 10^{-10}$  for every orientation of the lattice (the diagonal's  $3.65 \times 10^{-10}$ , the axis's  $3.44 \times 10^{-10}$ ; one number printed). Then  $\text{den} = 2/\omega_0^2 > 1.5 \times 10^{19}$  and  $\gamma_K - 1 = 1/\cos \omega_0 - 1 \approx \omega_0^2/2 < 6.7 \times 10^{-20}$  (S.19);  $\Gamma$  reads nothing of  $\omega_0$ ,  $\alpha$  fixing  $k/(\Gamma E_s)$  and not  $\Gamma$  (S.26). Were the frequency read as the energy,  $T\omega$ , the bound would be  $1.1 \times 10^{-9}$  along the diagonal and  $4.5 \times 10^{-10}$  along an axis: the law's resonance adds frequencies (S.43) and its share adds sines (S.14), the two agreeing to  $\omega^2/6$  relative, below  $10^{-18}$  at nature's gap, and parting at the Link scale, where which of the two a click takes the law as written does not decide and no measurement of nature reaches; named, not claimed. Check:  $\hbar/(m_e c^2) = 1.0546 \times 10^{-34} / 8.187 \times 10^{-14} = 1.288 \times 10^{-21}$  s;  $\sqrt{3} \times 3.862 \times 10^{-13} = 6.69 \times 10^{-13}$  m;  $0.511/(1.4 \times 10^9) = 3.65 \times 10^{-10}$ ;  $2/(3.65 \times 10^{-10})^2 = 1.5 \times 10^{19}$ ;  $(3.65 \times 10^{-10})^2/2 = 6.7 \times 10^{-20}$ ;  $\sin 1.231 = 0.943$ ;  $[\pi - \omega(\pi - \epsilon)]/(\sqrt{3}\epsilon) = 0.5774$  at  $\epsilon = 10^{-2}$  and  $10^{-3}$  on the diagonal.

**Derivation S.28** (the content's index for every family, Section 8 (p)). Inputs: Eq. (4) at the vacuum's pace and at a content  $U$  under the composed paces,  $(p_0/\Gamma)^2 = N^2$  and  $(p_a/\Gamma)^2 = N^4$  with  $N = (1 - 1/\Gamma)^c = e^{-U}$  to  $O(1/\Gamma)$  (Derivation S.2); a stationary boundary conserves the rotation  $\omega$ . Steps: outside,  $\cos \omega = 1 - g - \frac{\text{num}}{3\text{den}}(1 - \cos k)$  with  $g = 1 - \text{num}/\text{den}$ ; inside,  $\cos \omega = 1 - gN^2 - \frac{\text{num}}{3\text{den}}N^4(1 - \cos k_{\text{in}})$ . Equating,  $\frac{\text{num}}{3\text{den}}N^4(1 - \cos k_{\text{in}}) = \frac{\text{num}}{3\text{den}}(1 - \cos k) + g(1 - N^2)$ , so  $1 - \cos k_{\text{in}} = [1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}]/N^4$ , exact in  $k$  on the domain where the right side lies in  $[0, 2]$ . With  $1 - N^2 = 2U - 2U^2 + \dots$  and  $N^{-4} = 1 + 4U + \dots$  the first order is  $[1 - \cos k + 6(\text{den} - \text{num})U/\text{num}]/(1 - 2U)^2$ , the quadratic forms' line, the mass term's second order  $-6(\text{den} - \text{num})U^2/\text{num}$ . For light,  $g = 0$ :  $1 - \cos k_{\text{in}} = (1 - \cos k)/N^4$ , so  $\sin(k_{\text{in}}/2) = \sin(k/2)/s$  with  $s = N^2$ , exact in the law's variable  $2\sin(k/2)$  where  $\sin(k/2) \leq s$ . The phase index is  $k_{\text{in}}/k = (2/k) \arcsin[\sin(k/2)/s] = 1/s + k^2(1/s^3 - 1/s)/24 + O(k^4)$ , which is  $1/s = e^{2U}$ , Derivation S.2's long-wavelength index, at  $k \rightarrow 0$  alone. At  $k = \pi/2$  and  $s = 0.8$  the exact ratio is 1.380 against  $1/s = 1.25$ , and to first order in  $U$  the index is  $1 + [4 \tan(k/2)/k]U$ , the coefficient 2 at  $k \rightarrow 0$  and  $8/\pi$  at  $k = \pi/2$ : the lattice's own chromatic dispersion, absent in the ray limit in which the bending, the delay and the lens are read. For slow matter,  $k, k_{\text{in}} \ll 1$ :  $k_{\text{in}}^2 = [k^2 + 6(\text{den} - \text{num})(1 - N^2)/\text{num}]/N^4$ , so  $n_m^2 = k_{\text{in}}^2/k^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)]/N^4$ , to first order  $[1 + 12(\text{den} - \text{num})U/(\text{num} k^2)]/(1 - 2U)^2$ . Check: with  $k^2 = m^* v^2$  and  $m^* = 3 \tan \omega_0$  (S.22), the  $1/k^2$  term is  $12(\text{den} - \text{num})U/(\text{num} m^* v^2) = [4(1 - \cos \omega_0)/(3 \cos \omega_0 \tan^2 \omega_0)]U/v^2$ , which as  $\omega_0 \rightarrow 0$  is  $(2/3)U/v^2 = 2(U/3)/v^2 = 2\Phi/v^2$  with  $\Phi = c^2 U$ ,  $c^2 = 1/3$ : nature's index for a slow body in a potential. The factor  $6(\text{den} - \text{num})/\text{num}$  over  $m^* 2$  is the matter band's own, the Kepler column of Eq. (14), and  $1/N^4$  the Link's double read.

**Derivation S.29** (the clusters' cosmology: the ball of dust, Section 7.5). The paper's claim from this derivation is the first order in  $z$ , Friedmann's dust equations and Hubble's law; the second order and the vacuum's write are recorded here and not claimed. Inputs: Newton's pull between bodies at Kepler's  $G_K$  (Section 8 (j), Derivation S.25); a finite ball of dust of radius  $a(t)$  and uniform density  $\rho$  on the static lattice; the Doppler period of a receding source, row (m), and the moving clock's factor, row (e). Steps: a shell at  $a$  feels the mass inside it,  $M(a) = (4\pi/3)\rho a^3$ , and nothing from outside (Newton's theorem on a sphere, which holds for the  $1/r$  level of Section 8 (j)), so  $\ddot{a} = -G_K M(a)/a^2 = -(4\pi G_K/3)\rho a$ . Multiplying by  $\dot{a}$  and integrating with  $\rho a^3$  constant gives  $\dot{a}^2/a^2 = (8\pi G_K/3)\rho - \kappa/a^2$  with  $\kappa$  the start's declaration, positive for a bound ball: Friedmann's two equations for dust, as Milne and McCrea derived them for the Newtonian ball. The deceleration parameter is  $q_0 = -\ddot{a} a/\dot{a}^2 = (4\pi G_K \rho/3)/H^2 = \Omega_m/2$  with  $\Omega_m = 8\pi G_K \rho/(3H^2)$ , positive for every  $\rho > 0$ . The redshift: the period of a receding source at the speed  $\beta c$  is read  $(1 + \beta)$  times longer at the resting NodeReader (row (m)) and the source's own tick is longer by the moving clock's factor  $1/\sqrt{1 - \beta^2}$  at  $c_m = c$  (row (e)), so  $1 + z = (1 + \beta)/\sqrt{1 - \beta^2} = 1 + \beta + \beta^2/2 + \dots$ , nature's relativistic Doppler form to the second order, the classical  $1/(1 - \beta) = 1 + \beta + \beta^2$  differing at the second order. At the computed pair  $c_m = 0.867c$  the second order differs, as every number of the computed pair does. Every click's duration is stretched by the same  $1 + z$ , the light curves' time dilation, and a surface's brightness falls by  $(1 + z)^{-4}$ , Tolman's dimming, both from the same two rows; light from a source at  $r$  falls into the ball's deeper centre, a blueshift  $\Delta U = (\Omega_m/4)(H_0 r/c)^2$  beside the Doppler, the second order that makes  $d_L(z)$  Friedmann's with  $q_0$  unchanged. The vacuum's write: a uniform source  $s$  in the massless row's rest,  $\nabla^2 L = 3s$  in the lattice's units ( $c^2 = 1/3$ ), has the solution  $L = L_{\text{faces}} - s(R^2 - r^2)/2$  on the ball, a bowl rising outward. Its pull on

a shell at  $a$  is  $+\Lambda c^2 a/3$  with  $\Lambda$  proportional to the write's rate  $|q_{\text{vac}}|/(E_v \Gamma L W)$ , so  $\ddot{a} = -(4\pi G_K/3) \rho a + \Lambda c^2 a/3$  and  $\dot{a}^2/a^2 = (8\pi G_K/3) \rho + \Lambda c^2/3 - \kappa/a^2$ , Friedmann's equations with a cosmological constant;  $q_0 = \Omega_m/2 - \Omega_\Lambda$ , and  $\ddot{a} = 0$  where  $\rho = 2\rho_\Lambda$ ,  $1 + z_t = (2\Omega_\Lambda/\Omega_m)^{1/3}$ ,  $z_t = 0.67$  at  $\Omega_\Lambda = 0.7$ . Light's band at  $(\pi, \pi, \pi)$  needs  $L \geq 0$  at every Node (the guard), so  $L_{\text{faces}} \geq sR^2/2$ , in  $U$  the faces' content  $\Lambda R^2/6$ ; at one astronomical unit the bowl's pull  $\Lambda r/3$  is  $10^{-22}$  of the Sun's.

**Derivation S.30** (the shadow's capture radius from light's index, Section 8). Inputs: light's index  $n(r) = e^{2U}$  with  $U = m/r$  (Derivation S.2 under the composed paces),  $m = GM/c^2$  in Links. Steps: in a static isotropic index a ray at the impact parameter  $b$  keeps  $n(r) r \sin \psi = b$  along its path, so a circular ray stands where  $d(nr)/dr = 0$ :  $d(re^{2m/r})/dr = e^{2m/r}(1 - 2m/r) = 0$  at  $r = 2m$ , the photon sphere, and the capture radius is  $b = n(2m)2m = 2em = 5.437m$ . Schwarzschild's is  $3\sqrt{3}m = 5.196m$ , so the shadow is larger by  $2e/(3\sqrt{3}) - 1 = 0.046$ . Check: with the quadratic index  $1/(1 - 2U)$  the same condition gives  $r = 4m$  and  $b = 8m$ , +54 percent, which the Event Horizon Telescope excludes; the composed index is the one the principle gives.

**Derivation S.31** (the two clocks under the principle, Section 10.2). Inputs: the one assumption of Section 3.4, a uniform level  $U$  a change of units, time by  $N$  and length by  $h$  with  $hN = 1$ ; light's speed  $N^2$  (S.2). Steps: a bound body in the level is the flat body in the local units, so its size is  $1/h = N$  Links. Its band at the Node is  $2 - 2\cos \omega = 2(1 - \nu)N^2 + (\nu/3)N^4 \sum_a s(k_a)^2$  with  $\nu = \text{num}/\text{den}$  and  $s(x) = 2\sin(x/2)$ , since  $(p_0/\Gamma)^2 = N^2$  and  $(p_a/\Gamma)^2 = N^4$ , and its lattice wave numbers are  $k = k_{\text{proper}}/N$ , so  $(\nu/3)N^4 \sum_a s(k_a)^2 = (\nu/3)N^2 \sum_a k_{\text{proper},a}^2 [1 - k_a^2/12 + \dots]$ : every mode of the record, the rest rotation and every level of its spectrum, scales by  $N$  in  $s(\omega)$ , exactly at long wavelength and to the lattice's  $k^2/12$  at finite  $k$ . In  $\omega$  the rest line's rate is  $-2\tan(\omega_0/2)/\omega_0 = -f(\omega_0)$  per unit  $U$ , Derivation S.15's number. A difference  $s(\omega'_2) - s(\omega'_1)$  scales by  $N$  exactly and is no observable. The temporal beat a NodeReader reads,  $\Delta' = \omega'_2 - \omega'_1$  in  $\cos(\Delta't)$ , with  $\omega' = 2\arcsin[N\sin(\omega/2)] = N\omega[1 - (1 - N^2)\omega^2/24 + O(\omega^4)]$ , is  $\Delta' = N\Delta[1 - (1 - N^2)(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)/24 + O(\omega^4)]$ : at  $N = 0.8$ ,  $\omega_1 = 0.4$  and  $\omega_2 = 0.9$  the exact beat is 0.3916 against  $N\Delta = 0.4$  and the series 0.3920. The relative correction,  $(1 - N^2)(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)/24$  for the beat,  $(1 - N^2)\omega^2/8$  at  $\omega_1 = \omega_2$ , is  $2.5 \times 10^{-38}$  at nature's gap at the axis bound ( $\omega$  of  $10^{-14}$ ,  $U = 10^{-9}$ ;  $3.4 \times 10^{-29}$  at the LHAASO bound) and 1.4 percent at  $\omega_b = 0.78$  and  $U = 0.1$ , the lattice's arithmetic and not the law's. A cavity whose mirrors are bodies stands  $L/h = LN$  Links apart and its round trip is  $2LN\sqrt{3}/N^2 = 2L\sqrt{3}/N$  intervals, the rate  $N$  exactly with no  $\omega$  correction, the atom's beat to its correction  $(1 - N^2)\omega^2/8$  above; a cavity of declared faces keeps its  $L$  Links and ticks at  $N^2$ . So under the principle no clock type is singled out, which is what the null redshift tests measure, while a cavity of declared faces is a NodeReader's region and no body of nature.

**Derivation S.32** (the conserved form's Link factor, two witnesses, Theorem 3). Inputs: Theorem 3's line without its division on a periodic board at the content 0 with every Link's factor  $q$ , so  $p_0 = p_i = \Gamma$ ,  $R = 2\text{num } q^2$ ,  $w = 6\text{den } \Gamma^2$  and  $S = 12\text{num } \Gamma^2 - 6R$ ; the form  $E$  of Eq. (6) and the same form with the plain Link term  $-2\text{num } \sum_i \text{now}_i S_6(\text{before})_i$ ; exact fractions, no run of the implementation. First witness: three Nodes along  $x$  with  $y$  and  $z$  folded,  $\text{num} = \text{den} = 1$ ,  $\Gamma = 4$ ,  $q = 2$  ( $R = 8$ ,  $w = 96$ ,  $S = 144$ ),  $\text{now} = (1, 0, 0)$  and  $\text{before} = 0$ : one step gives  $\text{next} = (11/6, 1/12, 1/12)$ ; the plain form goes from 6 to  $-21/4$  and Eq. (6) stays 6. Second witness:  $3^3$  Nodes at  $[2, 3]$ ,  $\Gamma = 10$ ,  $q = 5$  ( $R = 100$ ,  $w = 1,800$ ,  $S = 1,800$ ), a uniform  $\text{now} = 3$  over  $\text{before} = 0$ : the step gives  $\text{next} = 4$  at every Node with the remainder 0, so the integer step and the line agree; the plain form per Node goes from 162 to  $-54$  and Eq. (6) stays 162 (4,374 over the board). Both are an independent automated check's, recomputed here. Check: at  $q = \Gamma$  the factor is 1 and the two forms coincide, the domain of the plain current.

**Derivation S.33** (magnetism: the continuity identity and the covector, Section 8). Inputs: a plane record  $z = \text{re} + i\text{im}$  stepping by Rule3 under the holder of the sign (Section 5.5):  $e^{-i\theta_0} z_{\text{next}} + e^{i\theta_0} z_{\text{before}} = [S z_{\text{now}} + \sum_a R_a (e^{i\theta_a} z_{+a} + e^{-i\theta_a} z_{-a})]/w$ , with  $\theta_0$  the time turn and  $\theta_a$  the phase on the axis  $a$ 's Links, read at the interval's start; the Wronskian with the angle,  $W_i = \text{Im}(\bar{z}_{i,\text{now}} e^{i\theta_0} z_{i,\text{before}})$ ; the Link quantity  $g_{ij} = \text{Im}(\bar{z}_i e^{i\theta_{ij}} z_j)$ , antisymmetric,  $g_{ji} = -g_{ij}$ , since the phase is read  $+\theta_a$  through the  $+a$  Port and  $-\theta_a$  through the  $-a$  Port. Steps: (a) The identity, with the angle fixed over the step (when a level changes in time the booked  $W$  moves with the record's amplitude by  $\Delta W/W = \mp \cot \omega \Delta \theta$ , nothing at nature's  $10^{-13}$ , so the charge is exact for static levels and to first order in a level's change). Multiply the step by  $w \bar{z}_{\text{now}}$  and take the imaginary part; the  $S$  term is real and drops:  $w \text{Im}(\bar{z}_{\text{now}} e^{-i\theta_0} z_{\text{next}}) + w \text{Im}(\bar{z}_{\text{now}} e^{i\theta_0} z_{\text{before}}) = \sum_a R_a (g_{i,i+a} + g_{i,i-a})$ . The first term is  $-W_i(t+1)$  and the second  $W_i(t)$ , and  $g_{i,i-a} = -g_{i-a,i}$ , so

$$w [W_i(t+1) - W_i(t)] + \sum_a [R_a g_{i,i+a} - R_a g_{i-a,i}] = 0 :$$

an exact lattice continuity equation at every Node and interval, the density  $\rho_i = w W_i$  and the flux through the Link  $(i, i+a)$  equal to  $R_a g_{i,i+a}$ , for any paces, any phases and any levels, the per-Node identity; the summed conservation needs the weight  $1/p_i^2$ , **DM** symmetric (S.9); the integer step's remainders add the term of S.9's domain. (b) Flux over density is the group velocity. For  $z = A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ ,  $\rho = w A^2 \sin \omega$  and the flux is  $R_a A^2 \sin k_a$  (common sign), so flux over density is  $(R_a/w) \sin k_a / \sin \omega = d\omega/dk_a$  from the band  $2w \cos \omega = S + 2 \sum_a R_a \cos k_a$ : the current is  $\rho v$  exactly, and the law's Node current  $J_a = g_{i,i+a} + g_{i-a,i}$  obeys  $J_a = (6\text{den}/\text{num}) W v$  on a plane record, nature's

$j = \rho v$  with the band's constant and no  $\sin \omega_b$ . (c) The covector. Inserting the plane wave in the step with the phases (the main text's convention, Section 5.5) gives  $\omega = \omega_{\text{band}}(\mathbf{k} + \boldsymbol{\theta}) - \theta_0$ , the time shift's sign reversing with the step's convention (S.43): the pair  $(\theta_0, \theta_a)$  shifts  $(\omega, k_a)$  and transforms as  $(\omega, \mathbf{k})$  does; at long wavelength and small gap every band is the Lorentz form  $\omega^2 = \omega_0^2 N^2 + c_s^2 k^2$  with  $c_s^2 = \text{num}_s / (3 \text{den}_s)$  (from  $2 \cos \omega = S/w + (2 \text{num}/3 \text{den}) \sum_a \cos k_a$ ), so  $(\theta_0, \theta_a)$  is a covector under the band's boost: minimal coupling,  $A_0$  on the time step and  $A_a$  on the Link. (d) The halved current over the time level's wall. The sign holder under the rotation lays the time line and three odd axis lines sourced by the sign's current; the derivation's own is one kernel over one wall and the halving: Maxwell's pair with one kernel sources  $(\theta_0, \theta_a)$  from  $(\rho, \text{flux}/c^2)$  over one wall. The Node's flux on the axis  $a$  is the mean of its two Links',  $R_a(g_{i,i+a} + g_{i-a,i})/2 = R_a J_a/2$ , and  $R_a/w = \text{num}/(3 \text{den})$  at the vacuum's pace, so flux over  $c_s^2 \rho$  is  $[(\text{num}/3 \text{den}) J_a/2] / [(\text{num}/3 \text{den}) W] = (J_a/2)/W$ : the odd line on the axis  $a$  is sourced by the halved current  $J_a/2$  over the same wall as the time level's  $W$ , with no number, when the pair is read at the source family's own  $c_s$ ; read at light's  $c^2 = 1/3$  the factor  $c_s^2/c^2 = \text{num}_s/\text{den}_s$  appears, 1 for light,  $1 - 10^{-28}$  at nature's matter,  $2/3$  at  $[2, 3]$ . The halving is not a choice: the Node's current is the mean of its two Links as the continuity identity defines it, as the pace reads the tension on an axis as the mean of its two ends' tensions. (e) Coulomb over  $\gamma$ . For a source moving at  $v$  along the Link and a reader co-moving with it, the time level is the moving density's,  $\gamma$  times the rest level by the current's covector ((a), (c)) at fixed total  $W$ , within (c)'s domain, so the electric force across the beam is  $\gamma F_0$ ; the odd level is sourced by  $J_a/2 = Wv$ , the reader at  $v$  feels  $-v \partial \theta_a / \partial y$ , a magnetic force  $\beta^2 \gamma F_0$  toward the source; the total is  $\gamma(1 - \beta^2) F_0 = F_0/\gamma$ , with  $\gamma$  Lorentz's at nature's gap, where the band is Lorentz's to  $10^{-20}$  (S.21); at the computed pair  $[2, 3]$  the band's own factor differs from it, and  $\beta = 0.5$  lies beyond the axis's largest group speed, so the two numbers below are the form's at nature's gap. Check:  $1/\gamma = 0.99499$  at  $\beta = 0.1$  and  $0.86603$  at  $\beta = 0.5$ , the law's row (b2)'s prediction (the law's document, the rows against nature); the identity (a) holds in exact arithmetic on a chain with contents and phases at every Node (the same check as S.9's, the phases a unitary on each read). Domain: the covector is the band's at long wavelength and breaks at the lattice's anisotropy  $k^2/12$  ( $2 \times 10^{-4}$  at  $k = 0.05$ ,  $1.1 \times 10^{-28}$  at nature's Compton wavelength); the pairing holds at the source family's own  $c_s$ , so a source and a reader of one family see one light speed.

**Derivation S.34** (the composed paces, the acts' factors and the exponential metric, Section 3.4). Inputs: the model's one premise beside the line, that a level is a count of units and every act is measured in the Node's own units (Section 3.4); the clock  $p_0$  as Rule3's coefficient in  $S$ ; the band at a pace,  $S.1$ ; the one assumption  $p_a = p_0^2/\Gamma$ . Steps: (a) The composition. One unit of level is one act on the clock as it stands, so it slows the clock by one part in  $\Gamma$  of the clock as it stands:  $p_0(c+1) = p_0(c)(1 - 1/\Gamma)$ , whence  $p_0(c) = \Gamma(1 - 1/\Gamma)^c$ . The functional equation  $P(U+l) = P(U)P(l)$  is Cauchy's, whose continuous solutions are the exponentials and no other, so  $N = p_0/\Gamma = e^{-U}(1 + O(U/\Gamma))$  with  $U = c/\Gamma$ ; the quadratic form's subtraction  $p \rightarrow p - 2$  is as memoryless and is excluded only by the premise that the read is an act measured in the cell's units, like the write. Check: at  $\Gamma = 6,000$  and  $c = 300$ ,  $(1 - 1/\Gamma)^c = 0.951225$  against  $e^{-0.05} = 0.951229$ , a relative  $4 \times 10^{-6}$ , which is  $U/(2\Gamma)$ . (b) The ruler.  $p_a = p_0^2/\Gamma = \Gamma N^2$  gives the axis ruler  $h = p_0/p_a = 1/N$ :  $hN = 1$  (the one assumption, fitted by two clicks, Section 3.4). (c) The metric.  $S.1$  at the paces reads  $2(1 - \cos \omega) = 2(1 - \nu)N^2 + (\nu/3)N^4 \sum_a s(k_a)^2$  with  $s(x) = 2 \sin(x/2)$ ; at long wavelength  $\omega^2 = N^2 [\omega_0^2 + c_m^2 (k/h)^2]$  with  $k/h = kN$ : Klein-Gordon's on a static metric with the lapse  $N$  and the spatial scale  $h$ ,  $ds^2 = -N^2 dt^2 + h^2 d\mathbf{x}^2 = -e^{-2U} dt^2 + e^{2U} d\mathbf{x}^2$ , the exponential metric in isotropic coordinates (Yilmaz 1958), whose post-Newtonian expansion is  $g_{00} = -(1 - 2U + 2U^2)$  and  $g_{ij} = (1 + 2U + 2U^2)\delta_{ij}$ :  $\gamma = 1$  and, for this metric,  $\beta = 1$ , the spatial second order 2 against Schwarzschild's 1.5 (S.19). (d) The acts' factors. Rule3 measured in the cell's units: a count per Node is a count per proper volume  $h^3$ , a step per interval is a step per proper interval  $N$ , and the write is Rule3's second difference in time, so a count source carries  $N^2/h^3 = N^5 = p_x p_y p_z / (p_0 \Gamma^2)$  (check:  $\Gamma^3 N^6 / (\Gamma N \cdot \Gamma^2) = N^5$ ); the Wronskian, one time difference, carries  $N/h^3 = N^4 = p_x p_y p_z / (p_0^2 \Gamma)$  and its readers' turn the remaining  $N$ ,  $\theta = (\text{level}/\Gamma)(p_0/\Gamma)$ , the turn per proper interval; the current's one space difference gives  $h^4$  in coordinates against the Peierls phase's  $h$  per coordinate Link, the two cancelling, so the Link angle is plain. With a tension the three rulers differ,  $h_a = p_0/p_a$ , and the volume is  $h_x h_y h_z$ . (e) The truncation. The clock's quadratic form  $(\Gamma - c)^2 + c^2 = \Gamma^2(1 - 2U + 2U^2)$  is  $\Gamma^2 e^{-2U} = \Gamma^2(1 - 2U + 2U^2 - 4U^3/3 + \dots)$  to the second order, differing at  $-4U^3/3$ ; the Link's  $(\Gamma - 2c)^2 = \Gamma^2(1 - 4U + 4U^2)$  is  $\Gamma^2 e^{-4U} = \Gamma^2(1 - 4U + 8U^2 - \dots)$  to the first order only. Check: at  $U = 0.05$  the clock's form gives  $0.905000$  against  $e^{-0.1} = 0.904837$ , the difference  $1.63 \times 10^{-4}$  against  $4U^3/3 = 1.67 \times 10^{-4}$ ; the Link's gives  $0.8100$  against  $e^{-0.2} = 0.8187$ . The horizon of the quadratic forms at  $c = \Gamma/2$  is the truncation's zero of  $\Gamma - 2c$  and not the law's (S.3).

**Derivation S.35** (the preferred-frame parameter of a diagonal metric, Section 10.3). Inputs: the model's metric is diagonal in the lattice's frame by construction: the content enters the clock  $p_0$  and the Link pace  $p_a$ , both real and even in time and in space, the content's rest is Poisson's in the lattice's frame with no speed in its kernel, and no odd real Link read exists, since a real asymmetric hopping with a curl breaks the reciprocity the conserved form needs (Theorem 3); so  $g_{0a} \equiv 0$  in the lattice's frame for every motion of the source (Section 10.3). The parametrised post-Newtonian metric in the theory's preferred frame (Will's review, the paper's reference, in the standard gauge):  $g_{0j} = -\frac{1}{2}(4\gamma + 3 + \alpha_1 - \alpha_2 + \zeta_1 - 2\xi) V_j - \frac{1}{2}(1 + \alpha_2 - \zeta_1 + 2\xi) W_j$ , with  $V_j$  and  $W_j$  the two vector potentials of the matter current, related by  $V_j - W_j = \chi_{,0j}$  for the superpotential  $\chi$ . Steps: (a) The post-Newtonian gauge freedom  $x^0 \rightarrow x^0 + \lambda \chi_{,0}$  adds  $\lambda \chi_{,0j} = \lambda(V_j - W_j)$  to  $g_{0j}$ , moving weight between  $V_j$  and  $W_j$  and leaving the sum of their two

coefficients,  $(4\gamma + 3 + \alpha_1 - \alpha_2 + \zeta_1 - 2\xi) + (1 + \alpha_2 - \zeta_1 + 2\xi) = 4\gamma + 4 + \alpha_1$ , gauge-invariant. (b) A metric whose  $g_{0j}$  vanishes in its preferred frame for every source current has both coefficients 0 in some gauge, so the invariant sum vanishes:  $4\gamma + 4 + \alpha_1 = 0$ ,  $\alpha_1 = -4(1 + \gamma)$ , which is  $-8$  at the model's  $\gamma = 1$  (S.19 in U). (c) The frame-dragging precession of a gyroscope near a rotating body is  $(4\gamma + 4 + \alpha_1)/8$  of general relativity's Lense-Thirring value, 1 in general relativity and 0 in the model: Gravity Probe B's 0 against  $37.2 \pm 7.2$  milliarcseconds per year is the same fact as  $\alpha_1 = -8$ , read by two experiments of nature. (d) Had the model's read been covariant, the boost of its static metric  $N^2 = e^{-2U}$ ,  $h^2 = e^{2U}$  to the velocity  $v$  would give  $g_{0i} = \gamma^2 v_i (N^2 - h^2) = -4U v_i$  to first order, Einstein's, with no new coefficient; the diagonal read in the lattice's frame is what removes it, so the finding is the frame's and not a number's (the carrier's derivation, S.36). Check:  $-4.000$  for the boosted cross term at first order; nature's  $\alpha_1$  from the binary pulsar PSR J1738+0333 is  $|\alpha_1| < 4 \times 10^{-5}$  (Shao and Wex 2012) and from lunar laser ranging  $10^{-4}$ , so the finding stands at  $2 \times 10^5$  and  $10^5$ ; Gravity Probe B's drag at 5 standard deviations; the two are one finding, not two. Domain:  $\alpha_1$  alone is gauge-invariant here; with  $g_{0j} = 0$  in the lattice frame,  $g_{00} = -e^{-2U}$  and  $g_{ij}$  isotropic, the standard post-Newtonian gauge, the two  $g_{0j}$  coefficients vanish separately and give  $\alpha_1 = -8$  and  $\alpha_2 = -1$  together ( $\alpha_2 = -1$  with  $\zeta_1 = 2\xi$ , both 0 under the post-Newtonian conservation laws;  $\alpha_1 = -8$  needs no such condition), the sum  $4\gamma + 4 + \alpha_1 = 0$  the gauge-invariant statement, against  $|\alpha_2| < 4 \times 10^{-7}$  (Nordtvedt 1987, the Sun's spin) and  $< 1.6 \times 10^{-9}$  (Shao, Caballero, Kramer, Wex, Champion and Jessner 2013, solitary pulsars); both are conditional on the metric read for a static source, the velocity-dependent terms of  $g_{00}$  depending on how a moving record's count writes, which the law does not fix: named, not claimed.

**Derivation S.36** (the carrier: the five forms tried, Section 10.3). Inputs:  $g_{0a}$  is the term of the wave operator odd in time and odd in space together,  $2g^{0a}\partial_t\partial_a$ ; Rule3's step is explicit (next from now and before at the Node and its six neighbours), of one Link's radius, bijective (Theorem 1), and conserves the form (Theorem 3), whose proof needs the Link coefficients reciprocal; a real record has no phase; the dispersion of a term is read by inserting the plane wave  $a = A \cos(kx - \omega t)$ . Steps, the four ways on the synchronous clock: (i) The one-sided real mixed term  $C_a[(a_{+a} - a_{-a})_{\text{now}} - (a_{+a} - a_{-a})_{\text{before}}]$  added to the line gives  $w(e^{-i\omega} + e^{i\omega}) = S + 2 \sum_a R_a \cos k_a + C_a(2i \sin k_a)(1 - e^{i\omega})$ , and  $1 - e^{i\omega} = -2i \sin(\omega/2) e^{i\omega/2}$ , so the term is  $4C_a \sin k_a \sin(\omega/2)[\cos(\omega/2) + i \sin(\omega/2)] = 2C_a \sin k_a \sin \omega + 4iC_a \sin k_a \sin^2(\omega/2)$ : the  $g_{0a}$  part beside an imaginary part, a growth or decay of  $\omega$  that breaks the form and the exact inverse. Check: at  $\Gamma = 6,000$ , light's row,  $k = 0.3$ ,  $C/w = 10^{-2}$ :  $\omega = 0.16984 - 2.47 \times 10^{-4}i$  against the band's  $0.17277$  (the first-order prediction  $-2.56 \times 10^{-4}i$ ). (ii) The time-centred mixed term, with the neighbours' next, is conservative but implicit: it needs the neighbours' next, outside the one-Link dependency radius and the explicit inverse. (iii) A real asymmetric hopping with a curl,  $R_{i \rightarrow j}/R_{j \rightarrow i}$  not a ratio of Node weights, has no symmetric step kernel, so Theorem 3 fails. (iv) A Peierls phase on the Link carries  $g_{0a}$  for a plane record only, a real record having no phase, and it shifts  $k$  by the same angle for every record, a force independent of  $\omega_0$ : it couples to the charge and not to the mass; light and neutral matter are real lines. So none of the four ways tried, real, local, explicit and bijective lines on the synchronous clock, carries  $g_{0a}$ . The fifth way, and why it fails: clock the cube's two sublattices alternately, half an interval apart, each Node reading its neighbours in the two slices that bracket its own time, their mean into the hopping and their difference into an odd Link level  $C_{ij}$ :  $w(a_i(t+1) + a_i(t-1)) = S a_i(t) + \sum_j R_{ij}[a_j(t + \frac{1}{2}) + a_j(t - \frac{1}{2})]/2 + \sum_j s_{ij} C_{ij}[a_j(t + \frac{1}{2}) - a_j(t - \frac{1}{2})]$ . Its dispersion,  $2w \cos \omega = S + 2 \cos(\omega/2) \sum_a R_a \cos k_a + 4 \sum_a C_a \sin(\omega/2) \sin k_a$ , is real for every  $k$ , the step's eigenvalues lie on the unit circle (to  $10^{-15}$  in real arithmetic on a ring of 12 Nodes,  $C$  uniform or random Link factors), the mixed term is exactly  $g_{0a}$ , and it is real, local, explicit and bijective. But a uniform record sampled at the neighbours' half-slices is  $a(t) \cos(\omega/2)$ , so the hopping does not cancel at  $k = 0$  and the rest rotation in a content becomes  $1 - \cos \omega = (1 - \nu)N^2 + \nu N^4(1 - \cos(\omega/2))$  in place of  $(1 - \nu)N^2$ ; at small  $\omega$ ,  $\omega^2 = 8(1 - \nu)N^2/(4 - \nu N^4)$ , and for  $\nu \rightarrow 1$ ,  $\omega/\omega_0 = N\sqrt{3/(4 - N^4)} = N(1 - 2U/3 + \dots)$ : a clock in a well runs at  $\omega_0(1 - 5U/3)$ , a redshift  $5/3$  of general relativity's. Check:  $\sqrt{3/(4 - 0.9^4)} = 0.9472$ , the three-dimensional staggered band's  $0.948$  at  $N = 0.9$ ,  $\nu = 0.99$  on a  $13^3$  grid, its light speed  $c^2 = 4/9$  and its top  $2\pi/3$ ; Pound and Snider's  $0.9990 \pm 0.0076$  (Physical Review 140, B788, 1965) excludes the 67 percent at 88 standard deviations, Gravity Probe A's  $1.4 \times 10^{-4}$  (Vessot et al., Physical Review Letters 45, 2081, 1980) at  $5 \times 10^3$ . Demanding the exact redshift for every small-gap family,  $P_0(4 - \nu) = N^2(4 - \nu P_a)$  for all  $\nu$ , forces  $P_0 = N^2$  and  $P_a = 1$ : the Link may not read the content. So the fifth way fails the clock as the first three fail the inverse, the form or the radius and the fourth the real record, and the result stands on Rule3's clock for the five forms tried, a derivation and not a theorem over every form:  $\alpha_1 = -8$  and the drag 0 stand on the five forms tried (S.35), and the carrier remains an explicitly named hypothesis.

**Derivation S.37** (confinement as the indivisibility of a record, Section 10.3). Inputs: a family of dimension 3: three real lines at every Node stepped by the one line, with a rotation among the three on each Link (the colour holder's phases), as the plane is two lines with a rotation in its plane; the NodeReader's definition (Section 5.3): a click is one whole quantum of a record credited to a declared region over its window, drawn by the share  $Q_R/W_c$  of the inflow through the region's front; the share of a record of several lines is the sum of the lines' forms (Theorem 3 line by line, the Link rotation a unitary on the triple, so the step kernel stays symmetric and the summed form is conserved); the readings of Eq. (1) are the record's, never one line's: the plane's re and im are booked together in  $D$ ,  $F$  and  $W$ , and so are the three lines. Steps: (a) No click reports one line. The click's object is the record's quantum, the share summed over the lines; no reading of the law credits a line apart, and the credit draws regions, not lines (Section 5.3, "never a

Node”, the law’s heading “no click names a Node”, and the one form of the click). So a record of dimension 3 clicks whole or not at all: no NodeReader ever reports one colour line, as none reports the plane’s real part alone. This is confinement with no coupling and no number: the quark model’s “no free quark” is the dimension’s indivisibility under the click. (b) The partons. A hard click, a scattering read at the momentum transfer  $q$ , reads the record’s current at the wave number  $q$  through the region’s front; the lines’ forms sum to the record’s, so what it reads is three fractions summing to one, the lines’ shares: the quark model’s three is the dimension’s three, and the fractions are the record’s lay and its Link rotations. (c) Scaling. The levels are deterministic and the count is conserved, so no pair is created and no coupling runs (Section 10.3; S.56 (6) and (13), the one root); the three fractions therefore do not change with  $q$ : Bjorken scaling is exact in the declared row, by construction, a statement of the model and not of nature, where nature’s measurements show logarithmic violations of order ten percent across the measured decades of  $q^2$  (SLAC to HERA), an explicit finding with the same root as  $\alpha$ ’s running. (d) The colourless bodies. The holder’s sources are the three lines’ Wronskian-like currents; a body whose three senses sum to 0 writes no colour level outside itself, the colourless states, and a pair of opposite sense or three coplanar axes are the two ways to 0. (e) What it is not. The binding between two nucleons is between separable records (a nucleus parts into them) and needs the total form to fall at a kept count, which Theorem 3 forbids at fixed paces (S.38); so the word “strong force” parts in two, the confinement the click’s under the dimension-3 hypothesis and the nuclear binding open. Check: the conserved form under the Link rotations was verified in exact arithmetic on a chain with random rotations (the step kernel symmetric, the form’s drift 0); the credit’s unit is the check’s own definition, read in every checked file. Status: derived from the click’s one form under the dimension-3 family, an explicitly named hypothesis, two independent derivations; clicks (no free quark, the partons’ fractions); the scaling an explicit finding, one derivation (S.56 (13)).

**Derivation S.38** (the mass defect at a kept count and the binding bound, Section 10.3). Inputs: Theorem 3, the form conserved by Rule3 at fixed paces for any records; the invariant of S.14, a body of  $M$  units of the invariant at the rotation  $\omega_b$  having the share  $M \sin \omega_b$  (the energy per quantum  $\sin \omega_b$  in units of  $T$ , Planck’s line); the rest rotation in a content,  $\omega_0(U) = N\omega_0$  to the order  $\omega_0^2$  (S.15 with  $f \rightarrow 1$ ); the massless kernel of a holder at the weight  $W$ : the level at  $r$  Links from a count  $C$  is  $3CW G(r)/E$  with  $G$  the lattice Green’s function per unit source,  $G(0) = 0.2527$  (Watson) and  $3/(4\pi r)$  far,  $e^{-r/R}$  beyond a gapped holder’s range (S.11). Steps: (a) No energy from combinatorics. Two records that approach, overlap, exchange quanta through their tails, join into one standing record or part again are records stepped by Rule3; if the paces do not change, the total form before and after is the same number to the last bit (Theorem 3), and a click moves share from one record to another and conserves the total too. So no joining, parting or crediting of clusters lowers the share at a kept count: there is no energy released, and a nucleus bound that way would have no binding energy. (b) What the defect is. The one act that changes a body’s share at a kept count is a change of its paces, a content read; a body in its own content  $x_{\text{self}}$  rotates at  $N\omega_0$  with  $N = e^{-x_{\text{self}}}$ , so its click mass is  $M \sin(N\omega_0) = M \sin \omega_0 - M\omega_0 x_{\text{self}} \cos \omega_0 + O(x^2)$ : the mass defect is  $M\omega_0 x_{\text{self}}$  to the order  $\omega_0^2$ , the self-read’s redshift of the body’s own rotation and nothing else; this is the law’s “a body that binds is lighter in every holder it sources by its binding”. Check (a reading of the implementation beside the derivation, labelled so): a standing mode in a well deepened from 300 to 600 levels at  $\Gamma = 6,000$ : the share read at the vacuum’s coefficients (Section 4.4) fell 4.0 percent, the analytic 4.1, with the invariant kept to  $10^{-5}$ ; the share at the Node’s own paces carries the factor  $p_i^2(t_1)/p_i^2(t_2)$  of Eq. (13) and rises. (c) The bound. A body of count  $C$  over  $r$  Links, bound by a holder at the weight  $W$  and the divisor  $E$ , writes on itself at most the one-Node value, the lattice kernel at the source,  $x_{\text{self}} \leq 3G(0)CW/(\Gamma E) = 0.758 CW/(\Gamma E)$ , independent of  $r$ ; for a uniform ball of radius  $r$  the centre’s self-level is 1.5 times the whole count’s level at  $r$  ( $0.36 CW/(\Gamma E)$  at the centre) and the count’s share-weighted average 1.2 times it, the first-order shift of a mode under a non-uniform content, a profile-dependent scaling and not a bound, so

$$\frac{\text{defect}}{\text{rest}} \leq 0.76 \frac{CW}{\Gamma E}, \quad \frac{\text{defect}}{\text{rest}} \times r \approx 0.29 \frac{CW}{\Gamma E} \text{ for a uniform ball, } 0.36 \text{ at its centre.}$$

(d) The numbers. The deuteron’s defect is  $2.224/1875.6 = 1.2 \times 10^{-3}$  and the nucleon’s in iron  $8.8/939 = 9 \times 10^{-3}$ , so, with the ball’s estimate, they need  $CW/(\Gamma E r) \geq 4 \times 10^{-3}$  and  $3 \times 10^{-2}$ ; at  $r$  about  $6 \times 10^{10}$  Links for a nucleon (0.84 fm at the MAGIC Link,  $1.3 \times 10^{-26}$  m) this is  $W$  of order  $10^{19}$  to  $10^{22}$  at  $\Gamma = 10^{14}$ , taken for illustration, growing with  $\Gamma$ , which no click bounds (S.26), under the reading of the nucleon as a cloud, which S.45 replaces (the nucleon one quantum of its own family, its binding the Yukawa well with its declared  $W$ ); the bound’s form does not move with the reading of  $\Gamma$ , its numbers do. The credit’s quantum does not supply it: the deuteron’s 2.22 MeV is 4.35 quanta of 0.511 MeV and the nucleon’s 8 MeV 15.7, not wholes, and no act credits share to nowhere. (e) The one lattice lightening without content, the band’s  $\sin \omega_b$  falling to 0 at the band’s top, acts at rotations of order one and makes a body nearly massless, not 99 percent of its quanta’s mass: not a nuclear binding. Check of the kernel: Watson’s integral gives  $G(0) = 0.252731$ ,  $3G(0) = 0.7582$ ;  $G(1) = G(0) - 1/6 = 0.08607$ ,  $3G(1) = 0.2582$ , 1.08 of the far kernel  $3/(4\pi) = 0.2387$  at  $r = 1$ .

**Derivation S.39** (the gapped holder’s kernel, Section 10.3). Inputs: a held row’s static line in the vacuum, from the start’s condition (Section 3.5):  $6 \text{ den } L - \text{num } S_6(L) = 3 \text{ den } \sigma$ , that is  $\text{num } \Delta L - 6(\text{den} - \text{num}) L = -3 \text{ den } \sigma$  with  $\Delta = S_6 - 6$  the lattice Laplacian and  $\sigma$  the source per interval; the reach  $\kappa^2 = 6(\text{den} - \text{num})/\text{num}$  of S.11; the write

and the read of every rotation holder the same act,  $W$  over  $E_s T$  into the level and the level over  $\Gamma$  into the angle (Section 5.5), whatever the holder's pair. Steps: (a) Dividing by num:  $\Delta L - \kappa^2 L = -(3/\nu)\sigma$ , whose lattice Green's function is  $G_\kappa(\mathbf{r}) = \int \frac{d^3 k}{(2\pi)^3} \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{6-2\sum_a \cos k_a + \kappa^2}$ , far away  $e^{-\kappa r}/(4\pi r)$  per unit source: the level is  $3e^{-r/R}/(4\pi\nu r)$  per unit  $\sigma$  with  $R = 1/\kappa$ , Yukawa's kernel with the massless one's constant, the range the gap's and the strength at the Link the massless one's:  $G_\kappa(1) = G_0(1)(1 - 0.92\kappa + O(\kappa^2))$ , so the strength at one Link is the massless one to one part in  $R$  (and the factor  $1/\nu = 1 + O(\text{gap})$ ). Check: the exact infinite-volume integrals,  $G_0(1) = 0.08606$  (Watson) and  $G_{1/20}(1) = 0.08216$  by the exact lattice Yukawa integral, the ratio 0.9546 at  $R = 20$  Links (0.08209 and 0.954 at the first order in  $\kappa$ ,  $G_0(1) - \kappa/4\pi$ ), the slope  $1 - 0.9246\kappa$ ;  $G_{1/400}(1)/G_0(1) = 0.9977$  (the  $400^3$  grid's own pair, 0.0857 and 0.0822, has the ratio 0.959). (b) The coupling. The sign holder's force constant,  $\alpha_{\text{law}}$  (S.26), is the write's half, the kernel's  $3/(4\pi)$  and light's  $\sqrt{3}$ , none of them the holder's own; a gapped holder written and read by the same acts has the same three factors and the kernel above, so it couples at  $\alpha_{\text{law}}(1 - 1/R)$  at one Link,  $1.08\alpha_{\text{law}}$  on the lattice at  $r = 1$ , and  $\alpha_{\text{law}}e^{-r/R}/r$  beyond: every rotation holder Rule3 steps, massless or gapped, abelian or three-line, couples at the charge's  $\alpha$  at the Link. A coupling  $k/(\Gamma E_s)$  itself, the file's  $1/28.33$ , would need the factor  $8\pi/(3\sqrt{3}) = 4.84$  on the kernel, which no gap supplies; a contact strength  $3/\kappa^2$  appears only for  $\kappa > 1$ , a range below one Link, whose quanta would be above the band's top. (c) The numbers. The  $W$ 's range  $\hbar/(m_W c) = 2.45 \times 10^{-3}$  fm and the pion's 1.41 fm are  $1.8 \times 10^8$  and  $1.1 \times 10^{11}$  Links at the Link's bound  $1.3 \times 10^{-26}$  m (S.24), so  $1/R$  is  $5 \times 10^{-9}$  and  $10^{-11}$ : the strength at the Link is the massless one to those parts. A pion-gap holder at its range with the law's coupling has the well's depth  $\alpha\hbar c e^{-1}/R = 1.44 \text{ MeV fm} \times 0.368/1.41 \text{ fm} = 0.38 \text{ MeV}$  at  $r = R$  and binds no state: a screened Coulomb well binds only for  $\mu g R/(\hbar c)^2 \geq 0.840$ , and at  $g = \alpha\hbar c$ ,  $R = 1.414 \text{ fm}$ ,  $\mu = 469.5 \text{ MeV}$  the number is 0.0245, 34 times under the threshold, so the deuteron's 2.22 MeV stands against no binding at all; a  $W$ -gap holder with the law's  $\alpha$  and nature's  $m_W$  gives  $G_F = \pi\alpha/(\sqrt{2}m_W^2) = 2.5 \times 10^{-6} \text{ GeV}^{-2}$  against nature's  $1.166 \times 10^{-5}$ , short by 4.7, or  $m_W = 37 \text{ GeV}$  from  $G_F$  against 80.4. The nearness of  $1/28.33$  to nature's  $\alpha_3$  at the Link scale and  $\alpha_2$  at  $m_W$ , 0.7 percent apart from each other in nature, is one coincidence at a few percent probability, used for nothing. Check of Yukawa's identity: the range  $R = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$  equals the Compton length  $c/\omega_g$  of the holder's own travelling quanta to the gap's first order (S.11), with no input.

**Derivation S.40** (the GHZ correlation under the declared credit, Section 9.4). Inputs: the GHZ family, light's pair with four real lines laid as one event and never summed at a Node, every part laid alike and stepped by the same line, so the four are identical at every NodeReader (the determinism of equal parts, Section 9.4); three NodeReaders  $A$ ,  $B$ ,  $C$ , each with its setting  $(p, q) = (\cos a, \sin a)$  up to the norm and its pattern, one integer pair per part:  $A(p, p, q, q)$ ,  $B(p, q, p, q)$ ,  $C(p, -q, -q, -p)$ , the  $+$  port reading the part  $k$  as  $e_k(+) = \alpha_k p + \beta_k q$  and the  $-$  port as  $e_k(-) = \alpha_k(-q) + \beta_k p$ ; the credit, the pair's own over three sides:  $J(\pm, \pm, \pm) = \left(\sum_{k=1}^4 e_k(A)e_k(B)e_k(C)\right)^2$  on equal parts, the common factor of the parts' amplitudes leaving every ratio. Steps: (a) With  $c_a = \cos a$ ,  $s_a = \sin a$  and the same for  $b$  and  $c$ , the  $+$  ports read  $A$  as  $(c_a, c_a, s_a, s_a)$ ,  $B$  as  $(c_b, s_b, c_b, s_b)$  and  $C$  as  $(c_c, -s_c, -s_c, -c_c)$ , so  $\sum_k e_k(A)e_k(B)e_k(C) = c_a c_b c_c - c_a s_b s_c - s_a c_b s_c - s_a s_b c_c = \cos(a + b + c)$ .  $A$   $-$  port replaces  $(p, q)$  by  $(-q, p)$ :  $A$ 's  $-$  port reads  $(-s_a, -s_a, c_a, c_a)$ , and the sum is  $-s_a c_b c_c + s_a s_b s_c - c_a c_b s_c - c_a s_b c_c = -\sin(a + b + c)$ ; by the symmetry of the three patterns every port triple with an even number of  $-$  ports gives  $\pm \cos(a + b + c)$  and with an odd number  $\pm \sin(a + b + c)$  (for instance  $(-, -, +)$  gives  $-\cos$ ,  $(-, -, -)$  gives  $\sin$ ). (b) The eight joint shares are therefore  $\cos^2(a + b + c)$  at the four even triples and  $\sin^2(a + b + c)$  at the four odd, summing to 4 at every setting; the normalised shares are  $\cos^2/4$  and  $\sin^2/4$ . (c)  $E_3$ , the shares signed by the product of the three signs, is  $4 \cdot \cos^2/4 - 4 \cdot \sin^2/4 = \cos 2(a + b + c)$ . A pairwise  $E_{AB}$  sums over  $C$ 's two ports first, which gives  $(\cos^2 + \sin^2)/4 = 1/4$  for each of the four sign pairs of  $A$  and  $B$ , so  $E_{AB} = (1 - 1 - 1 + 1)/4 = 0$ , and the same for the other two pairs; each single marginal is  $1/2$ : no signalling and no correlation in any two, the correlation standing in the three alone. (d) Mermin's combination  $M = E(a, b', c') + E(a', b, c') + E(a', b', c) - E(a, b, c)$  at  $a = b = c = 0$  and  $a' = b' = c' = \pi/4$  is  $\cos \pi + \cos \pi + \cos \pi - \cos 0 = -4$ ,  $|M| = 4$ , where every local assignment of signs to the three ports, and every mixture of them, gives  $|M| \leq 2$  (Mermin 1990). (e) With integer settings  $(p, q)$  the norms  $p^2 + q^2$  multiply every share alike and cancel, so  $x = (1, 0)$  and  $y = (1, 1)$  give the eight shares exactly, 0 and  $1/4$  on the three files with one  $x$  ( $E_3 = -1$ ) and  $1/4$  and 0 on  $(x, x, x)$  ( $E_3 = 1$ ),  $M = -4$  exactly. Check: the trigonometric identity  $\cos(a + b + c) = c_a c_b c_c - c_a s_b s_c - s_a c_b s_c - s_a s_b c_c$  and  $\sin(a + b + c) = s_a c_b c_c + c_a s_b c_c + c_a c_b s_c - s_a s_b s_c$ . Nature's angles: the law's  $E_3 = \cos 2(a + b + c)$  is nature's  $\cos(\theta_A + \theta_B + \theta_C)$  at  $\theta = 2a$ , nature's  $x$  and  $y$  at  $\theta = 0$  and  $\pi/2$  the law's bases  $(1, 0)$  and  $(1, 1)$ , as Bell's angles are (S.10). Status: derived under the declared credit, the credit the one assumption (Section 9.4).

**Derivation S.41** (the prediction, the check and the open question, Section 10.2). (1) A clock in two places. Inputs: a bound body's beat between two of its modes,  $\Delta = \omega_j - \omega_i$ , scales by the clock  $N = e^{-U}$  at the body's Node (S.31, exactly at long wavelength, to  $(1 - N^2)\omega^2/24$  at finite  $\omega$ ); a record laid on two paths is one record (Section 5.1), and the body's internal state is carried on each path by its own rotation. Steps: on two paths at the contents  $U_1$  and  $U_2$  for  $T$  intervals the internal beats accumulate the phases  $\Delta N_1 T$  and  $\Delta N_2 T$ , differing by  $\Delta(N_1 - N_2)T = \Delta \Delta U T$  to first order (write  $\Omega$  for the beat  $\Delta$ ); for a body prepared in an equal superposition of the two modes the internal states on the two paths overlap as  $|\cos(\Omega \Delta U T/2)|$ , which multiplies the path fringes' visibility. Check: strontium's clock line  $429.228 \text{ THz}$ ,  $\Omega = 2\pi \times 4.29228 \times 10^{14} = 2.697 \times 10^{15} \text{ s}^{-1}$ ;  $\Delta U = gh/c^2 = 9.807 \times 1/(2.998 \times 10^8)^2 = 1.091 \times 10^{-16}$  for

one metre;  $T = 1$  s:  $\Omega \Delta U T = 0.294$  rad, the visibility  $\cos(0.147) = 0.989$ ; the correction  $(1 - N^2)(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)/24$ ,  $(1 - N^2)\omega^2/8$  at  $\omega_1 = \omega_2$ , is  $2.5 \times 10^{-38}$  at nature's gap at the axis bound,  $3.4 \times 10^{-29}$  at LHAASO's ( $U = 10^{-9}$ ). The semiclassical expectation (Zych, Costa, Pikoński and Brukner 2011) has the same magnitude; the law's content is that the one record's beat reads the content at its Node, so the dephasing is the clock's and no new line. (2) The gravity of light. Inputs: the one write: a holder of the content gains the form  $D$  of every record over its wall (Section 3.5); Theorem 3 and S.5: a travelling light record's form summed over its Nodes is its share, its count  $n$  (energy  $nT$ ), and  $D = A^2 \sin^2 \omega$  per Node for a travelling wave as for a standing one; the tension sources the axis parts alone, which enter the Link's factor and not the clock (Section 3.4). Steps: a light packet of count  $n$  writes  $n$  quanta per interval into the massless row as a static body of count  $n$  does, so its far level is  $3n/(4\pi r)$  over  $E_g$  and the clock's well  $U = 3n/(4\pi\Gamma E_{gr})$ , Eq. (20)'s  $G_{\text{clock}}$  times  $n$ : a static body falls toward the packet at  $G_{\text{clock}} n/r^2$ , that is  $1 \times E/c^2$  for the packet's active mass. In general relativity a pencil of light pulls a static body at  $2 \times E/c^2$  (Tolman, Ehrenfest and Podolsky 1931), the pressure  $p = \rho c^2/3$  along the beam entering  $\rho + 3p/c^2$ ; in the model the stress of the wave, the Link's booking  $2 \text{ num } b^2 \sin^2 k$  and the Node's tension  $\text{num } b^2 \sin^2 k$  (S.7), enters the Link's factor on its axis and the clock reads the form alone, so the factor is 1: a prediction for a file, no measurement of nature. (3) The black hole's quanta. Inputs: the composed paces (S.34):  $p_0 = \Gamma(1 - 1/\Gamma)^c$ ,  $p_i = p_0^2/\Gamma$  rounded to the unit, every Link coefficient  $R_a = 2 \text{ num } p_i^2 q^2/\Gamma^2$ ; Theorem 1, the step a bijection; a collapse ends in Nodes whose content exceeds the frozen content (Section 7.2). Steps:  $p_i$  rounds to 0 where  $\Gamma e^{-2U} < 1/2$ , that is  $U > \frac{1}{2} \ln(2\Gamma)$ , the frozen content  $c_f = (\Gamma/2) \ln(2\Gamma)$  (4.70  $\Gamma$  levels, 28,206 at  $\Gamma = 6,000$  with  $p_0$  rounded to the unit before squaring, the law's rounding); there the Node's own six coefficients are 0, so it reads nothing of its neighbours, while its neighbours' coefficients, each the reading Node's own, still read it: the cut is one-way, the frozen Node dark by the share's declaration and not cut; what leaves a frozen region is the levels frozen in it, read by the surface's active shell at its pace, matter's a near-static source of a static well, light's a level drifting linearly that no line of the law closes; whether the law's black hole radiates, and what, is not established by the stencil and is open,  $T_H = 0$  not claimed (a reading of the stencil withdrawn by two derivations). Hawking's temperature for one solar mass, the number the open question is set against (a comparison only, as Bekenstein's entropy beside it), is  $\hbar c^3/(8\pi G M k_B) = 6.17 \times 10^{-8}$  K. The region's entropy as a count of the state is the count of its kept levels, the width's logarithm per frozen Node; what it means for radiation is open. Check:  $\hbar c^3/(8\pi G M_\odot k_B)$  with  $M_\odot = 1.988 \times 10^{30}$  kg gives  $6.17 \times 10^{-8}$  K;  $\frac{1}{2} \ln(12,000) = 4.696$ .

**Derivation S.42** (what a body writes into the holder of the sign, Section 7.4). Inputs: the continuity identity of S.33: at every Node  $w[W_i(t+1) - W_i(t)]$  equals the net of the Link fluxes  $R_{ag}$ , so  $\sum_i W_i$  over a body is exactly conserved for a static self-level and a single Node's  $W_i$  changes only by a current through its Links; in particular at one isolated Node,  $z_{t+1} + z_{t-1} = 2 \cos \omega_t z_t$  with any real  $\omega_t$  keeps  $W = \text{Im}(\bar{z}_t z_{t-1})$  exactly ( $W_{t+1} = \text{Im}(2 \cos \omega_t |z_t|^2 - \bar{z}_{t-1} z_t) = W_t$ ). The holder's line at the vacuum's pace,  $\ell_{t+1} - 2\ell_t + \ell_{t-1} = \frac{1}{3} \Delta \ell + s_t$ , with  $s$  the write per interval (the Wronskian over  $E_s T$ ): the wave equation at the speed  $1/\sqrt{3}$  with the retarded solution  $\ell(\mathbf{r}, t) = 3s(t - \sqrt{3}r)/(4\pi r)$  for a point source; a static record has  $D = 0$  and  $F = 0$  and carries no quanta (Section 3.5); the share of a travelling wave of the holder's record is  $A^2 \sin^2 \Omega/T$  quanta per Node (S.5 at  $[1, 1]$ ) moving at  $1/\sqrt{3}$ . Steps: (a) A steady rotation. A body at rest with the standing plane record  $z_i = \phi_i e^{-i\omega_b t}$ ,  $\phi$  real up to one phase, has  $W_i = \phi_i^2 \sin \omega_b$  constant at every Node and the Link quantity  $g_{ij} = \text{Im}(\bar{z}_i z_j) = 0$ : the source is static, the holder's level is its rest, no travelling event is born, no click anywhere. (b) Uniform motion. A uniformly moving source writes no travelling light in the long-wavelength band,  $\omega = c|k|$ , where no sub- $c$  source is phase-matched: its retarded level is the uniformly moving rest, which carries no energy flux to infinity, as for a uniformly moving charge in Maxwell's theory. On the lattice light's phase speed falls to 0.392 Links per interval at the zone's edge along an axis, so a source faster than that is matched to zone-edge light (the forced recurrence's denominator  $2 \cos(k \cdot v) - 2 \cos(k)$  vanishing there), which a body smooth over many Links sources only by its Fourier weight at that wave number, below  $e^{-100}$  for every body of the paper; at the computed pair no body reaches 0.392 (matter's largest group speed 0.385 along the diagonal), from  $[9, 10]$  on a body along an axis can, an approximation with a named domain and not an exact exclusion (a finding of two derivations). (c) A breathing body. Breathing is a change of the amplitudes  $A_i(t)$  across the body by currents between its Nodes; the total  $\sum_i W_i$  is kept, so the monopole of the source is constant and radiates nothing. The far level of a spherically symmetric source distribution  $\sigma_i(t)$  of constant total  $S$  is, by the retarded integral with the retardation expanded, the known multipole expansion of the holder's wave equation's solution, standard mathematics of the law's line ( $\sigma(t - \sqrt{3}|\mathbf{r} - \mathbf{r}'|) = \sigma(t_r) + \sqrt{3}(\hat{\mathbf{r}} \cdot \mathbf{r}')\dot{\sigma} + \frac{3}{2}(\hat{\mathbf{r}} \cdot \mathbf{r}')^2\ddot{\sigma} + \dots$ , the first order averaging to 0 over the sphere and  $\langle(\hat{\mathbf{r}} \cdot \mathbf{r}')^2\rangle = r'^2/3$ ),

$$\ell(r, t) = \frac{3S}{4\pi r} + \frac{3}{4\pi r} \cdot \frac{1}{2} \ddot{Q}(t - \sqrt{3}r), \quad Q(t) = \sum_i \sigma_i(t) r_i^2,$$

so a body of mean square radius  $\langle r^2 \rangle$  breathing with its second moment  $Q(t) = Q_0(1 + 2\delta \cos \Omega t)$ ,  $\delta$  defined by it with the monopole  $S$  constant and  $Q_0 = S\langle r^2 \rangle$ , and writes the travelling level  $\ell_1 = 3s_{\text{eff}}/(4\pi r)$  at the frequency  $\Omega$  with  $s_{\text{eff}} = \delta \Omega^2 S\langle r^2 \rangle$ : the light born carries the breathing's frequency  $\Omega$  and not the rotation  $\omega_b$ , as Section 7.4 says, with the amplitude  $S \cdot 2\delta$  of Section 7.4 multiplied by  $\Omega^2 \langle r^2 \rangle/2$ , the second moment's factor. The quanta radiated per interval through a far sphere are the share density times the speed times the area,  $4\pi r^2 \cdot \ell_1^2 \sin^2 \Omega/(\sqrt{3}T) = (3\sqrt{3}/4\pi) s_{\text{eff}}^2 \sin^2 \Omega/T$ , of order  $\delta^2 \Omega^6 S^2 \langle r^2 \rangle^2/T$ : the scalar analogue of the quadrupole formula, and a far NodeReader's click rate is that current's

share through its front. Section 7.4's three conclusions stand: no light from a steady rotation, none from a uniform motion in the long-wavelength band, light at the beat  $\Omega$  from a breathing body. Check:  $9/(4\pi\sqrt{3}) = 3\sqrt{3}/(4\pi) = 0.4135$ ; the static kernel  $3/(4\pi r)$  is  $S.25$ 's; the retarded expansion is the standard multipole one at the speed  $1/\sqrt{3}$ .

**Derivation S.43** (the two-mode line under the rotation, Section 7.4). The convention: *this derivation's time turn*  $U_t = e^{-i\theta_t}$  is the conjugate of S.33's and the main text's; under the main text's step the reader's shift is  $\omega_b - \theta$ , under this one the sign of  $\theta$  reverses, and the covector conclusion is the same. Inputs: the plane record's step under the holder of the sign (Section 5.5),  $U_t [2 \cos \omega\text{-operator}]$  in the form  $z_{t+1} = U_t (w^{-1} [S z_t + \sum_a R_a (e^{i\theta_a} z_{t+a} + e^{-i\theta_a} z_{t-a})] - U_{t-1} z_{t-1})$  with  $U_t = e^{-i\theta_t}$  the time turn between  $t$  and  $t+1$  and  $\theta_a$  the Link phases, both the holder's level over  $\Gamma$  at the interval's start; light as the holder's own record, a level  $b_L \cos \omega_L t$  at the body with the gradient  $\nabla b_L$  across it; the body's bound modes  $\phi_m$ , real eigenvectors of the symmetric  $\mathbf{M}$  with  $(\mathbf{M}\phi_m)_i = 2w \cos \omega_m \phi_{m,i}$ , orthonormal in the Node weights. Steps: (a) The uniform turn is a gauge. Put  $z_t = e^{-i\Phi_t} u_t$  with  $\Phi_{t+1} = \Phi_t + \theta_t$ . Then  $U_t z_t = e^{-i\Phi_{t+1}} u_t$ ,  $U_t U_{t-1} z_{t-1} = e^{-i\Phi_{t+1}} u_{t-1}$  and  $z_{t+1} = e^{-i\Phi_{t+1}} u_{t+1}$ , so  $u$  obeys the plain line  $u_{t+1} = \mathbf{M}u_t/w - u_{t-1}$  exactly, for any  $\theta_t$ : a time turn uniform over the body changes the record's phase and nothing of its moduli, its form or its modes; at a single Node light read as a turn drives no resonance and no absorption. (b) A constant turn. For  $\theta_t = \theta$  the plane solution  $z = e^{-i\Omega t}$  has, under this section's step,  $2 \cos(\Omega - \theta) = 2 \cos \omega_b$ , so the sense  $+$  rotates at  $\omega_b + \theta$  and the sense  $-$  at  $-\omega_b + \theta$ : both shift by  $+\theta$  (under the main text's step of Section 5.5,  $e^{-i\theta} z_{\text{next}} + e^{i\theta} z_{\text{before}}$ , the shift is  $-\theta$ , the sign reversing with the convention as noted above), their magnitudes split by  $2\theta$ , the energy shift following the record's own sense, Section 5.5's line. (c) The Link phases couple the modes. To first order in  $\theta_a$  the Link read adds  $Vz = i \sum_a R_a [\theta_{i,a} z_{i+a} - \theta_{i-a,a} z_{i-a}]$  at the Node  $i$ , an  $i$  times a real antisymmetric matrix, Hermitian. Write  $z_t = \sum_m c_m(t) \phi_m e^{-i\omega_m t}$  with  $c$  slowly varying;  $w(z_{t+1} + z_{t-1}) - \mathbf{M}z_t = -2iw \sum_m \sin \omega_m c'_m \phi_m e^{-i\omega_m t}$ , and projecting  $Vz$  on  $\phi_j$  with  $\theta_{i,a}(t) = \theta_L \epsilon_a(i) \cos \omega_L t$  ( $\epsilon_a$  the light's profile across the body,  $\theta_L = b_L/\Gamma$ ) gives  $\langle \phi_j, V\phi_m \rangle = i\theta_L \cos(\omega_L t) K_{jm}$  with

$$K_{jm} = \sum_{\text{Links } (i,i+a)} R_a \epsilon_a(i) [\phi_j(i)\phi_m(i+a) - \phi_m(i)\phi_j(i+a)], \quad K_{mj} = -K_{jm}, \quad K_{mm} = 0,$$

the inter-mode Link current of S.33's  $g$  taken between two modes. So

$$-2iw \sin \omega_j c'_j = i\theta_L \cos(\omega_L t) \sum_m K_{jm} c_m e^{-i(\omega_m - \omega_j)t};$$

the product carries the frequencies  $\omega_m - \omega_j \pm \omega_L$  and resonates at  $\omega_L = |\omega_j - \omega_m|$ , the difference line, for drives below the sum; at  $\omega_L = \omega_j + \omega_m$  the same matrix element couples  $a_m \phi_m e^{-i\omega_m t}$  to the other sense  $b_j \phi_j e^{+i\omega_j t}$ , since a second-order recurrence has both senses as solutions and the one-sense record is initial data:  $a'' = +[\theta_L K_{jm}/(4w \sqrt{\sin \omega_j \sin \omega_m})]^2 a$ , a growth at the rate  $\Omega_R$  (S.16's  $\epsilon_{ij}$  is  $\theta_L K_{jm}/w$  here, one rate in two notations) with S.16's detuned exponents  $\pm \sqrt{\Omega_R^2 - \Delta^2/4}$ , the sum line S.16 itself names; with  $K$  real  $\frac{d}{dt}(|a|^2 \sin \omega_m - |b|^2 \sin \omega_j) = 0$ , the Wronskian kept while the share grows, so the sum line creates content of the opposite sense in pairs, a pair line at  $\omega_j + \omega_m \approx 2\omega_0$ , in nature's names  $2mc^2$ , named and not claimed (the Floquet multipliers at  $\Omega = \pi/2$  for  $\omega = \pi/6, \pi/3$ ,  $\theta_L K/w = \epsilon = 0.01$ : 0.9849 and 1.0153, the growth halving with  $\epsilon$ ; at the difference, all four of modulus 1); the one-mode line  $2\omega_b$  of the pace read (S.16) is the degenerate case; and  $K_{mm} = 0$  says a Link phase shifts no single mode at first order, the gauge statement of (a) for the Links. (d) Rabi's rate. At  $\omega_L = \omega_j - \omega_m$  the resonant terms are  $c'_j = -\theta_L K_{jm} c_m/(4w \sin \omega_j)$  and  $c'_m = +\theta_L K_{jm} c_j/(4w \sin \omega_m)$ , so  $c'_j = -\Omega_R^2 c_j$  with

$$\Omega_R = \frac{\theta_L |K_{jm}|}{4w \sqrt{\sin \omega_j \sin \omega_m}},$$

a full transfer between the two modes every half period at resonance, one transition per click; detuned by  $\Delta$  the transfer generator  $[[-i\Delta/2, -g], [g, i\Delta/2]]$  has the eigenvalues  $\pm i\sqrt{g^2 + \Delta^2/4}$ , so  $P(t) = [g^2/(g^2 + \Delta^2/4)] \sin^2(\sqrt{g^2 + \Delta^2/4}t)$ , Rabi's law at the generalised frequency, oscillatory at every detuning with no threshold, the envelope's full width  $4\Omega_R$  in the detuning at half height (the parametric exponents of S.16 belong to the sum line of (c), not here; a finding of two derivations). (e) The matrix element is the dipole. For a uniform gradient,  $\epsilon_a = 1$ , the identity  $\sum_i x_a(i) [\phi_j(\mathbf{M}\phi_m) - (\mathbf{M}\phi_j)\phi_m]_i = 2w(\cos \omega_m - \cos \omega_j) \langle \phi_j, x_a \phi_m \rangle$ , evaluated with  $\mathbf{M}$ 's hopping and  $x_a(i+a) - x_a(i) = 1$ , gives  $K_{jm} = 2w(\cos \omega_j - \cos \omega_m) d_{jm} = -4w \sin \frac{\omega_j + \omega_m}{2} \sin \frac{\omega_j - \omega_m}{2} d_{jm}$  with  $d_{jm} = \langle \phi_j, x_a \phi_m \rangle$  the dipole: the velocity form equals the length form times the line's frequency, the discrete Ehrenfest relation; for a small line  $\Omega_R = \theta_L \omega_L |d_{jm}|/2$ , nature's  $d \cdot E_0/(2\hbar)$  with the field  $E_0 = \omega_L A_0$  and  $\theta_L$  the phase per Link of the vector potential, the rotating-wave factor  $1/2$  included; a static Link phase ( $\omega_L = 0$ ) mixes the modes without a transition, Stark's shift. Check: the slowly varying method is S.16's;  $K_{mm} = 0$  for every real mode by antisymmetry; the identity in (e) follows from  $\mathbf{M}$ 's hopping terms alone, the  $S$  term cancelling. What rests on it: the body's click of S.14, one transition at the line  $\omega_j - \omega_i$  taking the share  $T(\sin \omega_j - \sin \omega_i)$ , stands under the rotation read as under the pace read, the difference line being the absorption line under the rotation and the sum a pair line; the transfer rate is the law's  $\Omega_R$  above with the current's matrix element; the fraction absorbed, with the body's write back, stays open as Section 7.4 says. Status: derived, one derivation; the pace read's one-mode line (S.16) has no analogue under the rotation.

**Derivation S.44** (the two forces between quanta and their ratio between families, Section 10.1). Inputs: *the sign holder's coupling between two quanta*,  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$  (S.26); *gravity's level from a source of  $s_g$  wells per interval*,  $3s_g/(4\pi r \Gamma E_g)$  far away, Eq. (20) of the paper (S.25), and *the clock's rate  $1 - U$  in it* (S.2); *the quantum of a family, its energy  $T \sin \omega_s$ , its count  $\sin \omega_s$  and its Wronskian  $T/2$* . Steps: (a) *The two held rows are one kind of row, the kernel  $3/(4\pi r)$  and the same acts of write and read. The sign holder is written by a quantum's Wronskian,  $1/(2E_s)$  per interval for every family, and the content holder by its form,  $\sin \omega_s/E_g$  per interval, so a quantum of the family  $s$  at  $r$  writes the level  $U(r) = 3 \sin \omega_s/(4\pi r \Gamma E_g)$ . (b) A reader quantum of the family  $s'$  has its rotation redshifted by  $U$ , its energy by  $\delta E = -T \sin \omega_{s'} U$ , so the pull is  $F_g = 3T \sin \omega_s \sin \omega_{s'}/(4\pi \Gamma E_g r^2)$ ; the electric force between two quanta by S.26 (c) is  $F_e = 3kT \cos \omega/(8\pi \Gamma E_s r^2)$ ,  $\cos \omega \rightarrow 1$  at a small gap. In the form of a coupling, with light's  $c = 1/\sqrt{3}$  as in S.26 (b), gravity's between two quanta of one family is  $\alpha_g(s) = (3\sqrt{3}/4\pi) \sin^2 \omega_s/(\Gamma E_g)$ , the square of the quantum's energy over  $T$ . (c) The ratio at one family:  $F_e/F_g = \alpha_{\text{law}}/\alpha_g(s) = k E_g/(2E_s \sin^2 \omega_s)$ , the unit of charge  $(T/2)$  over the quantum's energy  $(T \sin \omega_s)$ , squared, times the file's two weights  $k$  and  $E_g$  over  $E_s$ . (d) Between two families the weights cancel:  $(F_e/F_g)_e/(F_e/F_g)_p = \sin^2 \omega_p/\sin^2 \omega_e = (m_p/m_e)^2$  with no number, the masses being the energies  $T \sin \omega_s$  (S.27), an identity forced by charge universality and  $F_g \propto m^2$ , no test. Check: nature's  $e^2/(4\pi\epsilon_0 G m_e^2) = 4.17 \times 10^{42}$  and  $e^2/(4\pi\epsilon_0 G m_p^2) = 1.24 \times 10^{36}$ , the ratio  $3.37 \times 10^6 = 1836.15^2$ ; at the MAGIC bound  $\omega_e < 2.0 \times 10^{-14}$  (S.24),  $1/\sin^2 \omega_e > 2.5 \times 10^{27}$  and  $k E_g/E_s = 2 \times 4.17 \times 10^{42} \sin^2 \omega_e$ , about  $3 \times 10^{15}$  there (about  $10^{24}$  at the slack LHAASO bound,  $\omega_e < 3.7 \times 10^{-10}$ , S.27), its largest value, and smaller at a smaller  $\omega_e$  (what grows at a smaller gap is  $1/\sin^2 \omega_e$ , the ratio the weights must supply per unit); on the board the two rows carry two integers and one kernel, and the  $10^{42}$  appears only when a NodeReader compares the two forces on one quantum, its charge read as its invariant and its mass as its energy. Status: derived from the definitions, an identity and no test (Section 10.1's mark); clicks.*

**Derivation S.45** (the nuclear holder's well: the threshold, the binding at a size, the capture and its reverse, Section 10.3). Inputs: *a gapped holder of the content with the reach  $1/\kappa$*  (S.11) and *a declared weight  $W$ , read by the nucleon families alone and sourced by them at the same weight; the far level of a source with the reach,  $3W \sin \omega_p e^{-\kappa r}/(4\pi r)$  per quantum* (S.25, S.39); *the clock's read of a level,  $\omega \rightarrow \omega e^{-x}$  with  $x = L/\Gamma$*  (S.2); *the matter band's inertia  $m^* = 3 \tan \omega_p$ ,  $3\omega_p$  at a small gap* (S.22); *the second moment's light* (S.42) and *the two-mode line under the rotation* (S.43). Steps: (a) *The well and its threshold. Each quantum reads the other's level into its clock, a potential  $-\omega_p x$  per quantum in rotation units, so the pair's potential is  $V(r) = -g^2 e^{-\kappa r}/r$  with  $g^2 = 3W \omega_p \sin \omega_p/(4\pi \Gamma)$ ,  $3W \omega_p^2/(4\pi \Gamma)$  at a small gap (one quantum writes the holder at the write weight  $W$ , the far level with the reach is  $3W \sin \omega_p e^{-\kappa r}/(4\pi r)$ , and the reading quantum's clock reads it at the weight 1,  $\omega \rightarrow \omega e^{-x}$  with  $x = L/\Gamma$ , a potential  $-\omega_p x$  in rotation units; the weight enters once, and a negative write weight writes a hill and is repulsive by the same line), and each record moves at its own inertia  $\mu = m^* = 3 \tan \omega_p$  about the pinned centre, two records being two fields with no joint amplitude (postulate 2). The slow limit's own equation in Yukawa's well binds if and only if  $2\mu g^2/\kappa \geq 1.6798$ , the known threshold of that equation (mathematics), that is  $W \geq 1.17 \Gamma \kappa/\omega_p^3$ : 591 (the small-gap formula at [2, 3]; 502 at the exact inertia) in the rule's own universe ([2, 3] as the nucleon, the binding holder [2400, 2401] at the range 20 Links,  $\Gamma = 6,000$ ), and  $6.5 \times 10^{11} \Gamma$  at nature's deuteron range 1.41 fm and the LHAASO Link (the Link  $2.5 \times 10^{-22}$  m at  $\omega_e = 3.7 \times 10^{-10}$ , S.27,  $\kappa = 1.76 \times 10^{-7}$  per Link,  $\omega_p = 6.8 \times 10^{-7}$ ). Under the pair hypothesis alone (the pair of two bound records, the law's line under its own name) the relative part carries  $\mu = m^*/2$  and the threshold doubles,  $W \geq 2.35 \Gamma \kappa/\omega_p^3$ , 1,183 at [2, 3] (1,004 at the exact inertia), nature's number doubling with it. (b) *The binding at a body's size. A body of radius  $R$  Links of quanta of the gap  $\omega_s$  standing in any well of the  $1/r$  kind that fits its size has, by the virial theorem of the slow limit's own equation (a known form, mathematics of that equation), the binding fraction  $b = 1/(6\omega_s^2 R^2) = (\lambda_s/R)^2/2$  with  $\lambda_s = 1/(\sqrt{3}\omega_s)$  its Compton length, the kinetic rotation over the rest rotation, whatever the holder's weight; the factor is the convention's: by the virial theorem the binding is the kinetic rotation  $\langle k^2 \rangle/(2m^*)$  with  $m^* = 3\omega_s$ ,  $1/(6\omega_s^2 R^2)$  under  $kR = 1$  with  $R$  the Bohr-like radius, and  $1/(2\omega_s^2 R^2)$  under  $R$  the form-weighted root-mean-square radius, where the hydrogenic ground state has  $\langle k^2 \rangle \langle r^2 \rangle = 3$ ; one convention, its factor, the order of the estimate, a finding, the profile entering through  $\langle k^2 \rangle \langle r^2 \rangle$  at the next order: the proton at 0.84 fm  $3.1 \times 10^{-2}$ , the deuteron at 2.13 fm  $4.9 \times 10^{-3}$ , iron at 3.75 fm  $1.6 \times 10^{-3}$ , against nature's  $1.19 \times 10^{-3}$  and  $9.4 \times 10^{-3}$ . (c) *The capture. Two slow packets entering the well are a superposition of the bound mode and the continuum; the relative coordinate breathes at the beat  $\Omega$  between them, the proton's charge rides it, so the body writes a dipole  $d = S r_p$  into the holder of the sign,  $S = k_w/(2E_s)$  and  $r_p = r/2$ , the first order of S.42's expansion since the charge is not symmetric about the centre, and radiates at the beat in the dipole's form, the time level carrying half of Larmor's power in form, the odd lines' part depending on their wall and coupling, the fork open; the body's own paces move with the level it writes, so it settles into the bound mode deterministically, the binding leaving as light at the beat; no rate and no settling time is claimed. (d) *The reverse. A light record of rotation  $\omega_L \geq B$  at the body drives the bound mode into the continuum by the two-mode line under the rotation* (S.43), *the matrix element the inter-mode current, the dipole; the bound mode's tail  $e^{-\sqrt{2\mu B} r}/r$  and the continuum's density give the shape  $(\omega_L - B)^{3/2}/\omega_L^3$ , Bethe and Peierls's, peaking at  $2B$ . (e) *The size law. A cloud of  $n$  quanta in one well has  $R^* \propto 1/n$  and  $b \propto n^2$ , iron 28 times smaller than the deuteron and bound 800 times harder, against nature's  $A^{1/3}$  and its saturation; the second holder, of shorter range and negative weight, is the saturating declaration. Check:  $1.6798 \times 4\pi/9 = 2.345$ ; at [2, 3],  $\omega_p = 0.8411$  and  $\kappa = 1/20$ :  $2.345 \times 6,000 \times 0.05/0.595 = 1,183$ ;*****

$\lambda_p = 0.210 \text{ fm}$ ,  $(0.210/2.13)^2/2 = 4.9 \times 10^{-3}$ . Status: *one derivation under the hypothesis, the normalisation of the radiated share for a second; nothing run.*

**Derivation S.46** (the atom with the nucleus's angle alone, the record's own level, and the holders the electron does not read, Section 7.3). Inputs: *the sign holder's angle of a nucleus of charge  $Z$  at  $r$ ,  $\theta_0 = Z\alpha_{\text{law}} c/r$  by the far kernel (S.26),  $c = 1/\sqrt{3}$  (S.1); a bound electron record of the matter family read into Schrödinger's slow limit (Section 3.3) with the band's inertia  $m^* = 3 \tan \omega_s$  per interval (S.22) and light's  $c = 1/\sqrt{3}$ ; the one write of every record into the one holder (Section 3.5). Steps: (a) Bohr's form with the nucleus's angle alone. The slow limit is the hydrogen-like problem with the coupling  $Z\alpha$ , the inertia  $m^*$  and the speed  $c$ : the levels  $E_n = -m^*(Z\alpha)^2 c^2/(2n^2) = -m^*(Z\alpha)^2/(6n^2)$  per interval, Rydberg's  $1 - 1/n^2$  between them, and the radius  $a_0 = 1/(m^* Z\alpha c) = \sqrt{3}/(m^* Z\alpha)$  Links, which at nature's gap ( $m^* = 3\omega_s$ ) is  $\lambda_e/(Z\alpha)$ ; the lines are the beats of two levels (S.43); the electron's rotation is redshifted by the binding,  $(Z\alpha)^2/2$  of its energy at a small gap, the mass defect as the click layer reads a frequency. (b) The orbital moment. The kinetic term of a plane record in a plaquette flux  $\Phi$  (S.33's covector) gives the orbital splitting  $\Phi m_\ell/(2m^*)$ , one Bohr magneton per unit of angular momentum, Zeeman's known form derived here under S.33's covector with the orbital  $g = 1$ ; the spin term's  $g = 2$  belongs to the spin route of Section 10.3, an explicitly named hypothesis. (c) The record's own level. The holder of the sign is one row, and a record cannot tell its own write from the nucleus's, so the bound electron reads the nucleus's angle  $Z\alpha/r$  and its own, of the opposite sense and the strength  $1/Z$  of the nuclear one, spread over its own radius: the Hartree self-term, repulsive and at full strength. With  $x = a_0/a$  for a hydrogen-like trial record the mode's frequency over  $Z^2$  times Rydberg's unit is  $\epsilon = x^2 - 2x + (1.25/Z)x$ , the kinetic, the nuclear and the self-term  $J$ ; its minimum is not a stationary state of the law's iteration. The stationary mode, Hartree's fixed point, is the minimum of  $E_{\text{tot}} = x^2 - 2x + (0.625/Z)x$  at fixed norm,  $x = 1 - 0.3125/Z$ , and its frequency, which the click reads, is  $\epsilon = E_{\text{tot}} + (0.625/Z)x$ :  $E_{\text{tot}}/(Z^2 R) = -(1 - 0.3125/Z)^2$  and  $\epsilon/(Z^2 R) = -(1 - 0.3125/Z)^2 + 0.625(1 - 0.3125/Z)/Z$ , with the radius  $a/(a_0/Z) = 1/(1 - 0.3125/Z)$ . At  $Z = 1$ :  $E_{\text{tot}} = -0.473$ ,  $\epsilon = -0.043$ , the radius 1.45; at  $Z = 2$ :  $-0.712$ ,  $-0.448$ , 1.19; at  $Z = 10$ :  $-0.938$ ,  $-0.878$ , 1.03; at  $Z = 30$ :  $-0.979$ ,  $-0.959$ , 1.01. The Rydberg ratios return as  $1 - 1.25/Z$ , 95.9 percent at  $Z = 30$ . So a record reading its own sign row is reflected on the board in Bohr's form at large  $Z$  only, and not hydrogen, the state before line (1) of the model's decision, which answered it: a record reads every sign row but its own, and hydrogen's lines are S.60's. (d) The holders the electron does not read. A nuclear holder the electron reads is excluded by the spectrum: a one-Node proton by the finite-size part of hydrogen's 1S level, 1.1 MHz for nature's radius (the finite-size shift's known form, derived here as the slow limit's own perturbation; the form factor and the muonic radii the comparison), which a point nucleus would lack at 500 times its 2.2 kHz uncertainty, and by the form factor, 1 at every momentum transfer where nature's falls; a 1 fm cloud bound by a gapped holder at the binding's content 8/938 by muonic hydrogen, 5.2 meV (the measured Lamb shift's own coefficient,  $5.2275 r_p^2$  meV with  $r_p$  in fm, at  $r_p = 1$  fm) against the 0.09 to 0.15 meV agreement of the electronic and muonic radii, by 35 to 58; an ungapped holder by the spectrum at nine orders. So the nuclear holders are read by the nucleon families and not by the electron, the selective read of Section 7.3 (the numbers computed by hand from the law's lines, one derivation). Check:  $\alpha^2/2 = 2.66 \times 10^{-5}$ ;  $(1 - 0.3125)^2 = 0.473$ ,  $0.473 - 0.625 \times 0.6875 = 0.043$ ; at  $Z = 30$  the self-term is 4.1 percent of the binding. Status: (a) and (b) derived in form, two derivations; (c) the finding and its numbers two derivations; (d) derived, one derivation. Under the two lines of S.60 the self-term of (c) is absent and (a) is the law's.*

**Derivation S.47** (the equivalence principle between families, Section 8 (i)). Inputs: *the fall of a free packet of the pair [num, den] in a pace gradient,  $a = c^2 [2\nu/(1 + \nu)] \nabla U$  with  $\nu = \text{num}/\text{den}$  (S.18), the proton and the neutron two families of the gaps  $\omega_p$  and  $\omega_n$  under the families' reading. Steps:  $2\nu/(1 + \nu) = 1 - \tan^2(\omega_s/2)$  exactly, so two families fall apart by  $[\tan^2(\omega_n/2) - \tan^2(\omega_p/2)]$ ,  $(\omega_n^2 - \omega_p^2)/4$  at small gaps. Two bodies differ in their fall by that times the difference of their neutron fractions: beryllium (5/9) against  $^{48}\text{Ti}$  (26/48) differ by 0.0139, and with  $\omega_n^2 - \omega_p^2 = (1838.68^2 - 1836.15^2)\omega_e^2 = 9,297\omega_e^2$  the Eötvös parameter is  $\eta = 0.0139 \times 2,324\omega_e^2 = 32.3\omega_e^2$ , below  $4.4 \times 10^{-18}$  at the LHAASO bound  $\omega_e < 3.7 \times 10^{-10}$ , under the beryllium against titanium bound of  $10^{-13}$  from the torsion balance (MICROSCOPE's titanium against platinum,  $10^{-15}$ , Touboul et al. 2022; the conclusion holds at either); the electron against the proton falls apart by  $\omega_p^2/4 = 1.2 \times 10^{-13}$  at the same bound, which no click has read (the electron's free fall known to 10 percent). The caveat: S.18's fall is a free packet's; a nucleon bound in the nuclear holder falls as the body's record does, and the body's fall per family is not derived, so the row is the free-packet estimate of the composition dependence, an upper estimate until the bound body's fall is derived. Check:  $(1 - \nu)/(1 + \nu) = \tan^2(\omega/2)$  with  $\nu = \cos \omega$ ;  $2 \times 4.17 \times 10^{42}$  is not used here;  $(1838.68^2 - 1836.15^2) = 9,297$ ;  $9,297/4 \times 0.0139 = 32.3$ . Status: derived in form, two derivations, the number an upper estimate; clicks.*

**Derivation S.48** (the proofs of the theorems of Sections 4 and 7.2). Theorem 1, the direction. Forward,  $\sigma = 1$ , the line is the definition of div and mod. Backward,  $\sigma = -1$ : write  $Y$  for the right side's first two terms,  $\sum_j R_{ij} \text{arr}_j + S a_{\text{now}}$ . Then  $u = w a_{\text{next}} - Y + r'$  and  $z = -[u/w] = [(Y - w a_{\text{next}} - r')/w]$ . From the forward line,  $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$  with  $0 \leq r < w$ , whose ceiling divided by  $w$  is  $a_{\text{before}}$  exactly; and  $\rho' = u - w[u/w] = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$ . The theorem holds for the GameBoard's step at every interval; a NodeReader's write and a body's window, click or not, are the NodeReaders' acts outside the reversal, as the lay is; the back-in-time gate is exact between clicks and runs across a click from the output's click line (the write a declaration).

Theorem 2, what the acts can form. *Each act is a linear form of the levels or the floor of one, and  $||L(x)|| \leq |L(x)|+1$ ; so sums and compositions of the four acts, at declared coefficients, grow at most linearly in the levels, while a product  $\text{now}_i \text{before}_j$  grows as their square. So nothing bilinear comes from the acts alone.*

Theorem 3, the conserved form. *Without the rounding the line is  $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$  with  $\mathbf{M}$  the matrix whose diagonal is  $S_i/w$  and whose entries on the six Links are  $R/w$ . Let  $\mathbf{D}$  be the diagonal matrix of the weights  $1/p_i^2$ ; then  $\mathbf{DM}$  is symmetric iff  $d_i M_{ij} = d_j M_{ji}$  on every Link,  $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$ : with  $R_a(i \rightarrow j) = 2 \text{num} p_i^2 q_{ij}^2 / \Gamma^2$  the weight  $1/p_i^2$  cancels the Node's factor on both sides of every Link and the Link's factor  $q_{ij}^2$  is the same from both ends, so  $\mathbf{DM}$  is symmetric for any content and any tensions. For a symmetric  $\mathbf{K} = \mathbf{DM}$  the quantity  $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$  satisfies  $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$ : with  $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$ ,  $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$ , which is the other side. Multiplied by  $w$  this is Eq. (6) exactly.*

Theorem 4, the stable body.  $F(R) = K/R^2 - \sum_r \sigma_r W_r R^{-s_r}$ ;  $F'(1) = 0$  is  $-2K + \sum_r \sigma_r s_r W_r = 0$ ;  $F''(1) = 6K - \sum_r \sigma_r s_r (s_r + 1) W_r = \sum_r \sigma_r (3s_r - s_r^2 - s_r) W_r$ : Derrick's scaling, Pohozaev's identity on the GameBoard.

The momentum flux, Eq. (7). *Substitute the line into  $w P_a(t+1)$ : the  $S$  term cancels, the  $w$  term is  $w P_a(t)$  and the  $R$  term is the difference of the fluxes  $G_{ab}$  as Eq. (7) states; the integer step subtracts the remainder term  $\delta_i \Delta_a(\text{now})_i - \text{now}_i \Delta_a(\delta)_i$  with  $\delta = r - r'$  (in this orientation of  $P_a$ ). The guide's lay (S.61 row 17; two derivations). (1) A source  $s e^{-i\Omega t}$  at one Node of a width-one guide, whose band is  $\cos \Omega = (\text{num}/3\text{den}) \cos k$  (the transverse mode at  $\cos(\pi/2) = 0$ ), drives two outgoing waves  $C e^{ik|x|}$  with  $C(2 \cos \Omega - (2\text{num}/3\text{den}) e^{ik}) = s$  at the source Node, so  $|C| = 3\text{den} s / (2\text{num} \sin k)$ . (2) The source adds at the interval  $t$  the increment  $A_t \cos(\Omega t)$ ,  $A_t$  its amplitude; the two outgoing waves have  $B = 3A/(2 \sin k)$  on the width-one guide ( $\text{num} = \text{den}$ , the giving's case;  $(3\text{den}/2\text{num}) A / \sin k$  for a general pair), and the form of a travelling wave is  $D = B^2 \sin^2 \Omega$  exactly at every Node, with no time mean; each interval's emission occupies  $v_g = \sin k / (3 \sin \Omega)$  Links on each side, so the two trains' form is  $\sum D = 2(9/(4 \sin^2 k)) (\sum_t A_t^2) \sin^2 \Omega \sin k / (3 \sin \Omega) = (3/2) (\sum_t A_t^2) \sin \Omega / \sin k$ , the sum over the increments' squared amplitudes. (3) One quantum on a real line is  $\sum D = T \sin \Omega$  (Section 5.4's packet,  $\sum a^2 = T / \sin \Omega$ ), so one quantum is laid at  $\sum_t A_t^2 = (2/3) T \sin k$ ,  $A_t$  the amplitude of the increment added at the interval  $t$ ; the increments' own squares,  $\sum_t A_t^2 \cos^2(\Omega t)$ , sum to half of it,  $(1/3) T \sin k$  (the convention named; two derivations).*

Status: theorems, proved for every lattice; GameBoard.

**Derivation S.49** (the words of the method, as the paper uses them). *The definitions, in the law's order from the Node to the click, each with where the paper states it. (1) Node, level, Event. A Node is a place with no size; a level an integer at a Node; an Event a level split to its six neighbours in one interval. These are the GameBoard's points: one Node, one interval; the remainder of a division is kept at one Node. (2) Record. A family's levels over the GameBoard, now and before at every Node, with a remainder; a record of dimension  $d$  has  $d$  lines at every Node (1 a real line, 2 a plane, 3 a colour triple, 4 four real lines never summed at a Node, the GHZ's), stepped by one line of Rule3 with one set of coefficients. (3) Line of a record. One component of a record of several lines, present at every Node of the record together with the others, as the real and the imaginary part of a plane are; a line has no form, no count, no place and no click of its own, and a hard click reads its share of the record's form as a fraction. (4) The quantum of a family. One unit of the record's invariant,  $2A^2 \sin \omega_s = T$ : the share  $T \sin \omega_s$ , which a click reads,  $\hbar \omega_s$  its small-gap name, the Wronskian  $T/2$  for every family; what a counter clicks and what stands for a particle of nature, one photon, one electron, one muon, one proton. (5) The credit's unit. A record of count 1 is one quantum of the invariant,  $2A^2 \sin \omega = T$  per line (the law's document, the count is the record's share); its form is  $T \sin \omega$ , the unit of energy; a bolometer region credits it at  $W_c \sin \omega_0$  of inflow,  $\text{isqrt}(9T^2(\text{den}^2 - \text{num}^2))$ ,  $3 \text{den} T$  for light; the implementation lays a record of a family with a gap by  $A^2 = \text{isqrt}(\text{count}^2 T^2 \text{den}^2 / (4 \text{laid}^2 (\text{den}^2 - \text{num}^2)))$ , one root of the whole product (count-unit), and reads a massless record's quantum at its wave's own frequency; not a particle. (6)  $T$ . The unit of action,  $\hbar$  per interval; not the electron's energy. (7) Message. The file's laid travelling record, a record in no body: a record in no body, free, laid by the file or written by a click, in flight until something takes it, if anything does; it spreads by the band's inertia. (8) Body. Its count and its record over its Nodes, standing in the well of a holder (a record of count one may stand at one Node, a reader with Nodes alone never): the atom in the sign holder's angle, the nucleus in the nuclear holder's well, a cloud in the binding holder's; a body has a place and a size, and energy parts it. (9) The criterion. What nature splits with energy is a body on the GameBoard; what no energy splits is a quantum of a family: the electron and the nucleons are quanta, the nucleus, the atom and the molecule are bodies, all by the one mechanism. (10) Compact body; frozen Node. A body of one Node, the pixel, is the lattice's compact body at the Link's energy and no particle, and exists for no quantum of nature's gaps; a Node whose content passed  $(\Gamma/2) \ln(2\Gamma)$  levels reads nothing, is read, and is dark by the share's declaration. (11) Compton length.  $1/(\sqrt{3} \omega_s)$  Links, the least size of a quantum of the gap  $\omega_s$ : the electron's 386 fm, the proton's 0.21 fm. (12) Holder. A held row of the content or of the sign, written by the one write and read into the paces or the turn; its form derived (Poisson's with the kernel  $3/(4\pi r)$ , Yukawa's with a gap), its strength a declared weight, its range its gap; the nuclear holder a gapped content holder at the weight  $W_1$ , read by the nucleon families alone, with a core holder of shorter range at  $-W_2$ . (13) The read. Every family reads the holders its declaration names at the weights it names and sources each at the weight it reads with; gravity and the binding row by every family at 1, the nuclear holders by the nucleon families alone; a read, never a change of Rule3. (14) NodeReader. The one declaration kind of the model: a declared connected region of Nodes with its seed, two or more where it reads*

with Nodes alone and one or more where the file gives it a record of its own; it reads the Nodes it was given, the inflow through their Ports over its window, and writes the clicks that happened, one click line per quantum taken (or given), naming the window's index, the NodeReader, the family and the part, the count before and after, and its own clock, and never a Node; a NodeReader whose file gives it a record of its own (parts, transitions, a lifetime) is the same kind and gives as well; after a realised click it writes once, at the one Node the quantum reached it from, the Node the draw picks by the share (Section 5.4). (15) Credit. The counting: the inflow over the window over the whole is the share, a real number; the clicks are whole quanta drawn by the shares with a declared seed; for a pair the joint share pairs the parts through the root and the partner is drawn from the conditional; the one non-local place of the description, Born's rule as the NodeReader's declaration. (16) Click. One whole quantum of a record credited to a region at an interval,  $(R, t)$ : it reads waves (a region and a window, never a Node), only wholes (never a line or a fraction), only what crosses a front (a body at rest clicks nowhere; no click without a taker); its write takes the levels that arrived at the Node of the exchange, the record's count down by one and the taker's up by one, the quantum conserved between them, and the taker's transition, the taker's change spreading from there by Rule3; the other paths a cancelled future at three levels, the record's count at 0 for every click to come, their share leaving through a face where the file declares one, and the front erasing them from the click's Node outward inside the click's light cone, an empty wave until it reaches them, a diagnostic; it never happens by itself. (17) The conversion. A change of family, a click, a hypothesis under its own name, one list of the one click act; the neutron's end, not the deuteron's forming, which is a body's. Status: the law's words; the paper's Sections 1, 3, 5 and 7. (18) Port, Link. A Node has six directional Ports,  $\pm x, \pm y, \pm z$ ; a Link is the pair of Ports between two neighbours, and a level reaches a neighbour only through a Port (Section 3.1). (19) The faces, the wall. An axis is periodic, open (a level beyond it read as 0) or folded (one layer, its two Ports returning the Node itself); an inner face is a plane of Nodes declared closed but for its gaps, which is the wall with its slits; a receding face grows the box by a layer of zeros whenever a nonzero level reaches the Node next to it, so that no event meets it (Section 3.1). (20) The lay. The act of the Universe24 file before the first interval: a message's levels at the declared amplitude, envelope and wave number, or a body's count at its Nodes; the loader's act, outside Rule3 and outside the reversal, as a face's is (Sections 3.2 and 4.1). (21) The current, the tension.  $F_{ij} = \text{num}(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$ , the current into the Node  $i$  from its neighbour  $j$ , antisymmetric, the Link term of the conserved form, what a NodeReader reads through its front; the tension on an axis the momentum flux averaged over the Node's two Links on it, read into the Link's pace and the source of a holder's axis part (Sections 3.5 and 4.3). (22)  $W_c, W_d$ . The count wall  $W_c = 3 \text{ den } T$ , one unit of count of the inflow through a region's front over a passage; the NodeReader's whole  $W_d$ , the share one click is,  $T$  for a bolometer and  $T \sin \omega$  for a counter (Section 5.3). (23) The taking, the giving. The two acts of a click at a body: the taking, the arriving record's levels and remainder at the drawn Node to 0 for one quantum's share, the hole, its count down by one, and the taker's part laid up by one; the giving, the body's upper part down by one and one quantum of light laid at the Node drawn by the NodeReader's own record's share over its region, at the resonance  $\Omega = \omega_e - \omega_g$  (Section 5.4). (24) The resonance. A body's transition between two of its modes, at their difference  $\omega_j - \omega_i$ ; a click at a body is the meeting of the arriving quantum with that transition, at resonance (Sections 5.2 and 7.4). (25) The hazard. The rate  $1/\tau$  per proper interval at which a body in the dark gives, drawn at the now, which makes the giving's times exponential;  $\tau$  the law's number per transition, declared (Sections 5.3 and 10.3). (26) What lays. What lays a record into Universe24: the file's lay of a message, or a NodeReader that gives; a NodeReader that gives is one with a record of its own, the one declaration kind, its seed and its Nodes (Section 5.3). (27) The blind, the gate. The blind is the expectation written from the law's lines before a run, kept in the file's folder; a gate is a check that a run returns its blind exactly or reverses to the bit (the back-in-time gate, MATCH); a reading that misses its blind is a named finding, never smoothed (Sections 1 and 3.7). (28) Cluster. Bodies standing in one another's wells; between two bodies there is nothing but the click, the light of the holder of the sign and the two real rows (Section 7.5). (29) The three layers. The board's rules (Rule3 fixed, the file's rows free), the numbers (all free, as few as the rule allows) and the click (its one form fixed, the NodeReader's declarations free) (Section 1, Fig. 1). (30) Mark, fence. Every claim carries one of the eight marks (theorem, derived, computed, assumption, hypothesis, experiment, fitted, inspiration) and one of the two fences (GameBoard, a reading of the lattice no NodeReader reports; clicks, a formula of what a NodeReader reports) (Section 1). (31) Register. A number kept at a Node beyond the law's own, a table, a formula or a draw, which postulate 2 forbids: the NodeState holds two levels and one remainder per part and nothing else (Sections 3.1 and 3.6). (32) Universe24; the file. Universe24 is the world the method computes: the clicks the NodeReaders write from the GameBoard, and nothing else; never the GameBoard itself, whose numbers are diagnostics, and never nature. The NodeReader is the gate between the GameBoard and Universe24: it reads the board and writes the clicks. The paper's one claim is that Universe24 helps compute things in nature, each quantity with its status and its number against measurement, the method how it computes; the clicks are the mechanism's own, the model's name for what a detector of an experiment registers, defined by the model (one quantum credited to a NodeReader) and not by nature, so that the two are not the same thing. A file declares one run of it, the experiment or a gate, and the implementation runs it: the families with their pairs and numbers, the holders with their weights and ranges, the bodies and the messages laid, the NodeReaders with their Nodes and seeds and the world's draw window, and the faces; the two slits is the paper's one run, with its blind row and its gate; Bell's and the GHZ's values are derived. The engine's documents call a file's run a world; the paper keeps the word world for the world we live in, which is seen in clicks alone (Sections 1 and 9). (33)

Conventions. The deflection of light is toward the source, and Eq. (15) of the paper and S.20 print its magnitude; the sense +1 of a plane record is the declared rotation of (re,im) that the generator lays, under which S.43 (b) reads.

**Derivation S.50** (the click and the NodeReader: the passage from the GameBoard to the clicks, Section 5). The click, algebraically. Let  $X$  be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step  $S_t$  is Rule3 over the interval with the paces read from the holders at its start, a bijection of  $X$ , exact by the division with its remainder; the line without its division is linear in the levels at fixed paces, the linearity the backward reading uses, the integer map not affine; its forward orbit is the future and its inverse orbit the past. The readings  $D$ ,  $F$  and  $W$  are bilinear forms on  $X$ , covariant under the symmetry group of Section 3.1 and never written. A NodeReader  $R$  is a declared region (two Nodes or more with Nodes alone, one or more with a record of its own) with its front Ports and its coefficients, which define  $Q_R$ : for one line a quadratic form of the levels, the inflow through its front Ports accumulated over the window; for a pair, the window's sum of the squared cross-side sum over the parts, quadratic in the joint amplitude, the product of the two sides' levels, and so quartic in the levels. A click is the pair  $(R, t)$ , the credit of one whole quantum to the NodeReader  $R$  at the interval  $t$ , drawn with the probability  $s_R / \sum_{R'} s_{R'}$ , where  $s_R = Q_R(x_t) / W_c$  is the NodeReader's form on the forward orbit over the quantum's measure of a current. In the meeting's words, the inverse orbit from the credited state coincides with the forward orbit on the realised orbit (in the state space, the record over every path at once), and the share is the bilinear form's value at the coincidence. The click's decision is the one map in the model that is not a bijection, from the vector of shares to a unit vector, a scalar under the group; its write is one act on  $X$ , at the one Node of the region from which the quantum reached the NodeReader, which the NodeReader knows and we do not: the exchange: the arriving record's levels and remainder at that Node to 0, the hole, its count down by one, and the taker's realised part laid there at its count up by one, the taking, the phase the product's and the remainder at the lay's origin, the quantum conserved between them, the taker's change spreading by Rule3 as an event and the record that gave erased from that Node outward at the causal bound, inside the click's light cone, nothing else written (ASSUMPTION, the NodeReader's declaration). Every observable is  $Q_R / W_c$  on the forward orbit or a ratio of two such. The one act at three storeys. Rule3 on the levels divides the six weighted arrivals with the Node's own terms and the kept remainder by the wall  $w$ , the quotient the next level, the remainder kept at the Node, the inverse a bijection by it. The hold's write divides the sources' forms by  $E_s T$ , the quotient the held level's gain, the remainder kept at the Node, exact by it. The click divides the inflow through the front over the window, with the books' remainder, by  $W_c$ , the quotient the clicks and the remainder kept in the books for the next window, the two records at the Node its neighbours, the inverse crossed from the click line by the remainder and the count. The count is the level of the top storey; the cancellation lives in the count because the count is the quotient; the exchange at the Node is the writing of the quotient into the two records; the reversibility between clicks is Rule3's; the three holders are the middle storey and the conversion the top storey's one click; the one thing in the click that is not Rule3 is the draw (DERIVED for the form, two derivations; CLICKS). The hole's booking identity (Section 5.4): with  $Q(p, q) = \sum_i [w p_i^2 + w q_i^2 - p_i (S q_i + \sum_j R_{ij} q_j)]$  the two-level form of a pair of consecutive levels, Rule3's line without the remainder,  $w(\text{next}_i + \text{before}_i) = S \text{now}_i + \sum_j R_{ij} \text{now}_j$ , gives  $Q(\text{next}, \text{now}) = Q(\text{now}, \text{before})$  and, by  $R_{ij} = R_{ji}$ ,  $Q(\text{now}, \text{before}) = w \sum_i D_i$ ; a face writing  $x$  in place of the level  $v$  the step would write at one Node, the leaving level  $b$  standing, changes  $Q$  by  $(x - v)[w(x + v) - (S \text{now}_i + \sum_j R_{ij} \text{now}_j)] = w(x - v)(x - b)$ , the line giving  $S \text{now}_i + \sum_j R_{ij} \text{now}_j = w(v + b)$ ; exact in the line and within  $|x - v|$  levels in integers, and the most one write removes is  $w(v - b)^2 / 4$  at the midpoint (three derivations, the reviewer's, the mathematician's and the advisor's). The credit rounds the top storey half up per window; the carried remainder is the law's form (S.61, row 12) with its own blind on  $N$ . The window is declared, the world's draw window set by name to the whole passage of a packet, or the credit floors at 0 (Section 4.4); a Node whose pace rounds to 0, a frozen Node, stands in no NodeReader's front and is never clicked (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS).

One form for every click. At the credit the parts of a record are read with the NodeReader's coefficients, each part paired with the same part through the common origin, summed and squared; the share is that density, and the draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's check is the case through the common origin: a pair born as one event is two trees with one root, and the backward reading from the first click passes the root and continues forward on the partner's tree (Section 9.3). The joint share is computed from the forward records alone, by the linearity of the rule, the inverse orbit existing by Theorem 1 (DERIVED, CLICKS).

The NodeReader is the one declaration kind

Assumption (the NodeReader's declaration). A NodeReader with Nodes alone is one connected region of two Nodes or more declared in the file, never one Node, at least half the wavelength of the family it reports across the beam (4 Links at  $\lambda = 8$ ); a narrower region, a single Node or a region in pieces is refused at load. It is opaque, backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns. Its click is its report, per interval and family: the net current into the region through its front boundary Ports, the Ports leading in from the declared board, summed in integers, with the region's name and the family and never a Node; one wall of the family,  $W_c \sin \omega_0$  (3 den  $T$  for a massless family), of the inflow over a passage is one unit of count, the bolometer's whole ( $W_d = T$ ; a counter's whole is  $T \sin \omega$ , S.49 (22)), so a screen saw  $N$  units where its regions' inflows sum to

$NW_c$ . The credit: one quantum to one NodeReader, drawn by the shares of the regions' inflows, read outside the GameBoard over every declared NodeReader of the file, one credit per quantum and never two; the NodeReader draws with its declared seed, the GameBoard draws nothing and the run is deterministic per seed. After a realised click the NodeReader writes once, at the one Node from which the quantum reached it. A NodeReader whose file gives it a record of its own (parts, transitions, a lifetime) is the same kind and gives as well: its region is one Node or more, resolving the place and not the wave, held by its declaration and never bound by a well, and its one write is at the Node the draw picks by the share, the taking's hole at the Node drawn by the arriving record's inflow over the region and the giving's quantum at the Node drawn by its own record's share. The credit is the one non-local place of the description and no line of the law.

Why a region: a one-Node record is every wave number at once, and the region's half-wavelength is the NodeReader's declaration; so no click names a position finer than a region, and every per-Node number is a diagnostic.

What produces a click is deterministic: the meeting, the current of the amplitudes through a region's boundary. What is not known is which of the possibilities becomes the click: when one possibility is spread over several NodeReaders, the GameBoard gives the shares, no line of the law gives which NodeReader fires, and the NodeReader draws. Exclusivity, one click per quantum, is derived under the credit's one draw, the NodeReader's declaration, from the count conserved; Born's rule, the credit's chance equal to the intensity's share, is a declaration of the NodeReader and not a finding of the model. The uncertainty relation is not a separate rule: it is the NodeReader's declarations, the region (its Nodes), the world's draw window and the draw (its seed), the write exact inside at a Node and an interval that are the board's alone and never reported, the report uncertain outside (ASSUMPTION). What each declaration costs: a region of  $n$  Nodes reads a fringe of period  $\Lambda$  with the factor  $\sin(\pi n/\Lambda)/(n \sin(\pi/\Lambda))$ ; a frequency is read to  $\Delta\omega \geq 2\pi/N$  over  $N$  intervals; and  $N$  quanta drawn by the shares scatter as  $\sqrt{Np_i(1-p_i)}$ , so a pattern is in the histogram and never in one click (DERIVED, CLICKS; S.8).

The click decides in the credit and writes at one Node

Assumption (the click's decision and its write). The GameBoard is deterministic and local: Rule3 with no draw, Theorem 1's bijection at every interval. The click is the meeting of two quanta at the NodeReader, the incoming record's quantum and the NodeReader's own transition; the NodeReader, declared in the Universe24 file, draws with its declared seed whether the meeting is realised, and the randomness is the NodeReader's, the board has none. Forward, to the future, one write: the exchange at the one Node from which the quantum reached the NodeReader, which the NodeReader knows and we do not (the uncertainty principle: for us the click is a region and a window), the one Node drawn in the same one draw by the window's inflows per Node, the region's share their sum, the Node a GameBoard diagnostic beside the click line; the exchange: the arriving record's levels and remainder at that Node to 0, the hole, its count down by one; the taker's realised part laid at that Node at its count up by one, the taking, with the phase the product's and the remainder at the lay's origin, the part it leaves laid at its count down by one; the quantum conserved between them, and the taker's change spreads by Rule3 as an event, one Link per interval; no act faster than one Link per interval: the front erases the record that gave from that Node outward at the causal bound, one shell of Link distance per interval, the hole's act on each shell, the clicked record alone; the other paths are a cancelled future at three levels: for every click there will be, the record's count 0; at every NodeReader they reach where the file declares a face behind it, their share leaving the board there with no credit; on the board the front erasing them inside the click's light cone, beyond it their levels remaining, still sourcing the holders they write to, until it arrives; a record's front starts when its count reaches 0. Backward, to the past, a reading and never a write: the credit's reading through the root, which chooses which tree is the past and gives Bell at a distance, the one non-local place; it cannot be a write, the board being deterministic forward.

A single quantum's count stands at 0 after its click, the count conserved and read by the credit through the root, the NodeReaders' shared other side on the GameBoard, the Nodes' levels carrying the shares and never the count: antibunching and one quantum one click per record between far NodeReaders, the credit's one draw over all the NodeReaders that read one record (ASSUMPTION for the one draw, DERIVED from it for the rest; CLICKS; the pair's 478/169 and the GHZ's  $-4$  unchanged, one click per record). Nothing at one NodeReader changes the amplitudes at another except by Rule3 from the Node of the exchange; no signal passes. A NodeReader is a body of the GameBoard and stands nowhere outside it: it has two sides, the front where the incoming quantum meets its own transition and the other side where its own record connects to the rest of the board, the apparatus's bodies and, through the record it took, that record's root, the root's event, so that two NodeReaders reading one record are connected through the root, their common past; it can be entangled on both sides. A bound body clicks by quanta from outside meeting its own transition, never by clicks from outside, and only where it is declared a NodeReader: the future is light's forward record at the body's boundary, the past the body raised by one quantum read backward to the source, and the meeting is the resonance of the body's rotation with the light's (DERIVED, CLICKS; Section 7.4); a NodeReader with a record of its own, declared over its region, writes its window's outcome at the Node the draw picks by the share, click or not (S.59), a spread body's click is the exchange at its entry Node and its settling a reading, named open, and the giving lays the light quantum at the Node drawn by the NodeReader's own record's share, at the resonance  $\omega_e - \omega_g$ , the share's difference booked and stated explicitly (S.14, S.43); a body in the dark is only the future and still gravitates; emission is the same meeting with the roles reversed, two local writes and one draw of the credit binding them. The Node stays generic, knowing no family and no click; the write is the NodeReader's act (ASSUMPTION, the NodeReader's declaration, the law's line of

the click's write; the decision's form DERIVED). The GameBoard holds only the future, every possibility, and has no arrow: Rule3 is a bijection at every interval (Theorem 1). The clicks have the arrow: each is forward only, the one act outside the reversal; at each click what was future becomes past, the present being the meeting; the chain of clicks is the past advancing into the future, written once and never undone. The direction of time is the clicks' and not the board's (THEOREM for the step; the clicks' arrow the NodeReader's declaration). The two slits' run was made with NodeReaders that report and write nothing, and the write at the window's end leaves its numbers unchanged.

The passage from the board to the clicks. Every formula of Section 8 is derived in three steps: the law's line in the GameBoard's words; each word traded for a NodeReader's report of it ( $\omega$  for a clock's rate,  $U$  for the ratio of two clocks' rates); and what remains, a relation among clicks alone, set beside nature's. The credit's bookkeeping is that the shares sum to what the screen saw, one whole per click, the whole the NodeReader's own transition; a frequency of nature is read as the inflow's beat within the window or as the resonance of the NodeReader's own record, and the GameBoard's quantities, the form, the content, the level, the rotation, are read by no click except as the share's inflow. Status: the click's algebraic definition and the NodeReader's declarations the law's; the credit a declaration; GameBoard for the record, clicks for every number of nature.

**Derivation S.51** (why there is entanglement, and its scope, Sections 5.2 and 9). Bell: the file. The entanglement the model computes is one event with two trees. The pair is laid as the pair family, two real lines of light never summed at a Node, with equal parts. Each side's NodeReader, a region at least half a wavelength across, reports the two parts' signed level sums over its region and knows nothing of angles; the ports enter at the credit. The reader applies the declared setting, a pair of integers  $(p, q)$ , with the port coefficients  $e(+) = (p, q)$  and  $e(-) = (-q, p)$ , exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are  $(1, 0)$ ,  $(1, 1)$ ,  $(12, 5)$  and  $(5, 12)$ , at 0, 45, 22.62 and 67.38 degrees. The credit is the one form of the click: the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \sum_{\text{window}} \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (\text{S.51a})$$

with  $\alpha_k$  and  $\beta_k$  the part  $k$ 's amplitude at the port on each side and  $e_k(p)$  the port's integer coefficient on the part  $k$ , the sum over the window's intervals standing outside the square, on both members of the level pair. The joint share is never the square of a local sum, which one NodeReader could form alone and which gives  $S$  at most 2. One draw is made per pair, the NodeReaders' draw with the declared seed, the GameBoard drawing nothing: the first side from its marginal and the partner from the conditional; each side's write at the Node of its exchange spreads by Rule3 and changes no number of the row, one click per record (S.50) (ASSUMPTION, CLICKS).

Bell: what the model says it must give

(1) Rule3 gives each part's amplitudes at the ports: the two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, identical bit for bit (THEOREM, GAMEBOARD). (2) The meeting with the linearity of Rule3 gives the joint amplitude: the realised click at  $A$ 's port weights the parts by  $A$ 's analyser, is read back to the root where the two parts were one event and forward to  $B$ , where  $B$ 's analyser weights them, and the root sums the two branches,  $\sum_k c_k e_k(a) e_k(b) = \cos(a - b)$  with  $c$  the parts' laid weights,  $(1, 1)$  for the symmetric lay (DERIVED, CLICKS). (3) That the share is the density of the meeting, its square, is Born's rule for a pair: nothing in Rule3 forces it; everything after it, the fractions,  $S$ , GHZ's  $M$ , is derived in the algebra (ASSUMPTION, FITTED; CLICKS). From the three, with the pair's common phase  $\theta$  uniform over the pairs and  $A$ 's + port at  $a$  and  $B$ 's at  $b$ ,

$$\begin{aligned} \text{the meeting:} \quad & P(++) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\ \text{a local credit by the shares:} \quad & P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\ \text{a local credit by the sign:} \quad & E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\ \text{the credit by the contrast:} \quad & E = \cos 2(a - b) \text{ at the efficiency } 2/\pi, \quad S = 2\sqrt{2}, \end{aligned} \quad (\text{S.51b})$$

The second, third and fourth lines follow by averaging over the pair's phase  $\theta$  (DERIVED, CLICKS; S.10). So  $E(a, b) = \cos 2(a - b)$  under the meeting. For the declared pairs it is exact, Lagrange's identity of the reader on equal parts:  $E = ((p_{APB} + q_{APB})^2 - (p_{APB} - q_{APB})^2)/((p_A^2 + q_A^2)(p_B^2 + q_B^2))$ , so at the four settings  $E = 119/169, -119/169, 120/169$  and  $120/169$  and  $S = 478/169 = 2.8284$  exactly, an efficiency of 1 by construction and the marginal one half on each side whatever the partner's setting (COMPUTED, CLICKS; S.10). For unequal parts,  $r$  their products' ratio at the ports, the same credit gives  $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$  with  $\rho = 2r/(1 + r^2)$  and  $S = \sqrt{2}(1 + \rho)$  at the ideal angles (DERIVED; CLICKS). The maximum of  $S$  over the four settings for  $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$  is  $2\sqrt{1 + \rho^2}$ , Tsirelson's bound  $2\sqrt{2}$  at  $\rho = 1$  and never the algebraic 4, and the declared pairs'  $478/169 = 2.82840$  sits below  $2\sqrt{2} = 2.82843$  by the rational angles (DERIVED, CLICKS).

What the GameBoard gives without the declaration is the second and third lines of Eq. (19) of the paper, two results on local credits on the same file (DERIVED under the uniform phase, CLICKS): a credit of each side by the shares of its ports gives  $S = \sqrt{2}$ , and a credit by the sign of its larger port gives the sawtooth, whose sum is exactly 2, Bell's

bound. Bell's value therefore sits in the credit rule, a declaration, and is no independent evidence that the rule accounts for entanglement. Nature beside the derivation: the two loophole-free photon tests of 2015 for the pair's row, and  $S = 2.42 \pm 0.20$  in the test of that year on two spins joined at a click, the swap kind, which the law names as met by the same form and which is not derived here. The same credit on four lines gives the three-quantum GHZ correlation  $E_3 = \cos 2(a + b + c)$ , every pairwise  $E_0$  and Mermin's  $M = -4$  where every local credit gives at most 2 (DERIVED under the declared credit, CLICKS; S.40). No signal passes between the sides, each side's marginal being one half; no settings are chosen by the GameBoard's own past, so a superdeterministic world is not this one.

The scope. Entanglement in the law is a property of the lay: parts laid alike as one event (the pair family, the GHZ family with  $2^{n-1}$  parts for  $n$  NodeReaders, S.40) share one root, and the credit's joint share pairs their parts through it; the run preserves the lay bit for bit and Rule3, being linear, creates none. What the law meets is therefore the pair laid alike as one event and two multi-part records meeting at one click, where the click is the root through which their parts are paired, the swap of entanglement; no other entanglement is named as met. Status: the joint share a declaration; the correlations under it derived, two derivations; clicks.

**Derivation S.52** (bodies, the atom and the nucleus as bodies, and the criterion, Sections 7 and 10.3). What a body is, and how it is computed. A body is its count and its record over its Nodes and the quanta that record carries; a record of count one may stand at one Node. Its count is the share of its record (Section 4.4), a reading and not a state; its form  $D_i \text{ div } T$  is written into the holders it sources, and the holders enter Rule3 through the paces alone. The record is bound by the wells its form sources, by the binding holder's screened sum with the range  $R$ , while gravity's  $1/r$  is weak by  $E_g$  and enters no body's binding. No law is written for a body: it is the support of a standing mode of the one rule, and its record moves by Rule3 alone. Four statements make a body (DERIVED from the line, with the decisions named): (a) the body is the fixed point of its binding holder, standing where the sources return themselves; (b) a holder's level is the rest the start solves from the bodies' forms, after which Rule3 steps it; (c) a body's quanta are its record's share, carried wherever its record's current carries it; (d) a free packet or a cloud falls as  $a = \Delta |d\omega_0/d\ell| |\nabla \ell|$  with  $\Delta = 1/m^*$  (Section 7.1 (d)), bodies with the same  $\Delta d\omega_0/d\ell$  falling alike (Section 8 (i)).

How a body is computed. The held rows' source is the record's form  $D_i \text{ div } T$ , the law's line; the implementation's start laid the count, and the generator's lay under a level is declared and shipped in the standing body's file, the write's factor  $N^4$  read in the level and no physical return (the open list; GAMEBOARD). By the rule's own iteration and no other formula: start from a spread of the declared count over the Nodes; write the wells that spread lays into the holders and let them rest; find the record's top mode in those paces by Rule3's own read act, iterated; scale its two levels so that the share carries the declared count; write the wells that record lays; and repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: a wide start reaches the cloud and a narrow one the compact body, two fixed points at one count.

The criterion and the two bodies. By the criterion of S.49 (9), the nucleus, the atom and the molecule are bodies by the one mechanism the law has for a body, a record standing in the well of a holder (S.49 (8)): the atom in the sign holder's angle (S.46), the nucleus in the nuclear holder's well (S.45), a cloud in the binding holder's (S.13). The atom's state, as the findings stand. (a) With the nucleus's angle alone the law's stationary mode has Rydberg's form and  $a_0 = \sqrt{3}/(m^* Z \alpha)$ ,  $\lambda_e/(Z \alpha)$  at nature's gap (two derivations). (b) A record that read its own sign row, the self-force repulsive at full strength (S.46 (c), the state before line (1) of the model's decision; two derivations), had hydrogen's 1s as the stationary mode at  $-0.043$  of Rydberg's unit at  $1.45 a_0$ , and the Rydberg scaling returning at large  $Z$  as  $1 - 1.25/Z$  ( $E_{\text{tot}}$  as  $(1 - 0.3125/Z)^2$ ), 95.9 percent at  $Z = 30$ : the law's form at large  $Z$  only, and not hydrogen, the finding that line (1) answered; under line (1) a record reads every sign row but its own (ASSUMPTION, FITTED by the paper's own key, an input chosen with a known result in view), and hydrogen's lines are this derivation's; the holders of the content keep the self-read (line (2), ASSUMPTION). (c) No lifetime number: the radiated share's coefficient is open (S.26). Status: the body's definition and its computation the law's; the criterion the model's decision; the atom's state as marked.

**Derivation S.53** (the masses as families, Section 10.3). GameBoard readings, not findings. The rule's own universe at  $[2, 3]$  is no finding against nature, which measures at its own pair (GAMEBOARD; Section 8, (e)). The masses are families. What the click's structure derives of the masses (two derivations), from Rule3's vacuum dispersion  $\cos \omega = (\text{num}/(3\text{den})) \sum_a \cos k_a$  alone: the rest rotation  $\cos \omega_0 = \text{num}/\text{den}$ , the inertial mass from the band's curvature  $m^* = 3 \tan \omega_0$  and the family's own light speed  $c_m^2 = \omega_0/(3 \tan \omega_0)$ , the band's kinetic scale (Section 8, (e); S.18, S.21, S.22;  $0.2508$  at  $[2, 3]$ ,  $c_m/c = 0.867$ ), so  $m^* c_m^2 = \omega_0$  exactly, the rest rotation, an identity of the dispersion, the share's energy per quantum  $T \sin \omega_0$  a second reading of the same quantum, the two one as the gap closes; the mass ratio  $\sin \omega_p / \sin \omega_e$ , two declared rotations, and at one denominator  $m^2$  proportional to the gap  $\text{den} - \text{num}$  to first order; the ceiling  $m_p/m_e \leq 1/\sin \omega_e$ , above  $1.4$  PeV at LHAASO's bound ( $1.34$  in the rule's universe); at a conversion the rotations add at the Node with each above its rest, so the excess is kinetic by construction and one constraint on three wave numbers leaves the electron's spectrum continuous with its endpoint at the excess, at the level of the law's words; dimension 3 adds no mass, one quantum being one unit of the invariant over all its lines. Not derived: the number  $1,836$ ; nothing in the click, the draw or the dimension fixes a family's gap, and the equivalence between families holds whatever the gaps. A cloud's click mass is its count at first order in  $C/\Gamma$  (DERIVED from the functional; GAMEBOARD),

and nature's mass ratios are not whole numbers,  $m_p/m_e = 1836.153$ ,  $m_\mu/m_e = 206.768$  and  $m_\pi/m_e = 273.13$ , so no particle of nature can be read as a cluster of the electron's quanta (DERIVED; CLICKS). So the masses are the families' declared integers, and the proton is one quantum of its own family (ASSUMPTION); hydrogen's clicks read it as one unit of  $W$  spread over 0.84 fm, four Compton lengths of its family (DERIVED, one derivation); Charge universality and the equivalence between families follow from the writes with no condition (DERIVED); if the families share one denominator, a declaration, the masses are the square roots of the gaps and the electron's gap is a declared integer and not 1 (HYPOTHESIS). The model's particle physics is families with declared masses and conversions by click, the conversion an explicitly named hypothesis.

Status: the mass ratios' reading derived under the hypothesis of Section 10.3; the families declared; clicks.

**Derivation S.54** (the properties of the lattice and the calibration series, in full, Section 10.2). What the lattice fixes with no number. Before any file the lattice fixes: the geometry, a cube of Nodes with an integer state, six Ports and the 48 symmetries with the reflections, so parity is kept and there is one preferred frame, hence no  $g_{0a}$ , no frame dragging and  $\alpha_1 = -8$  (DERIVED; S.35, S.36); the dynamics, one rule per family, explicit, local and bijective with the remainder kept, so reversibility to the bit, no noise but the remainders' and no loops, hence no running of  $\alpha$  and no Lamb shift (DERIVED; a finding against nature, S.56 (6)); the conservation, per family and no total across families: the form and so the energy per family, the Wronskian and so the charge under a static level, no spontaneous click and so no lifetime (DERIVED); the band, light at  $1/\sqrt{3}$  Link per interval under the causal bound 1, the anisotropy  $k^2/12$  at short wavelength, a gapped holder Yukawa's kernel with the massless constant (DERIVED); the paces, the exponential metric with  $\gamma = 1$  exactly and  $\beta = 1$  as the gap closes, the finest potential  $1/\Gamma$  and the frozen content  $\frac{1}{2} \ln(2\Gamma)$  (DERIVED under the one assumption); the holders, every held row Poisson's with the kernel  $3/(4\pi r)$ , its strength a declared weight and its range its gap (DERIVED); the click layer, the bilinear credit and so the angle it gives, one whole quantum per click and the draw by the shares (ASSUMPTION); and the scale, one scale  $\Gamma$  and pure numbers of order one, every large number a power of  $\Gamma$  or a declaration. In one sentence: the lattice is a machine of forms; it fixes every form and every ratio of order one, conserves, reverses and makes no noise; it fixes no strength, no scale and no large number; those are the file's.

The calibration series. Every measured number of an experiment is first written as a click pattern on a declared NodeReader against a declared clock. The families' pairs and the units  $\Gamma$  and  $T$  would be fixed by that match and by nothing read off the GameBoard, the shipped file's integers being the rule's own; the two ratios the model fixes without it are light's speed of  $1/\sqrt{3}$  Links per interval and the causal bound of 1 (DERIVED for the speed, GAMEBOARD; the bound is postulate (4)). The passage from the board to the clicks is S.50's. No coefficient of the file is derived from the lattice; the lattice dictates the form in which each enters and reads none of them. Each is read from one measurement of nature, Table 4: every coefficient of the file is read by one click on a bound body; the lattice fixes the form of the reading; the file holds the integer. The table's last column places each in the three layers (Section 1):  $T$  and the NodeReader's unit of credit are the click's, the gaps,  $k$ ,  $E_g$  and  $\Gamma$  the board's, read through bound bodies and given by no NodeReader. A number from the lattice itself, a fixed point of the whole selecting  $\Gamma$  or a ratio of the weights, the law has none; if it exists it is in the cosmology, the vacuum's write, an explicitly named hypothesis for the next paper (HYPOTHESIS). One derived check beside the table, the two clocks: a cavity of bodies and an atom's beat shift alike by  $N$  at long wavelength, as nature's clocks do to  $10^{-5}$ , a cavity of declared faces at  $N^2$  (DERIVED; CLICKS; S.31).

Status: each item as marked; GameBoard for the forms, clicks where a number of nature is named.

**Derivation S.55** (the conservation picture and the credit's bookkeeping, Section 3.5). What is conserved, and where. On the GameBoard the conservations are per family: the weighted form of a family while its reads are static (Theorem 3), the Wronskian of a plane family under a static level, exactly in the line and to first order in a level's change (Section 5.5), the quanta through the Ports, the wave number of a free record and reversibility bit for bit. Energy is conserved within a family by theorem, between a charged body and its light by the law's line (Section 10.1), and between a body and the holders of the content it is replaced by the mass defect (Section 10.3). The GameBoard conserves no total: the sum of the families' forms is not conserved and not assumed, the share moving with the paces with no current to compensate (Eq. (13) of the paper). The click layer has books of its own: a click credits one whole quantum at the NodeReader's own whole, drawn by the inflow of the record's share through its front over its window; a frequency of nature, the GameBoard's quantities and the credit's bookkeeping are read as S.50 states (Sections 5.3 and 7.4). In the clicks there are conservations other than the board's, and neither layer keeps a total energy (THEOREM for the board's, ASSUMPTION for the credit's).

The table.

| Layer           | Conserved                                                         | Scope                                                                        | By what                                                                       |
|-----------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| GameBoard       | the weighted form $E$ of a family                                 | per family, paces fixed                                                      | Theorem 3                                                                     |
| GameBoard       | the Wronskian $W$ of a plane family                               | per family, under a static level exactly, to first order in a level's change | S.9, S.33                                                                     |
| GameBoard       | the wave number $k$ of a free record                              | per family, paces fixed                                                      | the band's translation invariance                                             |
| GameBoard       | the sum of $D$ over families                                      | not conserved                                                                | Eq. (13) of the paper: the share moves with the paces, no current compensates |
| clicks          | the number of quanta, the charge per quantum, the whole per click | per whole quantum, across families                                           | the credit's integer unit; the energy line for a charged body and its light   |
| content holders | the body's share                                                  | not conserved: the loss is the mass defect                                   | S.38                                                                          |

On the GameBoard nothing is assumed: the conservations are theorems of Rule3's form. What is not conserved and not assumed is the sum of all families' energy; the holders are open to their readers and the law has no theorem equating one family's loss with another's gain. The energy line closes it for one sector: a charged body and its light conserve exactly at  $k_r E_s T \cos \omega_s = k_w \Gamma$  (S.26). For the content holders it is not closed and not to be: a body's count falls in a well, its potential being in its count, while the holder's waves gain a different share of the loss, the mass defect, the energy leaving the books, the law's own statement and nature's reading of bound bodies. Status: the board's conservations theorems; the credit's bookkeeping a declaration; the energy line the law's line, two derivations.

**Derivation S.56** (the explicit findings, with their numbers, Section 10.3). Each finding is a place where the law as written and a measurement of nature differ, named with its number and its root; none is smoothed. (1) The lattice's frame: a metric diagonal in the preferred frame,  $\alpha_1 = -8$  against  $|\alpha_1| < 4 \times 10^{-5}$  and, in the standard gauge,  $\alpha_2 = -1$  against  $|\alpha_2| < 4 \times 10^{-7}$  from the Sun's spin and  $1.6 \times 10^{-9}$  from solitary pulsars, conditional on the static read, no frame dragging (0 against Gravity Probe B's  $37.2 \pm 7.2$  milliarcseconds per year), no  $\times$  polarisation (S.35, S.36). (2) The nuclear binding by any derived body: bounded by  $0.76 \text{ CW}/(\Gamma E)$  per quantum at one Node (GAMEBOARD), against the deuteron's  $1.2 \times 10^{-3}$  (S.38); the size law of one well,  $R \propto 1/A$ , against  $A^{1/3}$  (S.45). (3) The neutron's  $878.4 \pm 0.5$  s: no spontaneous click, no rate; the conversion a declared list of the one click act at a declared rate, the table and the rate nature's numbers and nothing of the neutron's mass claimed (Section 10.3). (4) The  $Z$  and parity: no fourth rotation of a three-line record, the step commuting with the reflections. (5) The anomaly  $g - 2$  and Pauli's two per orbital: no loops, no vector giving the exclusion; the law does not support the reduced mass: it holds no amplitude over two positions, no wave of two records on pairs of Nodes (every record's levels stand at its own Nodes, jointly with the other records there and coupled to them by the read, and the pair's joint amplitude enters at the click, in the credit's pairing, S.51; postulates 1 and 2's locality), so a product of two bound records about one centre has no relative motion of the pair, and under the second of the two lines the nucleon's centre is one classical variable, so the isotope shifts and the muonic atoms are outside what it derives; the many-body correlation is not reproduced by a product of records (the ground levels, a GameBoard reading: helium 1.4 percent short, positronium 44 percent of nature's) (S.60). (6)  $\alpha$ 's running and the Lamb shift: no loops. (7) The mass ratios of nature not whole numbers, so no particle a cluster of the electron's quanta; the masses declared. (8) The record's own level: hydrogen's 1s at  $-0.043$  of Rydberg's unit at  $1.45 a_0$ , the Rydberg scaling returning at large  $Z$  as  $1 - 1.25/Z$  ( $E_{\text{tot}}$  as  $(1 - 0.3125/Z)^2$ ) (S.46), answered by line (1) of the model's decision (S.60). (9) The radiated share's coefficient: the time level carries half of Larmor's power in form, the odd lines' part depends on their wall and coupling, the fork open; no lifetime number (S.26, S.45). (10) At the second order in  $z$  the ball of dust alone decelerates,  $q_0 = \Omega_m/2 > 0$ , where the supernovae read  $q_0 \approx -0.5$  (flat  $\Lambda$ CDM); the paper's claim is at  $z \ll 1$  only, and the hypothesis that would close it, the vacuum's write, is in the law's documents and not claimed here (S.29). (11) The content holders' balance is not a line of the law: a body's loss in a well is the mass defect (S.38, S.55). (12) The electromagnetic binding energy's seat, open: the holders of the content gravitate with their binding, the sign holder's cross term sources nothing as written (Section 8 (i), S.38). (13) Bjorken scaling: under the dimension-3 family the three fractions a hard click reads do not change with the momentum transfer, exact scaling, where nature's measurements show logarithmic violations of order ten percent across the measured decades of  $q^2$ ; the root is the one that stops  $\alpha$ 's running, no pair created and no coupling running (S.37). (14) Bell violations between separately prepared bodies joined by a coupling or by a transferred quantum (gate-joined ions,  $S = 2.25 \pm 0.03$ ; transmons with a transferred photon,  $S = 2.0747 \pm 0.0033$ , loophole-free) exceed the local bound the law's credits give for such a join: on the law as written nothing after an exchanged quantum is bilinear in the two bodies' amplitudes, and every later credit is local,  $S \leq 2$ ; a finding of the first rank against the law as written, two derivations. The pair laid as one event and the swap of two multi-part records at a click the law meets (S.51). (15) The shelved ion's telegraph: bimodal counts and exponential dark periods, which the law without the NodeReaders' writes gives as Poisson at half the rate; read against their blinds by the body's own giving click at a declared rate (computed, the rate declared), the shelving at the Zeno-limited rate  $\Omega_w^2/R$ , the bright periods' mean  $R/\Omega_w^2$  and the dark  $\tau_D$ , the rate not derived; its file is the repository's (Section 10.3). Status: explicit findings; clicks where a number of nature is named.

**Derivation S.57** (one photon on two bodies: the anticoincidence, Sections 5.4 and 9). Inputs: a light record  $L$  of one whole quantum,  $2A^2 \sin \Omega = T$ , laid between two bound bodies  $A$  and  $B$  at equal distances, each with the modes  $g$  and  $e$  and the transition  $\omega_e - \omega_g = \Omega$ , a NodeReader on each body reading its  $e$ -mode, the whole the transition (Section 5.4); the share's conservation through the Ports (Theorem 3); the two-mode line's transfer at resonance (S.43); the credit's one draw per record over all the NodeReaders that read it (Section 5.3). Steps: (a) the inflow of  $L$  into the body  $X$ 's region over the window is  $\Phi_X$ , and by the share's conservation  $\Phi_A + \Phi_B \leq W_c$ , one quantum; the absorbed share is the two-mode line's transfer,  $s_X = \eta \Phi_X / W_c$  with  $\eta = \sin^2(\Omega_R \tau_w)$ ,  $\Omega_R = |\epsilon_{ge}| / (4\sqrt{\sin \omega_g \sin \omega_e})$  and  $\tau_w$  the passage, so by symmetry  $s_A = s_B = s \leq 1/2$ . (b) The credit draws once per record:  $P(A \text{ only}) = P(B \text{ only}) = s$ ,  $P(\text{both}) = 0$ ,  $P(\text{neither}) = 1 - 2s$ ; the anticoincidence parameter  $\alpha = P(\text{both}) / (P(A)P(B))$  is 0, where independent draws, classical light, give  $\alpha = 1$ . (c) The control of two quanta:  $P(\text{both}) = 2s^2$ ,  $P(A \text{ twice}) = P(B \text{ twice}) = s^2$ , and  $\alpha$  depends on its definition, both forms the independent draw's algebra and derived: by the mean counts,  $\langle N_A \rangle = 2s$ ,  $\alpha = 2s^2 / (4s^2) = 1/2$ ; by the probabilities per trial, Grangier's,  $P(A) = 1 - (1 - s)^2 = 2s - s^2$  and  $\alpha = 2s^2 / (2s - s^2)^2 = 2 / (2 - s)^2$ , the two meeting as  $s \rightarrow 0$  and differing at  $s = 1/2$  (8/9 against 1/2); the absorbed fraction of S.43 stays open, and the implementation's reading names which definition it is read under; three bodies:  $\sum_X s_X \leq 1$  and  $P(\text{two or more}) = 0$  for one quantum. Check: Grangier, Roger and Aspect's anticoincidence on a beam splitter,  $\alpha = 0.18 \pm 0.06$  with a source-limited single photon, and modern single-photon sources below 0.01 (the paper's reference). Status: derived from the credit's declaration, as Born's rule is, two derivations; clicks. Nothing of the click's write enters it: the anticoincidence rests on the one draw under the law as it stands; the file's readings are the repository's folder's document and no claim of this paper.

**Derivation S.58** (the push on a moving record, Section 8 (b), Eq. (16)). Inputs: the composed paces,  $p_0/\Gamma = e^{-U}$  and  $p_a/\Gamma = e^{-2U}$  (Eq. (3), the one assumption  $\hbar N = 1$ ); the band at a pace (Eq. (4)) at a small gap and long wavelength,  $E^2 = m^2 + P^2$  with  $E = \omega$ ,  $m = N\omega_0$  and  $P = cN^2 k$ , the exponential metric's dispersion ( $g_{00} = -e^{-2U}$ ,  $g_{aa} = e^{2U}$ ,  $\gamma = 1$ ); the force on a record as the gradient of its rotation at fixed wave number,  $\mathbf{F} = -(\partial\omega/\partial U)_k \nabla U$  (the fall of S.18 is its  $k \rightarrow 0$  case). Steps: (a)  $\partial(E^2)/\partial U = \partial(m^2 + P^2)/\partial U = -2m^2 - 4P^2$ , so  $-(\partial\omega/\partial U)_k = (m^2 + 2P^2)/E = (E^2 + P^2)/E$ ; (b) with  $\beta = P/E = v/(cN^2)$ ,  $v = \partial\omega/\partial k = c^2 N^4 k/\omega$  the group velocity and  $cN^2$  the local light speed,  $\mathbf{F} = E(1 + \beta^2) \nabla U$ ; (c) at  $\beta = 0$  the pull  $E \nabla U$ , over the band's inertia  $m^* = 3 \tan \omega_0$  the fall  $a = f(\omega_0) \omega_0 / m^* \nabla U = c^2 (U_K/U) \nabla U$ ,  $0.2667 \nabla U$  at [2, 3] (S.18, the same number); at  $\beta = 1$  twice the pull, the bending  $4U_b$  (S.20); (d) the units: the paper's momentum is the wave number, so the lattice form is  $(E^2 + c^2 \mathbf{p} \cdot \mathbf{p})/E$  and the covariant one  $(E^2 + P^2)/E$ ; the form  $E^2 + 3\gamma \mathbf{p} \cdot \mathbf{p}$  holds for  $\mathbf{p} = E\mathbf{v}$  alone, where  $3\mathbf{p} \cdot \mathbf{p} = E^2 \beta^2$  with  $c^2 = 1/3$ . Check: on the exact band at [2, 3] the rest push over  $E$  is  $f(\omega_0) = 1.0634$ , the redshift's factor, and the coefficient of  $\beta^2$  is  $1.3354 = \gamma_K - \alpha_K/2$  (two derivations); at [999, 1000]  $1.00017$  and  $1.00067$  ( $1 + \omega_0^2/12$  and  $1 + \omega_0^2/3$ ), both  $\rightarrow 1$ ; the angle's ratio to Newton's at  $\beta = 0.38$  is  $1 + \beta^2 = 1.144$ , the force's  $\gamma_L(1 + \beta^2) = 1.237$ . Status: derived from Eqs. (3) and (4) under the one assumption (fitted), two derivations; as physics the standard post-Newtonian factor  $1 + \gamma\beta^2$ , the law's own content being  $\gamma = 1$  derived and not declared and the finite-gap coefficient; the chain to Einstein's weak-field forms is the one dispersion read four ways (the bending at  $\beta = 1$ , Newton at  $\beta = 0$ , the redshift and Shapiro's delay by its time part and index, the perihelion by the orbit step of S.19 with its  $k^4$  caveat at the computed pair); clicks, the deflection angle's ratio  $1 + \beta^2$  to Newton's.

**Derivation S.59** (the quantum Zeno effect: the click writes the body, Section 5.4). Inputs: the law's line of the click's write for a body as a NodeReader, in the two derivations' wording: the NodeReader draws over its outcomes and writes the drawn one; for a light record the outcome "no taker" writes nothing, for a body's parts every outcome is a part and the body is re-laid in it at its count (the law's line of the click's write and its hypothesis (h3)); the decisions "the two hands close the line of the click's write" and "the Zeno clause"; one bound body of the matter family in its mode  $g$  (one plane, the modes  $g$  and  $e$ , no parts and no floor needed); a light level at the resonance  $\omega_e - \omega_g$  driving  $g$  and  $e$  by the two-mode line (S.43) as a  $\pi$  pulse, the full transfer from  $g$  to  $e$  in  $T_\pi = \pi/(2\Omega_R)$ ,  $\Omega_R = |\epsilon_{ge}|/(4\sqrt{\sin \omega_g \sin \omega_e})$ ; a probe of the  $g$ -mode's share in  $n$  equally spaced windows during  $T_\pi$ , the last at  $T_\pi$  (in nature a cycling transition from  $g$  whose fluorescence is counted), each window's draw writing its outcome,  $g$  or  $e$ , whichever it is, by the body's own giving click at the Node the draw picks by the share over its region of two Nodes or more, so that no external NodeReader is required for the write; the reading at the end the body's share in  $e$ . Steps: (a) between windows the drive rotates the body's amplitude angle by  $\pi/(2n)$ , so the flip probability per window is  $\sin^2(\pi/(2n))$  and the difference  $P(g) - P(e)$  is multiplied by  $\cos(\pi/n)$  at each window; after  $n$  windows  $P(e \text{ at } T_\pi) = (1 - \cos^n(\pi/n))/2$ : 1, 0.500, 0.375, 0.235, 0.133, 0.072 and 0.037 at  $n = 1, 2, 4, 8, 16, 32$  and 64 (column A, the formula's curve, the ideal probe's). (b) With a reading only and no write the drive transfers fully,  $P(e) = 1$  at every  $n$  (column C, a reading only and no write). (b') A body reading itself at its own period as its window would be a Zeno window of that period, suppressing every Rabi oscillation longer than it, which nature's measurements show at every duration; so a bound body's period is not its window, a body's window being the arriving message's duration (two derivations). (c) Were the click alone written, a window without a click leaving the record as it is, the curve would read 1, 0.625, 0.154, 0.005, 0, 0 and 0 (column B), and with the last probe a reading only 0.750, 0.286 and 0.046 at  $n = 2, 4$  and 8 (column B'): every version in which the null outcome leaves the body's record as it is falls. Beside it, against nature's measurement: Itano, Heinzen, Bollinger and Wineland, *Physical Review A* 41, 2295 (1990), beryllium-9 ions, a 256 ms  $\pi$  pulse with  $n = 1$  to 64 probe pulses, the measured transition probabilities (their Table I) following Itano's own corrected calculation, which carries the probes'

optical pumping and the finite pulses and lies below the bare formula of column A at the larger  $n$ ; the column here is the formula's, and the data's own column enters from that table and excluding B and B' (0.235 at  $n = 8$  against 0.005 and 0.046); a factor of 27 between A and C at  $n = 64$  and of 2 already at  $n = 2$ . The file's blind: with the write off the file reads 1 at every  $n$  and with the write on column A; every window's re-lay keeps the body's count and the board's inverse holds between the windows; a null outcome moves no quantum between families and moves form within the body's plane only. Status: derived under the law's line of the click's write with the window's outcome written, two derivations and the decision "the Zeno clause"; the file's readings are the repository's folder's document and no claim of this paper; in the supplement only, by the decision; CLICKS. Check:  $(1 - \cos^8(\pi/8))/2 = (1 - 0.92388^8)/2 = (1 - 0.5308)/2 = 0.235$ ;  $(1 - \cos^2(\pi/2))/2 = 0.5$ .

**Derivation S.60** (the atom under two lines: no record reads its own write of the sign; the nucleon a compact body). Inputs: the hydrogen-like problem of S.46 (a) with the nucleus's angle alone; the self-read of S.46 (c); two lines, the model's decision, recorded in the decisions' document: (1) every record that reads the holder of the sign is one quantum (count 1), writes its own held row of the sign (its Wronskian over  $E_s T$ , Rule3 at the paces of its reads) and reads every sign row but its own, the light born by its write living in its row; (2) the holders of the content keep the self-read, the nucleon families read the nuclear holders' rows including their own, so a nucleon is a compact self-bound body at the declared weight, its centre one classical variable in an outside well. Steps: (a) The chain, by the law's lines and nothing else. The nucleus, one quantum of its family, writes its Wronskian  $T/2$  into the sign holder's row every interval (the one write, Section 3.5; S.9), and the holder's rest far from it is the kernel's, the angle  $\theta_0(r) = 3kZ/(8\pi r \Gamma E_s) = Z\alpha_{\text{law}} c/r$  with  $c = 1/\sqrt{3}$  and  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$  (S.25, S.26 (a), (b)). Under line (1) the electron's record reads that row alone, so its Rule3 line is the plane record's step under the time turn  $\theta_0(r)$  (S.33's inputs; the sign holder rotates the two-part record),  $e^{-i\theta_0} z_{\text{next}} + e^{i\theta_0} z_{\text{before}} = [S z_{\text{now}} + \sum_a R_a(z_{+a} + z_{-a})]/w$ , and nothing else acts on it. Its stationary modes  $z = \phi e^{-i\omega t}$  obey, by the band at small wave number (S.1, S.22:  $\omega = \omega_s + k^2/(2m^*)$  with  $m^* = 3 \tan \omega_s$ ) and the turn's shift of the rotation (S.26 (c):  $\omega \rightarrow \omega - \theta_0$ ;  $2 \cos(\omega + \theta_0)$  equal to the band's  $2 \cos \omega(k)$ ), the law's own eigenvalue line on the lattice,  $-\Delta \phi/(2m^*) - \theta_0(r) \phi = (\omega - \omega_s) \phi$  with  $\Delta$  the six-Port second difference; its radial form is solved on a grid of 2,000 points with no formula put in: the bound modes sit at  $\omega_n - \omega_s = -m^*(Z\alpha_{\text{law}})^2 c^2/(2n^2) = -m^*(Z\alpha_{\text{law}})^2/(6n^2)$  per interval,  $Z^2$  and  $1/n^2$  exact in the small-gap limit, the radius  $a_0 = \sqrt{3}/(m^* Z\alpha_{\text{law}})$  Links; the levels are a GameBoard reading, the diagnostic behind the click, and the lines nature measures are the beats of two modes read by a click (S.14, S.43), the table below. Into nature's units: the interval of S.27 ( $\omega_e \hbar/(m_e c^2)$ ) and the law's  $\alpha$  read from Coulomb's force (S.26 (b)) turn the mode's depth into  $m_e c^2 \alpha^2/2$  at  $n = 1$ ,  $-1/2$  hartree exactly in the slow limit ( $-0.49995$  on the grid) and  $-1/8$  at  $n = 2$ . What the law computes is the coefficient  $1/2$  and the dependence on  $n$ ,  $Z$  and the family's gap  $m^*$ ; the two inputs,  $\alpha$  and the electron's unit, are the file's calibrations from clicks other than the spectrum (Coulomb's force, the electron's mass), so the chain is not circular: no formula of the atom enters, the absolute hartree is a comparison through the unit and the coefficients and the ratios are the derived content. (b) Without (2) the nucleus is a free record of inertia  $m_p^* = 1,836 m_e^*$  reading the electron's row, a well 27 eV deep and  $a_0$  wide: its stationary spread is  $0.21 a_0$  (the 1s cloud's curvature  $k = 4/3$  a.u.,  $\omega_p = \sqrt{k/m_p} = 0.027$  a.u.), and the self-consistent product of the two records sits at  $-0.4671$  Ha, 6.58 percent above Bohr, the 2s at  $-0.1147$ , the 1s-2s at 9.59 eV against 10.20; the spread scales as  $m^{-1/4}$ , so deuterium's 1s-2s differs from hydrogen's by 172 meV against nature's 2.8 meV. (c) Under (2) the nucleus does not spread, (a) holds, and the law does not support the reduced mass: it holds no amplitude over two positions, no wave of two records on pairs of Nodes (every record's levels stand at its own Nodes, jointly with the other records there and coupled to them by the read, and the pair's joint amplitude enters at the click, in the credit's pairing, S.51; postulates 1 and 2's locality), so a product of two bound records about one centre has no relative motion of the pair, and under (2) the nucleon's centre is one classical variable, so the isotope shifts and the muonic atoms are outside what the law derives; helium in the mean field without exchange  $-2.8617$  against  $-2.9037$  Ha; positronium, two free records of one gap about one centre,  $-0.22 R$  ( $-0.110$  Ha) against nature's  $-0.50 R$  ( $-0.25$  Ha, 6.80 eV): a product about a centre has none of the pair's relative motion, which carries the whole binding of two equal records; a named finding. Check: the pinned nucleus reproduces  $-1/(2n^2)$  to  $10^{-4}$  on the grid; the free-nucleus numbers by the self-consistent two-record Hartree iteration (60 iterations, mixing 0.5); the zero-point estimate  $\frac{3}{2}\omega_p = 0.040$  Ha against the self-consistent 0.0136 plus the smearing's 0.019. Status: derived under the two lines, the model's decision (line (1) ASSUMPTION, FITTED; line (2) ASSUMPTION); two derivations, with the two costs the second hand names: under (1) the rows at a Node number the file's charged records, so the bound of postulate 2 is the file's declaration; under (2) the nucleon's classical centre is the pinned nucleus by declaration, which gives the levels, the reduced mass unsupported by construction (no amplitude over two positions), the isotope shifts with it. What rests on it: S.46's finding is replaced in the open list by the statement that the law does not support the reduced mass and that the many-body correlation is not reproduced by a product of records, Pauli's exclusion having no line (S.56 (5)); CLICKS.

The atoms' table: the clicks the law derives, with how they were derived. Nature's measurements give lines and thresholds, the differences of levels, and each row is the beat of two modes read by a click (S.14, S.43), computed by hand from the modes of steps (a) to (c) with nothing run; a level alone is a GameBoard reading and stands in the text as the diagnostic behind the click. Every row is derived from the law's lines alone, no equation outside the law entering

(Bohr's, Schrödinger's and Hartree's forms in no column but nature's); nature's column is the measured value with its source, the comparison, entering no derivation. The inputs from nature, named once: the electron's unit of  $S.27$  ( $m_e$  and  $\alpha$ , read from the electron's gap and from Coulomb's force, clicks other than the spectrum; the hartree  $m_e c^2 \alpha^2$ ), so an absolute eV is a comparison through the unit and a coefficient or a ratio is the derived content, and the proton's inertia 1,836 as the family's declared pair. The rows are a supporting finding and no more, derived under the two lines, the model's decision, two derivations (the mathematician's chains and the advisor's circularity check, row by row), a gate file open and nothing of it claimed, CLICKS. Not in it: the slow limit of the band only, so the  $(Z\alpha)^2$  order, the Lamb shift and the nuclear size are not in it; the reduced mass unsupported by construction (step (c)); Pauli's exclusion without a line (two records in one mode need none).

| The click                                                | Nature measured, with its source                      | The law's click under the two lines             | How derived, by the law's lines                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrogen's 1s-2s line                                    | 10.20 eV, 2,466,061,413,187 kHz (Parthey et al. 2011) | 10.20 eV                                        | the beat of the $n = 1$ and $n = 2$ modes of step (a) read by a click, the share $T(\sin \omega_j - \sin \omega_i)$ (S.14, S.43): 3/4 of the 1s depth derived, the eV through the unit                                                                                                                                                                                                                                                                |
| The ionisation threshold                                 | 13.598 eV (CODATA 2018)                               | 13.6 eV                                         | the 1s mode's depth of step (a) read as a click's share (S.14): the coefficient 1/2 derived, the eV through the unit                                                                                                                                                                                                                                                                                                                                  |
| He <sup>+</sup> 's 1s-2s line                            | 40.81 eV (the He II spectrum)                         | 40.82 eV ( $Z^2 \cdot 3/8$ hartree at $Z = 2$ ) | step (a) at $Z = 2$ , the angle $Z\alpha_{\text{law}}c/r$ scaling the two modes by $Z^2$ , the beat 3/4 of the 1s: the $Z^2$ ladder derived as a click, the $(Z\alpha)^2$ order not in the slow limit                                                                                                                                                                                                                                                 |
| Helium's ionisation threshold                            | 24.587 eV                                             | 23.45 eV, 4.6 percent                           | two electron records in one mode about the pinned nucleus, each reading the nucleus's row and the other's and never its own (line (1)), iterated by the law's write and read to the fixed point, $-2.8617$ Ha (step (c)), against He <sup>+</sup> 's one record at $-2$ Ha exactly: the threshold their difference, the click; nature's total energy two thresholds summed and no click; the correlation the product misses sits wholly in this click |
| Positronium's 1S-2S line                                 | 5.10 eV, 1,233,607 GHz (Fee et al. 1993)              | 1.9 eV                                          | two free records of one gap about one centre, each reading the other's row, iterated to the fixed point (step (c)), the beat of the product's two modes ( $-0.110$ and $-0.039$ Ha, the diagnostic behind the click); a product about a centre has none of the pair's relative motion, the ground level at 44 percent of nature's, the line at 38                                                                                                     |
| Control: hydrogen's 1s-2s with the nucleus a free record | —                                                     | 9.59 eV                                         | without line (2) the nucleus's record of count 1 reads the electron's row and spreads by $0.21 a_0$ at the inertia 1,836, the two records iterated to the fixed point (step (b)): the control that separates line (2) from no line (2)                                                                                                                                                                                                                |

## The engine's derivation ledger (Section 3.7)

**Derivation S.61** (the engine's derivation ledger, Section 3.7). Inputs: the engine's acts as the engine's document lists them, one line per act, and the law's lines each act is derived from: Rule3's one line with its coefficients; the paces compose, the clock the Node's and the tension the Link's; the write per proper volume and per proper interval; the tension as the wave's own stress; the sign holder rotates the two-part record; the count is the record's share; the click is the meeting, the one act with its lists; nothing is kept at a Node beyond the law's numbers; the families' declaration. Statuses: D, derived from the law's lines with the proof shown; W, declared by the file (what the experimenter builds); L, declared by the law until a derivation, with the reason; A, a named assumption; O, open and not claimed; F, a named finding; X, an act fallen by the owner's word, withdrawn. The table: the two derivations' converged ledger, one row per act, each row pointing to the law's line it is derived from and to the engine's act; the proof where it can be given, and where it cannot the reason and the status. Declared by the law until a derivation, in one list with the reason: Rule3's line and its rounding (the model's rule);  $\Gamma$  and  $T$ , the units, what fixes  $T$  the calibration series and what fixes  $\Gamma$  open (row 3); the base  $1 - 1/\Gamma$  of the clock; the families' pairs and dimensions; the one rounding half up of the write (two derivations);  $\tau$  (no fluctuation at a Node); the seed and the generator (the law's one draw is the NodeReader's). Declared by the file: where (the Nodes), the reads and their weights,  $k_w$  and  $E_s$ , the transition's pair and weight  $k$ , the width  $w$  of a given packet, a continuous flow's window, the conversion's table and rate. Assumption: the direction drawn by the giver. Open and not claimed: what draws; where a click happens in a body of nature, heat included; momentum under the write (the recoil open); the one-in-flight limit. Named findings of the acts, F in their rows: the credit's window remainder dropped (row 12); the conversion's findings, in the folder's document (row 18). Result: no act of the engine was found with neither a derivation nor a stated reason, both hands' conclusion; the named findings are the acts with a known departure and a stated reason and no derivation (lines 6, 12, 18, 19 and 28). Nothing in the ledger is a claim of nature.

| #  | The act                                                                                                                                                                                                    | From the law's lines                                    | The derivation, the proof shown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Status                                                                                                                                                                         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | The read: the content and the Link factor $Q_{ij}$ , one integer per Link from both ends                                                                                                                   | the paces compose; the tension the Link's               | One number per Link read the same from both ends makes <b>DM</b> symmetric, $\mathbf{D} = \text{diag}(1/p_i^2)$ , since $M_{ij} = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$ carries the reading Node's own $p_i$ and <b>M</b> itself is symmetric at uniform paces alone; that symmetry is the whole of what lines 2 and 11 need (the conserved form, the flux, the modes' orthogonality).                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D; W (which holds a family reads, at what weight)                                                                                                                              |
| 2  | Rule3's step, the bijection, the band, the conserved form                                                                                                                                                  | Rule3's line                                            | $Q = a_t^\top \mathbf{D} a_t + a_{t-1}^\top \mathbf{D} a_{t-1} - a_t^\top \mathbf{D} \mathbf{M} a_{t-1} / w$ , the weighted form, Eq. (6) being $wQ: Q_{t+1} - Q_t = (a_{t+1}^\top \mathbf{D} a_{t+1} - a_{t-1}^\top \mathbf{D} a_{t-1}) - (\mathbf{D} \mathbf{M} a_t)^\top (a_{t+1} - a_{t-1}) / w = 0$ since $\mathbf{M} a_t = w(a_{t+1} + a_{t-1})$ and <b>DM</b> is symmetric by line 1; the plain form ( $\mathbf{D} = 1$ ) is conserved at uniform paces alone. The inverse recovers the level before bit for bit from the same integers (the direction). The band $\cos \omega = (\text{num} / (3\text{den})) \sum_a \cos k_a$ from $S = 0$ at the vacuum. Checked numerically on a well: the weighted $Q$ to the last digit over 200 intervals.                                                                                     | L (the line and its one rounding, the model's); D (the band, the direction, the form)                                                                                          |
| 3  | The composed paces: $p_0 = \Gamma(1 - 1/\Gamma)^c$ rounded once; $p_i = p_0^2 / \Gamma$ ; $p_a(i, j) = p_i q_{ij} / \Gamma$ per Link (Eq. (3)), the Node's axis pace their mean $p_i(q_+ + q_-)/(2\Gamma)$ | the paces compose                                       | Composition $f(c_1 + c_2) = f(c_1)f(c_2)/\Gamma$ forces the exponential, $f(c) = \Gamma r^c$ ; $r = 1 - 1/\Gamma$ is one level, one step of the clock. The Link twice: $p_i = p_0^2 / \Gamma$ from the Link joining two clocks of the same content. The axis pace the Node's pace times the mean of its two Links' factors, from the clock being the Node's and the tension the Link's.                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D (the forms, from the one assumption, A); L ( $\Gamma$ itself: what fixes $\Gamma$ open)                                                                                      |
| 4  | The guard at load: $p^2(\text{den} +  \text{num} ) \leq 2 \text{ den } \Gamma^2$ ; every Link's factor positive; the turn's tangent half-angle at most 1                                                   | Rule3's line; the sign holder rotates                   | Stability of the linear recurrence: the roots are unimodular if and only if $ \cos \omega(k)  \leq 1$ for every $k$ ; the extreme is the mode at $k = \pi$ ( $\text{num} \geq 0$ ) or $k = 0$ ( $\text{num} < 0$ ), which gives the inequality; the edge admitted (a repeated root grows linearly, light's checkerboard). The factor positive is $q = \Gamma - t > 0$ , the pace positive. The half-angle at most 1 is the reach of the three shears (line 9).                                                                                                                                                                                                                                                                                                                                                                              | D                                                                                                                                                                              |
| 5  | The start's rest, its miss bound $K$ and the refinement                                                                                                                                                    | Rule3's line; the write                                 | The static line $(2w - S)b = \sum R b_j + \text{source}$ ; the floored Jacobi iteration stops below the line by up to one fine unit over $(1 - \rho)$ , $\rho$ the neighbours' mean weight, so the miss bound $K$ is the least of two bounds of one quantity: the massless line's parabola at the unit source, $(6 \text{ div } 2)((n + 2) \text{ div } 2)^2 + 1$ along the longest open axis of $n$ Nodes, and, where the row has a gap $W - 6R > 0$ at every Node, $W \text{ div } (W - 6R) + 1$ , which is $\text{den div } (\text{den} - \text{num}) + 1 = 4$ at the vacuum's paces at $[2, 3]$ against the parabola's 508 on the 25-cube; a massless row on a board wrapping on every axis has neither and is refused by name; the residual read exactly and solved again by the same act (the fix, within 0.51 level on the 25-cube). | D                                                                                                                                                                              |
| 6  | The hold and the write's factor as one rounding                                                                                                                                                            | the write per proper volume and interval                | $N^a / (h_x h_y h_z)$ with $a = 2$ on a count and 1 on a Wronskian, a symmetric function of the three paces, so the cube's 48 images are kept to the bit (a symmetric function of the three paces). The bound of the factor from the paces' own definitions: $p_a \leq p_i = p_0^2 / \Gamma \leq P$ under $q \leq \Gamma$ , so the factor is at most $(P/\Gamma)^{5/2}$ on a count and $(P/\Gamma)^2$ on a Wronskian, both inside the room $(P/\Gamma)^3$ ; under a negative tension $q$ exceeds $\Gamma$ and the room is exceeded (1.60 against 1 on light's pair, 3.05 against 2 on a Wronskian of matter's pair): the room's factor of the ceiling $q_{\max}/P$ owed before any file with tensions of either sign.                                                                                                                       | D (the scaling, the symmetry, the bound under $q \leq \Gamma$ ); L (the one rounding half up, two derivations); W ( $k_w, E_s$ ); F (the room under a negative tension, named) |
| 7  | The tension: $T_{aa}(i) = -\text{num } h_a(i), h_a(i) = \text{now}_{i-a} \text{ now}_{i+a} - \text{now}_i^2$ ; the Link's book-<br>ing $-\text{num } G_{aa}(i) = T_{aa}(i) + T_{aa}(i + a)$                | the tension the wave's own stress                       | For a plane wave $b \cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$ , $-\text{num } h_a = +\text{num } b^2 \sin^2 k_a$ per Node: the wave's own pressure, light gravitating by its pressure and not by a hill; the sign the law's.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D                                                                                                                                                                              |
| 8  | The sign current: $J_a/2, J_a = \text{Im}(\bar{z}_i(z_{i+a} - z_{i-a})) = g_{i,i+a} - g_{i,i-a}$ with $g_{ij}$ the sign's Link current of S.33 (a) (S.33 (b))                                              | the sign holder rotates the plane                       | The spatial Wronskian of the plane; on a plane record $J_a = (6\text{den}/\text{num}) W v$ , so the odd level over the time level is $3(\text{den}/\text{num}) v = v/c_s^2$ and the magnetic over the electric force on a co-moving reader is $1/\gamma$ (two derivations); the half because the mean of the two Links is the Node's reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | D                                                                                                                                                                              |
| 9  | The turn by three shears                                                                                                                                                                                   | the sign holder rotates the plane                       | $\mathbf{R}(\theta) = \mathbf{S}_x \mathbf{S}_y \mathbf{S}_x$ at $\tan(\theta/2) = n/(2\Gamma)$ (the angle $2 \arctan(L/2\Gamma)$ of Section 5.5), an identity of $2 \times 2$ matrices, each shear a bijection on integer pairs: exact and reversible bit for bit; the angle per proper interval the Peierls phase at the turned family's own clock.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | D                                                                                                                                                                              |
| 10 | The reach, one Link per act                                                                                                                                                                                | Rule3's line                                            | Every read is a Port's.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D                                                                                                                                                                              |
| 11 | The reports: the inflow through a front                                                                                                                                                                    | the count is the record's share                         | The Link term of the weighted $Q$ is its flux: the change of a Node's $Q_i/p_i^2$ over a step is exactly the sum of its Ports' currents $\text{num}(q_{ij}^2/\Gamma^2) F_{ij}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D                                                                                                                                                                              |
| 12 | The credit: $N$ by the wall, the NodeReader's own units $W_d$ , the cap by the count                                                                                                                       | the count's line; one quantum one unit of the invariant | A plane wave's share per Node is $3\text{den} A^2 \sin^2 \omega$ , its action $2A^2 \sin \omega$ , so a quantum at $\omega$ reads $\sin \omega$ of the top's: $W_d = W_c \sin \Omega_d = \text{isqrt}(W_c^2(\text{den}_d^2 - \text{num}_d^2)) \text{ div } \text{den}_d$ ; the unit is the record's own, $W_{\text{rec}}$ , read once from the books' origin, and no transition is declared on a region; the cap: one count moves per click and the credit draws once over all the NodeReaders of one record. The two slits' 270 units are 626.5 photons at $\omega = 0.4456 (270/\sin \omega$ with $\cos \omega = (2 + \cos(\pi/4))/3$ ).                                                                                                                                                                                                  | D; L ( $T$ : what fixes it, the calibration series); D (the half-up rounding, a stated form); F (the window's remainder dropped, named)                                        |
| 13 | The window: a body's and a region's                                                                                                                                                                        | the two-mode line; nothing kept at a Node               | The window is declared, the world's draw window or the probe's lay, set by name to the arriving share's support, and no window closes on the wave itself (Section 5.4). What the support is: a body's, the turn is 0 exactly at $a_t = 0$ and the shears at 0 are the identity, so the labels change only on the arriving share's support (both levels); a region's, the inflow is 0 exactly outside a finite packet's passage; for a continuous flow the support is infinite and the cut is the declared window.                                                                                                                                                                                                                                                                                                                           | W (the declaration); D (the support it names)                                                                                                                                  |

| #  | The act                                                                                                                                                                             | From the law's lines                                         | The derivation, the proof shown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Status                                                                                                                                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14 | The turn and the shares, the two-mode line                                                                                                                                          | the two-mode line; the sign holder rotates                   | The law's form: first order in the drive, the change of $(u, v)$ over a window is $g \sum a_t e^{i\Omega t}$ , its modulus the angle and its argument the axis; for $a_t = A \cos(\omega_L t + \phi)$ the co-rotating sum has modulus $A \sin(\delta W/2)/\sin(\delta/2) /2$ ( $W/2$ at resonance at every $\phi$ , sinc off it); in integers two reference records in quadrature by Chebyshev's recurrence, $X$ and $Y'$ , $\theta_W = k \text{isqrt}(X^2 + Y'^2) \text{div } R$ , $R$ the largest power of two with $RAW < 2^{31}$ (two derivations). The shares: the shears conserve $u^2 + v^2$ , so the labels' squares sum to the count, Born's weights. The shipped $k$ rescales by $4/\pi$ under the signed form.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | D (the law's form); W (the pair and $k$ )                                                                                                                                                                      |
| 15 | The draw                                                                                                                                                                            | the click is the meeting; the one draw                       | The weights derived (lines 12 and 14); the generator's full period by Hull and Dobell. The pick: Rule3 is a bijection with no draw and the law's one draw is the NodeReader's; five refutations fence the cause.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D (the weights); L (the seed, with that reason); O (what draws)                                                                                                                                                |
| 16 | The taking's list and the taker's lay                                                                                                                                               | the count's line; the one act                                | The lay by the count, for a family with a gap, $A^2 = \text{isqrt}(\text{count}^2 T^2 \text{den}^2 / (4 \text{laid}^2 (\text{den}^2 - \text{num}^2)))$ , one root of the whole product, gives $2A^2 \sin \omega = T$ per quantum per line, the law's unit (the count is the record's share; two derivations); the phase passes with the quantum. The factor of two: a standing record declared a NodeReader is a plane (re, im), two real lines, each with the form $wA^2 \sin^2 \omega$ , so its share is $6\text{den}A^2 \sin^2 \omega$ where a travelling real line's is $3\text{den}A^2 \sin^2 \omega$ at the same level, both from one form, the engine's amplitude the reference.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D                                                                                                                                                                                                              |
| 17 | The giving: the list, $\Omega$ , $S$ , the carry, the hazard, the grain, the envelope                                                                                               | the sign holder's row; the invariant; nothing kept at a Node | $\Omega = \omega_e - \omega_g$ : the Wronskian's cross term is the only source of the row. $S = T/(2 \sin \Omega)$ as one root of the whole product; the source puts into the record the form it radiates, the radiated form $(3/2)(\sum_t A_t^2) \sin \Omega / \sin k = T \sin \Omega$ , one quantum, that is $\sum_t A_t^2 = (2/3) T \sin k$ on a width-one guide, $A_t$ the increment's amplitude, the increments' own squares summing to $(1/3) T \sin k$ (two derivations; S.48); the carry $A$ , $A + 1$ on $m$ intervals. The hazard: nothing at the body records the time since entry, so $h$ is constant and the survival $(1-h)^t$ , the exponential; $h = 1/\tau$ not derivable, a spontaneous giving being a draw with nothing arriving and the law having no fluctuation at a Node; in the dark the grain is the interval, one draw per interval at the now. The envelope: the amplitude the survival's root, $e^{-t/(2\tau)}$ , the Lorentzian of width $1/\tau$ , its length to the floor $2\tau \ln A_0$ . The open board: the lay along the drawn axis, $k_z = \arccos(3(\text{den}/\text{num}) \cos \Omega - 2 \cos(\pi/(w+1)))$ , the reach $(\pi w/\lambda) \sqrt{S/L}$ ; the packet's root at the lay one line at $\sum a^2 = T/\sin \Omega$ , or two lines a quarter turn apart at $S$ each, the branch's half quantum the finding (two derivations); the write at the one Node given up for the open board's giving. | D ( $\Omega$ , $S$ , the carry, the exponential, the grain, the envelope and its length, $k_z$ , the reach); L ( $\tau$ , with the reason); W ( $w$ ); A (the direction drawn by the giver, the floor's price) |
| 18 | The conversion: the record whole at $-1$ at its Node; each record out at $+1$ laid by the count at the massless pair, $A^2 = T \text{div } 2$ ; the weights [window, rate - window] | the count's line; the one act                                | The weights the hazard form of line 17 (the constant hazard, the geometric). The lay at [den, den]: the one-Node lay of a massless record carries no frequency (a delta over the band), its share $T$ by the count alone. The table and the rate are nature's numbers (the neutron's lifetime), not the law's. The engine's findings stand in the folder's document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | D (the weights' form, the lay); W (the table and the rate)                                                                                                                                                     |
| 19 | The null window                                                                                                                                                                     | the one act                                                  | The complement outcome: the record re-laid in its part at its count, the coherence ended; the projection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | D (the form); F (its cause line 15's)                                                                                                                                                                          |
| 20 | The face over two intervals                                                                                                                                                         | Rule3's line; the one act                                    | Rule3 solved for one Port's arrival so the step writes 0 with its remainder below one read coefficient; the second interval sets the before; the inverse presents the same values, MATCH across the clicks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | D                                                                                                                                                                                                              |
| 21 | The front                                                                                                                                                                           | the one act; line 10                                         | One Link per interval, so beyond the ball the NodeState is bit for bit the run without the front (the invisibility theorem); it erases the empty wave of a record at count 0, which the count's conservation requires.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | D (the form)                                                                                                                                                                                                   |
| 22 | The receding face: layers of zeros where a level reaches the layer before the face                                                                                                  | Rule3's line                                                 | A grown Node at the start's state of a Node with no level is exactly the state the infinite board would hold there, so the run is the larger board's bit for bit (theorem; one copy per layers of intervals).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | D                                                                                                                                                                                                              |
| 23 | The generator's lay                                                                                                                                                                 | line 5                                                       | The engine's own start on each record, one function.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | D                                                                                                                                                                                                              |
| 24 | The loader's refusals                                                                                                                                                               | Rule3's line; the count's line; the families' declaration    | A region narrower than half a wavelength across the beam refused: a share read over less than half a wavelength is not the count's (the plane wave's share per Node oscillates as $\sin^2$ and sums to the count only over a half period). The guard's refusals: line 4. A frozen content refused: the pace rounds to 0, the clock stops. The rest's cycle refused: line 5. The amplitude bound $A$ : the width's room with the turn's reach $2A + 2$ (line 9). A giving at $\cos \Omega$ below $1/3$ refused: no axis of light's band carries it, the diagonals' lay. The missing resonance key: the file's. The dimension bounded: the families' declaration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | D (the half wavelength, the guard, the frozen clock, the cycle, the bound, the band's top); W (the resonance key); L (the dimensions, the pairs)                                                               |
| 25 | The sign's rows                                                                                                                                                                     | the sign holder; the families' declaration                   | The row sourced by its owner's Wronskian over $E_s T$ (two derivations); one row per record the law's structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D (the sourcing); L (the structure)                                                                                                                                                                            |
| 26 | The back-in-time gate: MATCH across the faces, the clicks and the erasures by Rule3's inverse alone; the lays crossed by the host's tool                                            | lines 2, 9, 20                                               | Every number a step reads is one the state after the interval still holds, so the inverse returns every level and remainder; the draw is forward only, the one act outside the reversal, and its writes are crossed (theorem; the two slits' MATCH over 40 intervals at the pinned commit, $N = 270$ against the law's line's 269.3 at the same lay within the integer step's walk).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | D                                                                                                                                                                                                              |

| #  | The act                                    | From the law's lines                        | The derivation, the proof shown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Status                                                                                                                                                               |
|----|--------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27 | The NodeReader on a body's modes, no label | Rule3 linear; the conserved form; the start | The modes are the start's fixed points (the rest, and the next standing in the weighted sum, $\phi_n^\top \mathbf{D} \phi_n = 0$ , the plain sum at uniform paces alone; $\cos \omega_n = \phi_n^\top \mathbf{D} \mathbf{M} \phi_n / (2w \phi_n^\top \mathbf{D} \phi_n)$ , the transition read. The share: the mode's own invariant from the record's two time levels projected on the mode, $u_n = \phi_n \cdot z_{\text{now}}$ , $v_n = \phi_n \cdot z_{\text{before}}$ , $Q_n = [w(u_n^2 + v_n^2) - 2w \cos \omega_n u_n v_n] / (\phi_n \cdot \phi_n)$ , phase-free, $Q(z) = \sum Q_n$ with no cross term, $p_n = Q_n / Q(z)$ ( $p = 0.3685$ and $0.6315$ constant through 60 intervals, the sum 1 within 0.3 percent at levels rounded to 1/30). The draw's weights products of the levels, no root. Declared on a body: where, the seed, $\tau$ . | X (fallen by the owner's word: a bound body records nothing; a reader of Nodes alone is never on one Node, while a reader with a record of its own may stand on one) |
| 28 | The integer floor                          | the Port reads levels; the invariant        | A quantum carried as a wave occupies at most $T/(2 \sin \Omega)$ Nodes; a one-Node source's spherical shell is below one level beyond $0.28 A_t$ Links; a packet of width $w$ beyond $(\pi w / \lambda) \sqrt{S/L}$ . The method's bound stated explicitly (the law's document, what is open, item 49); nature's single photon in the open is not on the board as written.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | D (the bound); F (the method's reach)                                                                                                                                |

## The implementation's versions of the click's write and their readings (Sections 5.3, 5.4, 7.3 and 10.3)

**Derivation S.62** (the implementation's versions of the click's write and their readings, Sections 5.3, 5.4, 7.3 and 10.3). *The implementation's versions of the click's write and their readings, which this supplement recorded here before the paper and the implementation were separated, stand in the engine's document and in the folders' documents of the archive (the engine's document, the interval; the gates) and not in this paper, which reports one run, the two slits' (Sections 3.7 and 9 of the paper). What the paper takes of them is derived in S.50, S.57, S.59 and S.61, with no reading. Status: the paper reports no reading of the implementation but the two slits'.*

## The slow limit (Section 3.3)

**Derivation S.63** (Schrödinger's equation as the slow limit of the line, Section 3.3). *Inputs: Rule3's line at the vacuum's paces without the remainder,  $a_{\text{next}} + a_{\text{before}} = (\text{num}/3 \text{ den}) \sum_{\text{Ports}} \text{arr}$  (S.1);  $\cos \omega_0 = \text{num}/\text{den}$ ;  $\Delta$  the six-Port second difference,  $\sum_{\text{Ports}} a = 6a + \Delta a$ ; the rest rotation at a content,  $\omega_0(U) = \omega_0 - \omega_0 f(\omega_0) U + O(U^2)$  (S.15). Steps: write  $a_t = \text{Re}[\psi_t e^{-i\omega_0 t}]$  with  $\psi$  slowly varying,  $\psi_{t\pm 1} = \psi \pm \psi' + \psi''/2 + \dots$ ; then  $a_{\text{next}} + a_{\text{before}} = \text{Re}[e^{-i\omega_0 t} (2 \cos \omega_0 \psi - 2i \sin \omega_0 \psi' + \cos \omega_0 \psi'')]$ , and the right side is  $\text{Re}[e^{-i\omega_0 t} (2 \cos \omega_0 \psi + (\text{num}/3 \text{ den}) \Delta \psi)]$ , since  $(\text{num}/3 \text{ den}) 6 = 2 \cos \omega_0$ . The  $2 \cos \omega_0 \psi$  cancels, the band at  $k = 0$ ; dropping  $\psi''$ ,*

$$i\psi' = -\frac{\text{num}}{6 \text{ den} \sin \omega_0} \Delta \psi = -\frac{\Delta \psi}{2m^*}, \quad m^* = \frac{3 \text{ den} \sin \omega_0}{\text{num}} = 3 \tan \omega_0,$$

*Schrödinger's free equation on the lattice,  $\hbar = 1$  per interval, the inertia the band's (S.18, S.22: the band's own expansion  $\omega - \omega_0 = k^2/(2m^*)$  is the same statement). At a content  $U$  the same steps with  $\omega_0(U)$  add  $V\psi$  with  $V = -\omega_0 f(\omega_0) U$ , the well lowering the rotation. The three conditions, each with its order: (i) the remainder dropped, below one level per Node per interval (the budget clause), the hard bound  $n$  times it and the walk  $\sigma\sqrt{n}$ ; (ii) long wavelength, the band's  $k^4$  term dropped, of relative size  $k^2/12$  along an axis; (iii) slow variation,  $\psi''$  dropped against  $2 \sin \omega_0 \psi'$ , of relative size  $(\omega - \omega_0)/(2 \tan \omega_0)$ , below  $(\omega - \omega_0)/\omega_0 = k^2/(2m^*\omega_0)$ , the Klein-Gordon correction. Within the three the statement is exact algebra of the line; in nature's names it is the slow limit of the discrete Klein-Gordon equation, and the model's own content is  $m^* = 3 \tan \omega_0$ , the potential's coefficient  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$  and the lattice's departures named. Check (`schrödinger_check.py`): a Gaussian packet of amplitude width 40 Links at  $k_0 = 0.08$  on a chain of 4,000 Nodes with the two other axes folded, laid with the exact band's before level and stepped 3,000 intervals by the line at  $[2, 3]$  ( $m^* = 3.3541$ ): the envelope's centre at 71.4 Links against Schrödinger's  $k_0 T/m^* = 71.6$  (the exact band's group velocity times  $T$ , 71.4); the density's rms width 32.38 against Schrödinger's  $a\sqrt{1 + (T/(2m^*a^2))^2} = 32.40$  with  $a = 40/\sqrt{2}$ . Status: derived, the slow limit with its three conditions named, three derivations (the writer's, the advisor's and the reviewer's, R271); GameBoard.*